

* Couple $N=1$ sugra to chiral and vector multiplets.

-> Start with $(n+1)$ -chiral mult. Φ^i

$$(\bar{\chi}^{\bar{I}}, \Omega^I, F^I), \quad I=0, \dots, n$$

• One of these are the compensating mult.

Reserve $(\bar{\chi}^{\bar{a}}, \chi^a, F^a)$ for the physical chiral multiplets.

* In coupling $(n+1)$ chiral fields Φ^i we will obtain Two different Kähler metrics

(i) The embedding mfd, which has complex dim $(n+1)$ coord's $X^{\bar{I}}, \bar{\chi}^{\bar{I}}$ w/ signature $(-, +, +, \dots, +)$

- component corresponds to the scalar compensator

↳ This will be an extension of the m-linear σ -model

• We then fix the gauge for these.

n -chiral mult's will remain and represent the physical theory

• These have a different geometry: Projective Kähler mfd.
(Kähler-hodge mfd) w/ metric $(+, \dots, +)$

Gauge ^{vector} Multiplets: $(A_\mu^A, \lambda^A, D^A) \quad (V^T(x^\mu, \theta, \bar{\theta}) = V(x^\mu, \theta, \bar{\theta}))$

Superconformal Construction of chiral mult. coupling to gravity

(Weyl mult $\oplus$ $(n+1)$ - Φ^i) $\Rightarrow$ Embedding Kähler mfd w/ Conf. symm

$\Downarrow$ (Gauge fix $\oplus$ field equations)

(Poincaré sugra $\oplus$ n -chiral Φ^i) $\Rightarrow$ Projective Kähler for physical fields

* Recall:

Real Multiplet: lowest component is a real scalar (CM)

$$\delta C = \bar{\epsilon} \chi, \quad \chi \mapsto i \gamma_\mu \dot{\chi}$$

$$(L, S, H, B_m, \lambda, D)$$

Supercent Gauge Mult: $(A_m^A \oplus \lambda^A)$, D^A : Auxillary field
 $\rightarrow$ The internal symmetry which acts on A_m^A & λ^A is not part of the supercent. transformations. This means they must commute w/ dilatations and R-symmetry.

i.e. let $T^{(A)}$ denote the generator for these symmetries, then

$$[D, T^{(A)}] = 0 \quad \text{and} \quad [T^{(R)}, T^{(A)}] = 0$$

so the Weyl & R-weight of the vector potential A_m^A vanish. This then fixes the weights of the other fields in the multiplet.

$$h_\lambda = \frac{1}{2}, \quad h_D = 2$$

Q: why? If $h_\lambda = 0 \Leftrightarrow h_\lambda = \frac{1}{2}$

$$\begin{aligned} [D, Q] A_m^A &= D(-\frac{1}{2} \gamma_m \lambda^A) - Q D(A_m^A) \\ &= -\frac{1}{2} h_\lambda \gamma_m \lambda^A \\ &= \frac{1}{2} (Q A_m^A) = -\frac{1}{4} \gamma_m \lambda^A \end{aligned}$$

$$\Rightarrow (h_\lambda = \frac{1}{2}) ???$$

$$\rightarrow [D, Q] \lambda^A = D(Q \lambda^A) - Q(D \lambda^A) = \frac{1}{2} Q \lambda^A$$

$$\cdot D(Q \lambda^A) = h_\lambda Q \lambda^A = \frac{1}{2} Q \lambda^A$$

$$\cdot (h_{Q\lambda}) Q \lambda^A - h_\lambda Q \lambda^A = \frac{1}{2} Q \lambda^A$$

$$\therefore h_{Q\lambda} - h_\lambda = \frac{1}{2} \Leftrightarrow \boxed{h_{Q\lambda} = h_\lambda + \frac{1}{2}}$$

$$\Leftrightarrow A_m^A \text{ has chiral weight } \underline{h=1} \text{ not } h_\lambda = 0$$

with this, we have, omitting lowest:

$$\delta A_m^A = -\frac{1}{2} \bar{\epsilon} \gamma_m \lambda^A$$

$$\delta \lambda^A = \frac{1}{2} [\lambda_D + i \gamma_5 \lambda_-] \lambda^A + [\frac{1}{4} \gamma^{ab} \hat{F}_{ab} + \frac{1}{2} i \gamma_5 D^A] \epsilon$$

$$\delta D^A = 2 \lambda_D D^A + \frac{1}{2} i \bar{\epsilon} \gamma_5 \gamma^\mu D_\mu \lambda^A$$

$$F_{ab} := e_a e_b (\frac{1}{2} d(\omega_{ab}) + \frac{1}{4} \omega_{ab} \omega_{cd} + \frac{1}{2} \omega_{ab} \omega_{cd})$$

$$D_\mu \lambda^A := (\partial_\mu - \frac{3}{2} b_\mu + \frac{1}{4} \omega_\mu^{ab} \gamma_{ab} - \frac{3}{2} i \gamma_\mu A_\mu) \lambda^A + \lambda^B A_\mu f_{B\mu}^A$$

$$- [\frac{1}{4} \gamma^{ab} \hat{F}_{ab}^A + \frac{1}{2} i \gamma_\mu D^A] \psi_\mu$$

$\Rightarrow$ Further imply that $\delta_p^A A_\mu^A = h_A^\mu \lambda_0 A_\mu^A = 0$
 and so since $\mathbb{Z}$ has chiral weight $\frac{1}{2}$, A_μ^A must have
 chiral weight of $\underline{2}$

Superconformal Real multiplet:

$(C, \{, B_\mu, \mathcal{H})$: must determine the chiral weights.

$C \in \mathbb{R} \Rightarrow$ it cannot undergo a complex transformation.

$$\Rightarrow \boxed{h_C = 0}$$

* General feature: field $\phi(x)$ valued in $\mathbb{R}$ cannot have
 $h_\phi \geq 0$.

$$[D, Q]C = \frac{1}{2} QC \Leftrightarrow h_C = 0$$

$$\text{since } QC = \frac{1}{2} i \gamma_\mu \{$$

If C has $h_C = -\frac{1}{2}$, then

$$\delta_Q C = \frac{1}{2} i \bar{\epsilon} (\gamma_\mu \{$$

$$\frac{1}{2} \gamma_\mu \{ + \frac{1}{2}$$

$\Rightarrow (C, P_L \{, B_\mu, \mathcal{H}, D)$ form the Real multiplet and
not }

Take $P_L \{$:

$$[D, Q] P_L \{ = D(P_L (i \mathcal{H} \cdot \beta - i \mathcal{D} C) - i \omega C P_L \mathcal{H})$$

$$- Q h_\gamma P_L \{ = \frac{1}{2} Q P_L \{$$

• Putting these together:

$$\delta C = \frac{1}{2} i \bar{\epsilon} \gamma_\mu \{$$

$$\delta H = -i \bar{E} P_R (\lambda + \not{D} \zeta) + (h-2) i \bar{\eta} P_L \zeta$$

$$\delta B_c = -\frac{1}{2} \bar{E} (\gamma_c \lambda + \not{D}_c \zeta) + \frac{1}{2} (1+h) \bar{\eta} \gamma_c \zeta$$

→ (Superconformal transformations) \ (gets, Lorentz, dilatations,) $\nearrow$ T_R

$$\delta P_R \lambda = P_R \left[\frac{1}{4} \gamma^{ab} F_{ab} - i \frac{1}{2} D \right] \epsilon + \frac{1}{2} h P_R [i \not{H} - \not{D} + i \not{D} C] \eta - h \gamma_c P_L \zeta$$

$$\delta D = \frac{1}{2} i \bar{E} \not{D} \lambda + i h \bar{\eta} (\lambda + \frac{1}{2} \not{D} \zeta) - 2 h \lambda_c^c D_c C$$

where $F_{ab} := 2 D_{[a} B_{b]} + \epsilon_{ab cd} D^c D^d C$

- $\mathcal{D}_m$ are covariant under all symmetries of $S\text{-conf}(M^D)$

Weyl multiplet:

field:	e_m^a	$P_L \psi_m$	b_m	ω_m^{ab}	A_m	$P_L \phi_m$	f_m^a
h	-1	$-\frac{1}{2}$	0	0	0	$\frac{1}{2}$	1
c	0	$\frac{3}{2}$	0	0	0	$-\frac{1}{2}$	0

For this multiplet it is easy to see from $\{D, \cdot\}$ that

$$\delta(\epsilon) A_m^A = \mathcal{D}_m \epsilon^A + \epsilon^C A_m^B f_{BC}^A \quad \text{and}$$

$$1: P_c \Rightarrow e_m^c \text{ has } -1$$

$$\frac{1}{2}: Q \Rightarrow \psi: -\frac{1}{2} \leftarrow \text{actually } P_L \psi$$

$$0: D, M_{ab}, T_R \quad b_m = 0, \omega_m^{ab} = 0, A_m = 0$$

$$-\frac{1}{2}: S$$

$$-1: K_a$$

and so on. we can simply see if $\{D, T^{(i)}\} = 0$
in which case the Gauge fields of this symm.
also carry zero conf. weight.

Chiral Mult:

h	h	$h + \frac{1}{2}$	$h + 1$
c	h	$h - \frac{3}{2}$	$h - 3$

Real mult:

	C	$P_L \lambda$	χ	B_c	$P_R \lambda$	D
h	h	$h + \frac{1}{2}$	$h + 1$	$h + 1$	$h + \frac{1}{2}$	$h + 2$
c	0	$-\frac{3}{2}$	-3	0	$-\frac{3}{2}$	0

Gauge Mult:

	A_μ^A	$P_R \lambda^A$	D^A
h	0	$\frac{3}{2}$	2
c	0	$-\frac{3}{2}$	0

$$\delta A_\mu^A = -\frac{1}{2} \epsilon \gamma_\mu (P_L \lambda^A + P_R \lambda^A) = -\frac{1}{2} \epsilon \gamma_\mu P_L \lambda^A - \frac{1}{2} \epsilon \gamma_\mu P_R \lambda^A$$

Since

$$\mathcal{M}^D := (\text{Kähler Mfd})$$

Gauge transformations of superconf. Chiral Mult.:

- Gauge symm must be compatible w/ the Conf. struc.
So, Gauge & Superconf. must commute:

$$[(\text{Gauge}), Q] = [(\text{Gauge}), S] = 0$$

- For scalar fields X^I ; Recall a target space

$$X: \mathcal{M}^D \hookrightarrow M^D \leftarrow \text{Kähler}$$

$$X'^I = X^I + \theta^A V_A^I(X)$$

for a much more general Killing vector on $\mathcal{M}^D$

Naturally, θ^A do not carry a weight and

weight as χ^I

Notation:

$$\Omega^I := P_L \Omega^I$$
$$\Omega^{\bar{I}} := P_R \Omega^I$$

$$\delta \chi^I = \frac{1}{\sqrt{2}} \bar{\epsilon} \Omega^I$$

$$\delta \Omega^I = \frac{1}{\sqrt{2}} P_L (\not{D} \chi^I + F^I) \epsilon + \sqrt{2} h \chi^I P_L \eta$$

$$\delta F^I = \frac{1}{\sqrt{2}} \bar{\epsilon} \not{D} \Omega^I + \bar{\epsilon} P_R \lambda^A \psi_A^I(x) + \sqrt{2} (1-h) \bar{\eta} \Omega^I$$

↳ Does not include A_0 or λ_R , however, we now have the extra (general) Killing vectors $K_A^\pm(x)$. Here $\not{D}$ contains the Gauge conn. terms A_μ as well as b_μ , (which account for the "missing" terms above?).

Chiral Mult. spinors:

(i) omit P_L for Ω^I , understanding that the projection still exists. (why not just use $\pm$ and $\bar{\pm}$ of $SL(2, \mathbb{C})$ then?)

Ω^I : 4-component Dirac

$SO(4)$: $\bigcirc - \bigcirc$, $\frac{4}{2}$ spinor

• In the Weyl rep, Ω^I : Left-handed.

(ii) Chiral: (χ^I, Ω^I, F^I)

Anti-chiral: $(\bar{\chi}^{\bar{I}}, \Omega^{\bar{I}}, \bar{F}^{\bar{I}})$

$$(\chi^I)^* = \bar{\chi}^{\bar{I}}$$

$\Omega^{\bar{I}}$ is the conjugate of Ω^I

$$(\Omega^I)^c = \gamma^5 \Omega^I^* \equiv \Omega^{\bar{I}}$$

field ψ can be written in terms of Weyl spinors $P_L \psi$ and its complex conj.

$$(P_L \psi)^c = P_R \psi^c$$

$$\psi = P_L \psi + P_R \psi^c$$

or if $\chi = P_L \chi$, $\chi^c = P_R \chi^c$

$$\Rightarrow \psi = i \chi + \chi^c$$

we have a Majorana spinor:

$$\psi = \psi^c \Leftrightarrow \psi^I = \Omega^I + \Omega^{\bar{I}}$$

Explicitly: $\bar{\Omega}^I = (\Omega^I)^T C = \bar{\Omega}^I P_L$
 $\bar{\Omega}^{\bar{I}} = (\Omega^{\bar{I}})^T C = \bar{\Omega}^{\bar{I}} P_R$

Invariant actions

For a superconf. theory we cannot use $\mathcal{L}_{\text{free}}$ as it is not invariant. We use:

$$S_F = \int d^4x \text{Re} \left\{ F + \frac{1}{2} \bar{\psi}_\mu \gamma^\mu P_L \chi + \frac{1}{2} \bar{\chi} \gamma^\mu P_R \psi_\mu \right\}$$

$\rightarrow (Z, \chi, F)$: Chiral mult. of Weyl weight
 $\underline{h_F = 4}$

why?:

$\left\{ \begin{array}{l} e: \text{ has Weyl weight } -4, \text{ so } F \text{ must} \\ \text{ have } +4 \Rightarrow \underline{h_Z = 3} \end{array} \right.$

$\bullet F$: the F-term is the only invariant action

lowest weight $h_2 = 2$.

→ For $h=2$, $\mathcal{H}$ does NOT transform under S -susy. $\Rightarrow \mathcal{H}$ as the lowest comp. of an Anti-chiral mult.

$$\delta \mathcal{H} = -i \bar{\epsilon} P_R (\lambda + \not{D} \zeta) + (h-2) i \bar{\epsilon} P_L \zeta$$

$$\Rightarrow h=2$$

$$\delta \mathcal{H} = -i \bar{\epsilon} P_R (\lambda + \not{D} \zeta)$$

$$\Rightarrow (\tilde{\mathcal{H}}, \tilde{\chi}, \tilde{F})$$

$$\text{where } \left. \begin{aligned} \tilde{\chi} &= -i \chi_R - i \not{D} \zeta_R \\ \tilde{F} &= D + D^c C + i D_a B^a \end{aligned} \right\}$$

proof:

$$\tilde{Z} := -\bar{\mathcal{H}}$$

$$\text{Chiral: } \delta Z = 0 + \frac{1}{\sqrt{2}} \bar{\epsilon} P_L \chi$$

$$\delta P_L \chi = \frac{1}{\sqrt{2}} P_L (\not{D} Z + F) \epsilon + \sqrt{2} h Z P_R \chi$$

$$\delta F = \frac{1}{\sqrt{2}} \bar{\epsilon} \not{D}$$

Real mult, weyl $h=2$

$$\delta \mathcal{H} = -i \bar{\epsilon} P_R (\lambda + \not{D} \zeta)$$

$$\Rightarrow \delta \bar{\mathcal{H}} = i \lambda P_L \epsilon + i D_m \bar{\zeta} P_L \chi^m \epsilon$$

$$= +i \bar{\epsilon} P_R \lambda + i \bar{\epsilon} P_L \not{D} \zeta$$

$$= +i \bar{\epsilon} P_L (\lambda + \not{D} \zeta)$$

$$\therefore \delta(-\bar{\mathcal{H}}) = \epsilon (-i P_L (\lambda + \not{D} \zeta))$$

$$\frac{1}{\sqrt{2}} P_L \tilde{\chi} = -i P_L (\not{D} \zeta + \lambda)$$

$$\therefore P_L \tilde{\chi} = -i \sqrt{2} (P_L (\lambda + \not{D} \zeta))$$

$$(Z = -\bar{\mathcal{H}}, P_L \chi = -\sqrt{2} i P_L (\lambda + \not{D} \zeta), F = D + D^a C + i D^a B_a)$$

→ Real mult fields

* The general $\mathcal{N}=1$ field term will then be

$$S_F = \int d^4x e \text{Re} \{ F + \frac{1}{\sqrt{2}} \bar{\Psi}_\mu \gamma^\mu P_L \chi + \frac{1}{2} \bar{\chi} \gamma^{\mu\nu} P_R F_{\mu\nu} \}$$

$$= \int d^4x e \text{Re} \{ F + \bar{\Psi}_\mu \gamma^\mu (-\sqrt{2}i P_L (\lambda + \not{D}\zeta)) + \frac{1}{2} (-F) \bar{\Psi}_\mu \gamma^{\mu\nu} P_R \Psi_\nu \}$$

$$= \int d^4x e \text{Re} \{ D + D^\alpha D_\alpha C + i \not{D}_a B^a + (-i\sqrt{2}) \bar{\Psi}_\mu \gamma^\mu P_L \lambda - i\sqrt{2} \bar{\Psi}_\mu \gamma^\mu \not{D}\zeta - \frac{1}{2} \bar{\Psi}_\mu \gamma^{\mu\nu} P_R \Psi_\nu F \}$$

$$\not{D}_a B^a = \not{\partial}_a B^a - \frac{1}{2} \bar{\Psi}_a (\not{\partial}_a \lambda + \not{D}_a \zeta) + \frac{3}{2} \not{\partial}_a \zeta$$

$$S_D = \int d^4x D$$

• Not sure where $\not{\partial}_a B^a$ went.

$$D(\frac{1}{2}\epsilon) \sim \not{\epsilon} \alpha_{\bar{\beta}} (-\not{\partial}_\mu \bar{\chi}^\alpha \not{\partial}^\mu \bar{\chi}^\beta - \frac{1}{2} \bar{\chi}^\alpha P_L \not{\partial} \chi^\beta - \frac{1}{2} \bar{\chi}^\beta P_R \not{\partial} \chi^\alpha + F^\alpha \bar{F}^\beta) + \not{\partial}(\not{\epsilon} \alpha_{\bar{\beta}} \bar{\chi})$$

$$S_D = \int d^4x e [D + \not{D}^\alpha D_\alpha C + (\frac{1}{2}i \bar{\Psi}_\mu \gamma^\mu P_R (\lambda + \not{D}\zeta) - \frac{1}{4} \bar{\Psi}_\mu P_L \gamma^{\mu\nu} P_R \Psi_\nu F + \text{h.c.})]$$

These covariant der's are

$$D_a C := \partial_a C - 2b_a C - \frac{1}{2} i \bar{\Psi}_a \gamma_* \zeta$$

$$D_a P_L \zeta = (\partial_a + \frac{1}{4} \omega_a^{bc} \gamma_{bc} - \frac{5}{2} b_a + \frac{3}{2} A_a i) \zeta - \frac{1}{2} P_L (i\gamma - \not{B} - \not{\chi} \gamma_a) \zeta + 2i C P_L \not{\partial}_a \zeta$$

Conformal Replace: $\not{D}^C \equiv D^\alpha D_\alpha$

$$\square^c C := \partial^a D_a C - 3b^a D_a C + \omega_a^{ab} p_b C + 4C f_a^a \\ - \frac{1}{2} i \bar{\Psi}^a \gamma_a D_a \zeta + \frac{1}{2} i (\bar{\Psi} \cdot \gamma) \gamma_* \zeta$$

Claim: In terms of graviton & gravitino curvatures $R(\omega)$, $R'(\alpha)$, the Action takes the form

$$S_D = \int d^4 x e \left[D - \frac{1}{2} \bar{\Psi} \cdot \gamma i \gamma_* \lambda - \frac{1}{3} C R(\omega) \right. \\ \left. + \frac{1}{6} (C \bar{\Psi}_m \gamma^{\mu\nu\rho\sigma} - i \bar{\zeta} \gamma^{\rho\sigma} \gamma_*) R'_\rho(\alpha) \right. \\ \left. + \frac{1}{4} \varepsilon^{abcd} \bar{\Psi}_a \gamma_b \psi_c (B_d - \frac{1}{2} \bar{\Psi}_d \zeta) \right]$$

(Super-conformal extension of "D-term" for global supersymmetric action)
"Master formula"

Proof: Recall $\omega_\mu^{ab}(e, \psi) = \omega_\mu^{ab}(e, \gamma) + \frac{1}{2} \psi_\mu \gamma^{ab} \psi$
 $+ \frac{1}{4} \bar{\psi}^a \gamma_\mu \psi^b$

$$\gamma^\mu \psi_\mu = -\frac{1}{6} \gamma^{\mu\nu} R_{\mu\nu}(\alpha)$$

$$f_\mu^m = -\frac{1}{12} \hat{R} = -\frac{1}{12} (R(\omega) - \psi_a \gamma^{ab} \psi_b)$$

→ Insert cov. der. into previous S_D expression

$$S_D = \int d^4 x e \left[D + \square^c C + \frac{1}{2} (i \bar{\Psi} \cdot \gamma \underline{P_L (\lambda + \mathcal{D} \zeta)}) \right. \\ \left. - \frac{1}{4} \bar{\Psi}_m \underline{P_L \gamma^{\mu\nu} \psi_\nu \mathcal{H}} + \text{L.C.} \right]$$

$$\rightarrow (\partial^a - 3b^a + \omega_b^{ba}) (\partial_a C - 2b_a C - \frac{1}{2} i \bar{\Psi}_a \gamma_* \zeta) \\ + 4C (-\frac{1}{12} \hat{R}) + \frac{1}{2} i (\bar{\Psi} \cdot \gamma) \gamma_* \zeta \\ - \frac{1}{2} i \bar{\Psi}^a \gamma_* \left[(\partial_a + \frac{1}{4} \omega_{bc} \gamma_{bc} - \frac{5}{2} b_a + \frac{3}{2} i \gamma_a) \zeta \right. \\ \left. - \frac{1}{2} P_L (i \mathcal{H} - \mathcal{D} - i \gamma^a (\partial_a C - 2b_a C - \frac{1}{2} i \bar{\Psi}_a \gamma_* \zeta) \right. \\ \left. + 2 i C P_L \psi_a) \right]$$

→ Since $\omega_a^{ab} = \omega_c^{ab}(e) + 3b^c + \frac{1}{2}(4 \cdot 1) \psi^c$

$$\hookrightarrow \underline{\partial^a} + \omega_b^{ba}(e) + \frac{1}{2}(\bar{\psi} \cdot \gamma) \psi^b$$

Total derivative term $\rightarrow 0$

$$\rightarrow = \frac{1}{2}(\bar{\psi} \cdot \gamma) \psi^b D_b C - \frac{1}{3} C \hat{R} + \frac{i}{2}(\bar{\psi} \cdot \gamma) \gamma_4 \zeta \\ + \left(\frac{i}{2} \bar{\psi}^a \gamma_4 (D_a \zeta) + P_L (i \mathcal{H} - \cancel{B} - i \gamma^a D_a C) \psi_a \right)$$

$$= \frac{1}{2}(\bar{\psi} \cdot \gamma) \psi^b D_b C - \frac{1}{3} C \hat{R} + \frac{i}{2}(\bar{\psi} \cdot \gamma) \gamma_4 \zeta \\ - \frac{i}{2} \bar{\psi}^a P_L D_a \zeta + \frac{1}{2} \mathcal{H} \bar{\psi}_a \gamma_4 P_L \psi_a$$

$$\hookrightarrow \frac{1}{2} i (\bar{\psi} \cdot \gamma) P_R (\lambda + \gamma^a D_a \zeta)$$

→ Nothing to expand here

$$\therefore S_D = \int d^4x e [D + \dots + \dots] \gamma_4 \zeta \\ = \int d^4x e [D + \frac{1}{2}(\bar{\psi} \cdot \gamma) \psi^b D_b C - \frac{1}{3} C \hat{R} + \frac{1}{2} i (\bar{\psi} \cdot \gamma) \\ + (\frac{i}{2} \bar{\psi} \cdot \gamma P_R \lambda + \frac{1}{2} i \bar{\psi}_a \gamma^{ab} \underline{\hat{D}_b P_L \zeta}) - \frac{1}{4} \bar{\psi}_a P_L \gamma^{ab} \cancel{P}_4 \mathcal{H} \psi_b + \dots]$$

$$\text{Replace } \hat{R} = R(\omega) - \psi_c \gamma^{ab} \phi_b$$

$$\int d^4x e [D + \dots + -\frac{1}{3} C (R(\omega) - \bar{\psi}_a \gamma^{ab} \phi_b) + \dots]$$

$$\hat{D}_a P_L \zeta = D_a \zeta - \frac{1}{2} P_L (i \mathcal{H} - \cancel{B} - i \cancel{D} C) \psi_a \\ + 2i C P_L \phi_a$$

$$\rightarrow \frac{1}{2} i \bar{\psi}_a \gamma^{ab} D_a \zeta + \frac{1}{4} \bar{\psi}_a \gamma^{ab} P_L \mathcal{H} \psi_b \\ + \frac{1}{4} i \bar{\psi}_a \gamma^{ab} (\cancel{B} + i \cancel{D} C) \psi_b$$

→ Cancel

$$\int d^4x e \left[D - \frac{1}{2} (\bar{\Psi} \cdot \gamma) \gamma^\mu D_\mu \Psi - \frac{1}{3} C R(\omega) \right. \\
+ \frac{1}{3} \bar{\Psi}_a \gamma^{ab} \psi_b + \frac{1}{2} i (\bar{\Psi} \cdot \gamma) \gamma^\mu \gamma_\mu \gamma^\nu D_\nu \Psi \\
+ \left. \left(\frac{1}{2} i (\bar{\Psi} \cdot \gamma) \gamma^\mu \gamma_\mu \lambda + \frac{1}{2} i \bar{\Psi}_a \gamma^{ab} D_b \psi_a \right) \right. \\
+ \left. \left(\frac{1}{2} i \right) \left(\frac{1}{4} \right) \bar{\Psi}_a \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma (B + i D C) \psi_b \right] + h.c.]$$

$$\gamma^{ab} d_b = -\frac{1}{4} \gamma^{abc} R'_{bc}(Q)$$

$$\rightarrow -\frac{1}{12} \bar{\Psi}_a \gamma^{abc} R'_{bc}(Q)$$

$$\rightarrow -\frac{1}{8} \bar{\Psi}_a \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma B_c \psi_b = -\frac{1}{16} \bar{\Psi}_a (\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) \gamma^{ab} \gamma^c B_c \psi_b \\
\sim -\frac{1}{16} \bar{\Psi}_a \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^c B_c \psi_b$$

$$\gamma^{abc} = \frac{1}{2} (\gamma^a \gamma^b \gamma^c - \gamma^c \gamma^b \gamma^a) \\
= \frac{1}{2} (\gamma^a \gamma^b \gamma^c + \gamma^b \gamma^c \gamma^a + \gamma^c \gamma^a \gamma^b)$$

$$\Rightarrow \gamma^{abc} \gamma^d \sim a \gamma^{abc}$$

$$D=4: \gamma^{abc} \gamma_d = i \epsilon^{abcd} \gamma_d$$

$$\rightarrow -\frac{a}{16} i \epsilon^{abcd} \bar{\Psi}_a \gamma_d B_c \psi_b \quad \frac{i(\bar{\Psi} \cdot \gamma) \gamma^\mu \gamma_\mu \gamma^\nu D_\nu \Psi}{2}$$

$$S_b = \int d^4x e \left[D - \frac{1}{2} (\bar{\Psi} \cdot \gamma) \gamma^\mu \gamma_\mu \gamma^\nu D_\nu \Psi - \frac{1}{3} C R(\omega) + \right. \\
+ \frac{1}{6} C \bar{\Psi}_a \gamma^{abc} R'_{bc}(Q) + \frac{1}{2} i \bar{\Psi}_a \gamma^{ab} \gamma^\mu D_\mu \psi_a \\
+ \left. \frac{1}{4} \epsilon^{abcd} \bar{\Psi}_a \gamma_d \psi_b B_c \right]$$

$$\Rightarrow (a = -4)$$

$$\rightarrow \text{Claim: } \frac{1}{2} i \bar{\Psi}_a \gamma^{ab} \gamma^\mu D_\mu \psi_b = -\frac{1}{4} i \bar{\Psi} \gamma^{\rho\sigma} \gamma^\mu R'_{\rho\sigma}(Q) \\
= \frac{1}{8} \epsilon^{abcd} (\dots) + (\dots) \\
\text{total der.} \quad \nearrow$$

Proof: Since

$$LHS: \frac{1}{2} i \bar{\psi}_c \gamma^{ab} \gamma_\mu \left[(\partial_a + \frac{1}{4} \omega_a{}^{bc} \gamma_{bc} - \frac{5}{2} b_a + \frac{3}{2} i A_a) \right] \psi_c$$

$$- \frac{1}{2} P_c (i \gamma - \gamma - i \gamma C) \psi_c$$

$$+ 2 i C P_c \psi_c$$

→ Consider torsion terms?

$$\omega_\mu{}^{ab}(e, b) = \omega_\mu{}^{ab}(e) + 2 e_\mu^{[a} e^{b]\nu} b_\nu$$

$$\bullet \omega_\mu{}^{ab} \gamma_{ab} \psi_\nu = (\omega(e) \gamma_{ab}) + 2 e_\mu^{[a} e^{b]\sigma} b_\sigma \gamma_{ab} \psi_\nu$$

on LHS

$$\text{For } R: -\frac{1}{4} i \bar{\psi} \gamma^{\rho\sigma} \gamma_\mu (e_\mu^{[a} e^{b]\sigma} b_\sigma \gamma_{ab} \psi_\nu)$$

$$\omega_\mu{}^{ab}(\psi, b) \neq \omega_\mu{}^{ab}(e, b) + \frac{1}{2} (\bar{\psi}, \gamma) \psi^b$$

→ ψ^3 term:

$$i \omega_a{}^{cd}(\psi) [\gamma_{cd}, \gamma^{ab}] \gamma_\mu \psi_b$$

$$= i \frac{1}{2} (\bar{\psi}, \gamma) \psi^b [\gamma_{cd}, \gamma^{ab}] \gamma_\mu \psi_b$$

$$\Leftrightarrow i \omega_a{}^{cd} (\gamma_{cd} \gamma^{ab} - \gamma^{ab} \gamma_{cd}) \gamma_\mu \psi_b$$

$$i \omega_a{}^{cd} (\gamma_{cd} \gamma^{ab} + 4 \gamma_\delta - \gamma^{ab} \gamma_{cd} - 4 \gamma_\delta) \gamma_\mu \psi_b$$

$$= i \omega_a{}^{cd} 8 (\gamma_{[\delta}^a \gamma_{c]}^b) \gamma_\mu \psi_b$$

$$= i 4 (\omega_a{}^{cd} \gamma_\delta^a \gamma_c^b) \gamma_\mu \psi_b$$

$$= i 8 \omega_a{}^{cd} (\gamma_\delta^a \gamma_c^b - \gamma_c^b \gamma_\delta^a) \gamma_\mu \psi_b$$

$$= 2 i (\omega_a{}^{bd} \gamma_\delta^a - \omega_a{}^{ad} \gamma_\delta^b) \gamma_\mu \psi_b$$

$$= 8 i \gamma_c^d \gamma_\mu \omega^{[a}{}^{c}{}_{\delta} (4) \psi_b]$$

expanding in

$$\gamma_{abc} = i \epsilon_{abc}{}^d \gamma_d \gamma_\mu$$

$$\gamma_{abc} = i \epsilon_{abc}{}^d \gamma_d$$

$$\begin{aligned}
 2\gamma_{ab}\gamma_k &= i\epsilon_{abcd}\gamma^a\gamma^d \\
 &= i\epsilon_{abcd}[\gamma^{ad} + \eta^{ad}] \\
 &= i\epsilon_{abcd}\gamma^{ad}
 \end{aligned}$$

$$2\gamma_c^d\gamma_k = i\epsilon_c^d{}_{ab}\gamma^{ab}$$

$$\Rightarrow 8i\gamma_c^d\gamma_k = -4\epsilon_c^d{}_{ab}\gamma^{ab}$$

$$\rightarrow = -4\epsilon_c^d{}_{ab}\gamma^{ab}\omega_{[c}^{ca}(\psi)\psi_{b]}$$

After a Fierz rearrangement:

$$i\omega^{cd}[\gamma, \gamma]\gamma_k\psi_b = \epsilon^{abcd}\psi_d\bar{\psi}_a\gamma_b\psi_c$$

Plugging this in we finally have

$$\begin{aligned}
 S_0 = \int d^4x e & \left[1 - \frac{1}{2}(\bar{\psi}\cdot\gamma)\gamma_k\lambda - \frac{1}{2}C R(\omega) \right. \\
 & + \frac{1}{6}(C\bar{\psi}_m\gamma^{m\rho\sigma} - i\bar{\psi}\gamma^{\rho\sigma}\gamma_k)R^{\rho\sigma}{}_a{}^b(\psi) \\
 & \left. + \frac{1}{4}\epsilon^{abcd}\bar{\psi}_c\gamma_b\psi_d(B_2 - \frac{1}{2}\psi_d\bar{\psi}) \right]
 \end{aligned}$$

□

Construction of the Action:

Idea: Construct a general action of chiral & Gauge multiplets coupled to supergravity.

Ingredients:

(i) Kähler potential: $C := \frac{1}{2}N(\Phi, \bar{\Phi})$

(Text: $C \mapsto \Phi, \Phi^c \mapsto \bar{\psi}^i$)

of the embedding manifold.

(ii) Holomorphic Super Potential $W(\Phi)$
 (and its conjugate $\bar{W}$)
 [Text: $W(\Phi^i) \mapsto \eta(s^i)$]

(iii) Gauge Kinetic f^u : $f_{AB}(\Phi)$

$f_{AB}(\Phi)$: Chiral superfield transforming
 in $\text{Sym}^2(\text{Ad}_g)$

This f^u is found in the extension of

$$\int d^4x \text{Re} \left\{ \int d^2\theta \, \epsilon^{\alpha\beta} W_\alpha^A W_\beta^A \right\}$$

$$\rightarrow \int d^4x e \text{Re} \left\{ \int d^2\theta \, f_{AB}(\Phi) W_\alpha^A \epsilon^{\alpha\beta} W_\beta^B \right\}$$

Note: "e" here should be the super-gravity
 multiplet E .

• For Kähler, take $C(\Phi, \bar{\Phi}) \mapsto C(\Phi, \bar{\Phi} e^{2V})$

• Take $V(x, \theta, \bar{\theta})$ to be in the WZ-gauge

In the WZ Gauge

$$V_{WZ} \sim \theta\sigma\bar{\theta}V_m + \theta\theta\bar{\theta}\bar{\lambda} + \bar{\theta}\bar{\theta}(\theta\lambda) + (\theta\theta)(\bar{\theta}\bar{\theta})D$$

$$\Leftrightarrow V_{WZ}^3 = 0 \quad \text{or} \quad (V^\alpha V^\beta V^\gamma = 0 \text{ as in the text})$$

$$* \exp(V) = 1 + V + V^2$$

W-Z gauge: $\sqrt{2} \mathcal{K}(x) = -\phi(x)$
 $F(x) = -M(x)$

In this gauge:

$$\begin{aligned}
 V_{WZ}(x, \theta, \bar{\theta}) &:= V(x, \theta, \bar{\theta}) + \Phi + \Phi^\dagger \\
 &= C(x) - C(x) + \theta(\chi(x) - \chi(x)) + \bar{\theta}(\bar{\chi}(x) - \bar{\chi}(x)) \\
 &\quad + \theta\theta(M(x) - M(x)) + \bar{\theta}\bar{\theta}(M^*(x) - M^*(x)) \\
 &\quad + (\theta\sigma^\mu\bar{\theta})(V_\mu(x) + i\partial_\mu(\phi(x) - \phi^*(x))) \\
 &\quad + \bar{\theta}\bar{\theta}\theta(\lambda(x) - \frac{i}{2}\sigma^\mu\partial_\mu\{\bar{\chi}(x) - \bar{\chi}(x)\}) \\
 &\quad + \theta\theta\bar{\theta}(\bar{\lambda}(x) - \frac{i}{2}\bar{\sigma}^\mu\partial_\mu\{\chi(x) - \chi(x)\}) \\
 &\quad + \theta\theta\bar{\theta}\bar{\theta}[D(x) - \frac{1}{4}\square(C - C)]
 \end{aligned}$$

$$\therefore V_{WZ} = \theta\sigma^\mu\bar{\theta}(V_\mu(x) + i\partial_\mu(\phi - \phi^*)) + \dots$$

• V_μ : Gauge field, λ : Super-symmetry partner.
 D : Auxiliary field.

• We still have residual gauging as $\text{Im}(\phi)$ must still be fixed.

* Generally work in the Gauge $\text{Im}(\phi) = 0$

$$\begin{aligned}
 V_{WZ} &= \theta\sigma^\mu\bar{\theta}V_\mu + \theta\theta\bar{\theta}\chi + \bar{\theta}\bar{\theta}\theta\lambda + \theta\theta\bar{\theta}\bar{\theta}D \\
 \Rightarrow V_{WZ}^2 &\neq \text{term can only come from } \theta\sigma^\mu\bar{\theta}
 \end{aligned}$$

$$\begin{aligned}
 V_{WZ}^2 &= (\theta\sigma^\mu\bar{\theta}V_\mu)(\theta\sigma^\nu\bar{\theta}V_\nu) \\
 &= \theta^\alpha\sigma^\mu_{\alpha\bar{\alpha}}\bar{\theta}^{\bar{\alpha}}V_\mu\theta^\beta\sigma^\nu_{\beta\bar{\beta}}\bar{\theta}^{\bar{\beta}} \\
 &= \delta^\alpha_\beta\delta^{\bar{\beta}}_{\bar{\alpha}}(\theta^\gamma\sigma^\mu_{\gamma\bar{\alpha}}\bar{\theta}^{\bar{\alpha}}V_\mu\theta^\beta\sigma^\nu_{\beta\bar{\beta}}\bar{\theta}^{\bar{\beta}}) \\
 &= (\frac{1}{2}\epsilon^{\alpha\beta}[\theta\theta])(\frac{1}{2}\epsilon^{\bar{\alpha}\bar{\beta}}[\bar{\theta}\bar{\theta}]\sigma^\mu_{\alpha\bar{\alpha}}\sigma^\nu_{\beta\bar{\beta}})VV \\
 &= \frac{1}{4}\theta\theta\bar{\theta}\bar{\theta}V_\mu V_\nu\sigma^\mu_{\alpha\bar{\alpha}}\bar{\sigma}^{\nu\bar{\alpha}\alpha}V_\mu V_\nu \\
 &= \frac{1}{2}\theta\theta\bar{\theta}\bar{\theta}V^\mu(x)V_\mu(x)
 \end{aligned}$$

$$\therefore \exp(V_{WZ}) = 1 + V + V^2$$

$$= 1 + \theta \sigma^\mu \bar{\theta} V_\mu + \theta \theta \bar{\theta} \bar{\lambda} + \bar{\theta} \bar{\theta} \theta \lambda + \theta \theta \bar{\theta} \bar{\theta} (D(x) + \frac{1}{2} V^2)$$

* WZ Gauge is not supersymmetric

Conf. weights: Superconf. supergravity has
 $(\text{Weyl} \oplus \text{WZ} \oplus (n+1)\text{-chiral})$

Conf. weights are fixed to $h=3$ or 2 .

- For the physical Chiral mult. h not determined.
 for $n=4$ cannot have $h_{Z_i} = h_{Z_j} \forall i, j$.

Since $C(x)$ in

$$S_D = \int d^4x \, e [D + \dots + \frac{1}{3} CR(\omega) + \dots]$$

must have a defined chiral weight.

$$h_e = -4 \Rightarrow [CR(\omega)] \Leftrightarrow h_{[3]} = 4$$

$$R(\omega) = e_a{}^\mu e_b{}^\nu R_{\mu\nu}{}^{ab}$$

$$R_{\mu\nu}{}^{ab} = \partial_\mu \omega_\nu{}^{ab} - \partial_\nu \omega_\mu{}^{ab} + \dots$$

$$\omega_\mu{}^{ab}(e) = 2 e^\nu{}_{[a} \partial_{[\mu} e_{\nu]}{}^{b]}$$

$$\therefore \boxed{h_{R(\omega)} = 0}$$

$$\Leftrightarrow h(R(\omega)) = 2$$

$\Leftrightarrow$

- For the scalar field C

$$h_C = 2$$

* C , scalar field for the Real vect. mult.
 $(C, \mathbb{Z}_2, T, B_L, \lambda_R, D)$

After choosing X^0 to have non-zero

depend on the lowest value of h .

Geometrically:

$$X^I: M^4 \rightarrow \mathbb{R}^D$$

$\Rightarrow$ Depends on choice of coordinates on the Kähler manifold $\mathbb{R}^D$.

ex: X^0 : has cart. weight 1 ($h_0 = 1$)

Then each multiplet $(X^\pm, \Omega^\pm, F^\pm)$ will be affected.

$\hookrightarrow$ Take $h_1 := h$

Define $X'^{\pm} := (X^0)^{h'-h} X^\pm$

s.t. $h_1 \equiv h'$

For this to be non-degenerate $X^0 \neq 0$.

* In the simple Supergravity case,
I non-trivial gauge w/:

$$S_{SG} = - \int d^4x \, e \, \text{Re} [F \bar{F} + z D^c z + \dots]$$

• Gauge fix this by choosing D & T Gauge

$$z_{sg} := \sqrt{3}/R.$$

$$(z' := \bar{F}, \quad \mathcal{D}_\mu \chi' := \mathcal{D}_\mu \chi, \quad F' := D^c z)$$

"Compensating multiplet"

Conclude: Choice of weyl weight of chiral

Convention: $\boxed{w_{\underline{0}} \equiv 1}$ $\forall$ ^{physical} chiral-multiplets.

Reference multiplet: (X^0, Ω^0, F^0)
w/ value on the Kähler manifold $\mathbb{R}^D$

Super Conformal Invariant Action [Ungauged]

* The super potential must have chiral weight 3.

This gives rise to the "F" term.

S_F :

$$\boxed{\bar{X}^I W_I = 3 W(X)}, \quad W_I := \frac{\partial}{\partial X^I} W$$

Recall, this comes from the supersym. of the fields

$$\begin{aligned} \Leftrightarrow \delta_Q W(X) &= \frac{\partial W}{\partial X^I} \delta_Q X^I = \frac{\partial W}{\partial X^I} \left(\frac{1}{\sqrt{2}} \bar{\epsilon} \Omega^I \right) \\ &= \frac{1}{\sqrt{2}} \bar{\epsilon} \left(\frac{\partial W}{\partial X^I} \Omega^I \right) \end{aligned}$$

$$\Leftrightarrow \Omega' := W_I \Omega^I \quad (\text{on } \mathbb{R}^D)$$

Assuming $X^0 \neq 0$

$$\boxed{W(X) = (X^0)^3 \bar{W}\left(\frac{X^\alpha}{X^0}\right)}, \quad \alpha = 1, \dots, n$$

(Note, $\bar{W} \neq W$)

proof: $X^I = (X^0)^{3-h'} X'^I$

here, choose X'^I to have zero

conf. weight.

$$(X^0)^3 X'^I \bar{W}_I\left(\frac{X^I}{X^0}\right)$$

- Kinetic term for the chiral mult. In global by the D-term of $K(z, \bar{z})$. In the superconf. case K must have $w_d = 2$.
 $K(z, \bar{z}) \mapsto C(x, \bar{x}) := \frac{1}{2} N(x, \bar{x})$

$N(x, \bar{x})$ has w.r.t weight 2 since $\underline{C}$ must have 2 and $C = \frac{1}{2} N$.

- C , as defined is also a real scalar field. Thus, it is invariant under the superconformal U(1)-R symm.

$\therefore$

$$N(\lambda x, \bar{\lambda} \bar{x}) = (\lambda \bar{\lambda}) N(x, \bar{x}) \quad \forall \lambda \in \mathbb{C}$$

ex: (i) $N = a_{\bar{I}\bar{J}} x^{\bar{I}} \bar{x}^{\bar{J}}$
 (ii) $N = i \frac{\bar{x}^0 (x^1)^2}{(x^0)^2} - 3i \frac{\bar{x}^0 (x^1)}{(x^0)^3} - i \frac{x^0 (\bar{x})^3}{(\bar{x}^0)^2} + i \frac{(\bar{x}^1)^2 x^1}{\bar{x}^0}$
 (iii) $N = i \bar{x}^0 \sqrt{(x^1)^2 (x^0)^4} + 3i \bar{x}^1 \sqrt{x^0 x^1} - i x^0 \sqrt{(\bar{x}^1)^2 (\bar{x}^0)^4} - 3i x^2 (\bar{x}^0 \bar{x}^1)^{3/2}$

Homogeneity allows us to write

$$N(x, \bar{x}) = x^{\bar{I}} N_{\bar{I}} = \bar{x}^{\bar{I}} N_{\bar{I}} = x^{\bar{I}} \bar{x}^{\bar{J}} N_{\bar{I}\bar{J}}$$

From this

$$(i) \quad x^{\bar{I}} N_{\bar{I}} = \bar{x}^{\bar{J}} x^{\bar{I}} N_{\bar{I}\bar{J}}$$

$$N_{\bar{I}} = \bar{x}^{\bar{J}} N_{\bar{I}\bar{J}}$$

$$(ii) \text{ Claim } x^{\bar{I}} N_{\bar{I}\bar{J}} = 0$$

Proof: $N = x^{\bar{I}} N_{\bar{I}} = x^{\bar{I}} \frac{\partial}{\partial x^{\bar{I}}} (x^{\bar{J}} N_{\bar{J}})$

$$x^{\bar{I}} \delta_{\bar{I}}^{\bar{J}} N_{\bar{J}} + x^{\bar{I}} x^{\bar{J}} N_{\bar{I}\bar{J}} = N + x^{\bar{I}} x^{\bar{J}} N_{\bar{I}\bar{J}}$$

$$X^\pm X^\pm N_{\pm\bar{5}} = 0$$

$$\bullet N_K = \delta_K^{\bar{I}} N_{\bar{I}} + X^{\bar{I}} N_{\bar{I}K} \Leftrightarrow X^{\bar{I}} N_{\bar{I}\bar{5}} = 0$$

$$(i) N_{\bar{5}\bar{K}} \bar{X}^{\bar{K}} = N_{\bar{5}\bar{5}}$$

$$(ii) \underline{X^{\bar{K}} N_{K\bar{I}\bar{5}} = 0}$$

→ Although $N(x, \bar{x})$ is Kähler the expression through: $N = X^{\bar{I}} \bar{X}^{\bar{5}} N_{\bar{I}\bar{5}}$ does not allow for local Kähler trans.

$$N(x, \bar{x}) \mapsto N(x, \bar{x}) + f(x) + \bar{f}(\bar{x})$$

* Since $\bar{\lambda}^A P_L \lambda^B$ has weyl weight 3

→ Both from the Gauge multiplet

$P_L \lambda^B$: $3/2$, so $(\bar{\lambda}^A P_L) P_L \lambda^B$ has 3

• In the global case

$$\int d^4x \operatorname{Re} \{ \int d^2\theta f_{AB}(\Phi) W^\alpha W_\alpha \}$$

⇒ we obtain an F-term

$$F(f) = \frac{1}{4} f_{AB} (-2 \bar{\lambda}^A P_L \cancel{\not{\lambda}^B})$$

so f_{AB} has zero conf. weight?

$$\boxed{X^{\bar{I}} \int f_{A\bar{B}\bar{I}} = 0}$$

$$\text{Global: } \mathcal{S} = \mathcal{S}(\kappa) + \mathcal{S}(W) + \mathcal{S}(f)$$

$$\left[\begin{array}{l} \mathcal{S}(\kappa) = \int d^4x \mathcal{D}(\frac{1}{2}\kappa) = \int d^4x \mathcal{L}_{kin} \\ \mathcal{S}(W) = \int d^4x F(W) = \int d^4x \mathcal{L}_{pot} \\ \mathcal{S}(f) = \int d^4x F(f) = \int d^4x \mathcal{L}_{kin, gauge} \end{array} \right.$$

Superconf.:

$$\mathcal{L} = [N(X, \bar{X})]_D + [W(X)]_F + [f_{AB}(X) \bar{\lambda}^A \lambda^B]_F$$

Gauged Superconf. Sugra:

Recall: For H^D with a continuous isometry group
3 holomorphic Killing vects. $K_A^I(z)$, related to
the moment map $P_A(X, \bar{X})$

$$K_A^I(x) = -g^{I\bar{J}} \partial_{\bar{J}} P_A(X, \bar{X})$$

These are generated on H^D by

$$X'^I = X^I + \theta^A K_A^I(X)$$

Here $\delta X^I = \theta^A K_A^I$ and

$$\begin{aligned} \delta_Q X'^I &= \partial X'^I / \partial X^{\bar{J}} \delta_Q X^{\bar{J}} \\ &= (\delta_{\bar{J}}^I + \theta^A \partial_{\bar{J}} K_A^I(X)) \delta_Q X^{\bar{J}} \\ &= (\delta_{\bar{J}}^I + \theta^A \partial_{\bar{J}} K_A^I) \frac{1}{\sqrt{2}} \bar{\epsilon} \Omega^{\bar{J}} \\ &= \frac{1}{\sqrt{2}} \bar{\epsilon} \Omega'^I \end{aligned}$$

$$\Rightarrow \Omega'^I = \Omega^I + \theta^A \partial_{\bar{J}} K_A^I \Omega^{\bar{J}}$$

$$\therefore \delta(\theta) \Omega^I = \theta^A (\partial_{\bar{J}} K_A^I) \Omega^{\bar{J}}$$

For a superconf. theory, K_A^I must have the same
conf. weight as X^I : $w_K = 1$

- Since Gauge transformations (Kähler transf. on H^D)
are forbidden. Thus $N(X, \bar{X})$ is homogeneous in X
and is (super) gauge invariant.

• W is also gauge invariant

→ In terms of the theory, we recast these statements as:

$$\delta(\theta) N(X, \bar{X}) = 0 \quad \text{; } \delta(\theta) W = 0$$

The latter:

$$\delta(\theta) W = \theta^A K_A^I W_I + X^I \partial_I \delta W$$

$$\theta^A K_A^I \partial_I W \Rightarrow X^I \partial_I (\theta^A K_A^J W_J) = 0$$

$$\theta^A X^I (\partial_I K_A^J W_J + K_A^J W_{IJ}) = 0$$

but

$$X^I W_{IJ} = 2W_J$$

$$\theta^A X^I \partial_I K_A^J W_J + 2K_A^J W_J \theta^A = 0$$

$$\therefore X^I \partial_I K_A^J = K_A^J$$

Invariance of Kähler potential:

$$\left\{ \begin{array}{l} N(X, \bar{X}) = X^I N_I = X^I N_{\bar{I}} = X^I \bar{X}^{\bar{J}} N_{I\bar{J}} \\ N_I = N_{I\bar{J}} X^{\bar{J}}, \quad X^I N_{I\bar{J}} = 0 \\ N_{I\bar{J}} \bar{X}^{\bar{K}} = N_{I\bar{K}}, \quad X^K N_{K I \bar{J}} = 0 \end{array} \right\}$$

$$\Rightarrow \delta(\theta) = N = 0$$

$$\theta^A K_A^I N_I + X^I K_A^J \theta^A N_{IJ} + X^I K_A^{\bar{J}} \theta^A N_{\bar{I}\bar{J}}$$

$$N_I K_A^I + N_{\bar{I}} K_A^{\bar{I}} = 0$$

$$K^I N_I + X^I \partial_I (K^J N_J + K^{\bar{J}} N_{\bar{J}})$$

$$= K^I N_I + X^I \partial_I K^J N_J + X^I K^J N_{IJ} + \partial_I K^{\bar{J}} N_{\bar{J}} + K^{\bar{J}} N_{\bar{J}}$$

$$0 = X^I \partial_I K^J N_J + X^I \partial_I K^{\bar{J}} N_{\bar{J}}$$

∴

$$\partial_{\bar{J}} K_A^I = 0$$

Since $P_A := i(k^\mu \partial_\mu N - r(z))$

$$P_A = \frac{1}{2} i (N_I K_A^I - N_{\bar{I}} K_A^{\bar{I}}) = i N_I K_A^I = -i N_{\bar{I}} K_A^{\bar{I}}$$

$$\partial_{\bar{I}} P_A = i N_{\bar{I}} K_A^I$$

* For W : $K^I W_I + X^\pm \partial_I (K^I W_I) = 0$
 $X^\pm \partial_I K^I W_I + X^\pm K^I W_{I\bar{J}} + K^I W_I = 0$

$$\partial_I W = W_{\bar{I}} + \underline{X^{\bar{I}} W_{I\bar{J}}} = 0$$

$$\Rightarrow \underline{K_A^I W_I = 0}$$

$\hookrightarrow$ Only need to show: $X^{\bar{J}} \partial_{\bar{J}} K_A^I = K_A^I$;
 $X^{\bar{I}} X^{\bar{J}} (K^I N_{\bar{I}\bar{J}} + K^{\bar{I}} N_{I\bar{J}}) = 0$
 $\hookrightarrow$ applying ∂ : $X^{\bar{J}} \partial_{\bar{J}} K_A^I = K_A^I$

Gauge invariance of N & W implies:

$$\partial_{\bar{J}} K_A^I = 0, \quad X^{\bar{J}} \partial_{\bar{J}} K_A^I = K_A^I$$

$$N_I K_A^I + N_{\bar{I}} K_A^{\bar{I}} = 0$$

$$P_A = i \frac{1}{2} (N_I K_A^I - N_{\bar{I}} K_A^{\bar{I}}) = i K_A^I N_I = -i N_{\bar{I}} K_A^{\bar{I}}$$

$$\partial_{\bar{I}} P_A = i N_{\bar{I}} K_A^I$$

$$N_I K_A^I = 0$$

The full superconformal action:

$$S = \int d^D x \int d^4 \theta \left[N(x, \bar{x}) + W(x) \delta^2(\bar{\theta}) + \overline{W}(\bar{x}) \delta^2(\theta) + f_{AB}(\bar{x}) \bar{\lambda}^A \rho_L \lambda^B \delta^2(\bar{\theta}) \right]$$

$$S_0 = \int d^4x \left[D - i(\bar{\Psi} \gamma) i \gamma_5 \lambda + \frac{1}{6} (C \Psi_m^{\mu\nu\rho\sigma} - i \bar{\zeta} \gamma^{\rho\sigma} \chi_\mu) R_{\rho\sigma}(\omega) + \frac{1}{4} \varepsilon^{abcd} F_a \delta_b \psi_c (B_d - \frac{1}{2} \psi_d) \right]$$

* The kinetic term for the "D" term for a global susy system: (Paper uses $D(\kappa) \rightarrow D(\phi)$)
 $D(\frac{1}{2}\kappa) = K_{\alpha\bar{\beta}} (-\partial_\mu z^\alpha \partial^\mu \bar{z}^\beta - \frac{1}{2} \bar{\chi}^\alpha \rho_L \not{\partial} \chi^\beta - \frac{1}{2} \bar{\chi}^\beta \rho_R \not{\partial} \chi^\alpha + F^\alpha F^\beta)$
 $+ \frac{1}{2} [a_{\alpha\beta\bar{\gamma}} (-\bar{\chi}^\alpha \rho_L \chi^\beta F^\gamma + \bar{\chi}^\alpha \rho_L (\not{\partial} z^\beta) \chi^\gamma) + h.c.]$
 $+ \frac{1}{4} [K_{\alpha\beta\bar{\gamma}\bar{\delta}} (\bar{\chi}^\alpha \rho_L \chi^\beta \bar{\chi}^\gamma \rho_R \chi^\delta)]$

Global: $S = S_{kin,cl}(\kappa) + S_{pot,cl}(\omega) + S(f)$
 $S(f)$: Kinetic Gauge.

LONG ASIDE:

(Derivation of $S(\kappa)$, $S(\omega)$ and $S(f)$ terms)

See Superspace Notes

$$N(\bar{\Phi}, \bar{\Phi}^2 V) = N(\bar{\Phi}, \bar{\Phi}) + 2g N_{\bar{\Phi}}(\bar{\Phi} V)^{\bar{\Phi}} + 4g^2 N_{\bar{\Phi}\bar{\Phi}}(\bar{\Phi} V)^{\bar{\Phi}}(\bar{\Phi} V)^{\bar{\Phi}} + 2g N_{\bar{\Phi}}(\bar{\Phi} V)^{\bar{\Phi}} + 4g^2 N_{\bar{\Phi}\bar{\Phi}}(\bar{\Phi} V)^{\bar{\Phi}}(\bar{\Phi} V)^{\bar{\Phi}}$$

Give rise to superconformal terms.

$$S_D = \int d^4x e \left[D(\tfrac{1}{2}N) - \tfrac{1}{2} \bar{\psi} \cdot \gamma_i \delta_{\nu} \lambda - \tfrac{1}{3} C R(\omega) \right. \\ \left. + \tfrac{1}{6} (C \bar{\psi}_\mu \gamma^{\mu\nu} - i \bar{\zeta} \gamma^{\mu\nu} \delta_\nu) \mathcal{R}'_{\mu\nu}(\mathcal{Q}) \right. \\ \left. + \tfrac{1}{4} \varepsilon^{abed} \bar{\psi}_a \gamma_b \psi_c (B_d - \tfrac{1}{2} \bar{\psi}_d \zeta) \right]$$

• Now note that we must make the following identifications

$$\chi \lambda \mapsto P_R \lambda_R(\tfrac{1}{2}N)$$

$$C \mapsto \tfrac{1}{2}N$$

$$\zeta \mapsto \zeta(\tfrac{1}{2}N)$$

$$B_d \mapsto B_d(\tfrac{1}{2}N)$$

For which

$$P_R \lambda(\tfrac{1}{2}N) = \tfrac{1}{\sqrt{2}} i N_{I\bar{J}} P_R (\not{D} \bar{\chi}^{\bar{J}} \Omega^I - F^{\bar{J}} \Omega^{\bar{I}}) \\ + \tfrac{1}{2\sqrt{2}} N_{I\bar{J}} \kappa \Omega^{\bar{K}} \bar{\Omega}^{\bar{I}} \Omega^{\bar{J}}$$

$$\rightarrow \tfrac{1}{2\sqrt{2}} (\bar{\psi} \cdot \gamma) \left[i N_{I\bar{J}} P_R (\not{D} \bar{\chi}^{\bar{J}} \Omega^I - F^{\bar{J}} \Omega^{\bar{I}}) + \tfrac{1}{2} N_{I\bar{J}} \kappa \Omega^{\bar{K}} \bar{\Omega}^{\bar{I}} \Omega^{\bar{J}} \right]$$

$$= \tfrac{1}{2\sqrt{2}} (\bar{\psi} \cdot \gamma) \left[N_{I\bar{J}} F^{\bar{I}} \Omega^{\bar{J}} \cdot N_{I\bar{J}} \not{D} \bar{\chi}^{\bar{J}} \Omega^{\bar{I}} - \tfrac{1}{2} N_{I\bar{J}} \kappa \Omega^{\bar{K}} \right] \\ + N_I P_R \lambda^I K_A^{\bar{I}} \leftarrow ???$$

→ Each term now has additional components.

For the Kähler potential:

$$D(\tfrac{1}{2}N) = N_{I\bar{J}} \left(-D_\mu \chi^{\bar{I}} D^\mu \chi^{\bar{J}} - \tfrac{1}{2} \bar{\Omega}^{\bar{I}} \not{D} \Omega^{\bar{J}} - \tfrac{1}{2} \bar{\Omega}^{\bar{J}} \not{D} \Omega^{\bar{I}} \right. \\ \left. + F^{\bar{I}} F^{\bar{J}} \right) \\ + \tfrac{1}{2} [N_{I\bar{J}} \kappa (-\bar{\Omega}^{\bar{I}} \Omega^{\bar{J}} F^{\bar{K}} + \bar{\Omega}^{\bar{I}} (\not{D} \chi^{\bar{J}}) \Omega^{\bar{K}} + \text{h.c.})] \\ + \tfrac{1}{4} N_{I\bar{J}} \kappa \bar{\Omega}^{\bar{I}} \Omega^{\bar{J}} \bar{\Omega}^{\bar{K}} \Omega^{\bar{L}} \\ - i N_I K_A^{\bar{I}} D_A - \sqrt{2} \tilde{\lambda}^4 N_{I\bar{J}} (\Omega^{\bar{I}} K_A^{\bar{J}} + \Omega^{\bar{J}} K_A^{\bar{I}})$$

These come from the two extra superspace terms in the expansion:

$$\int d^4x \int d^4\theta N(\Phi, \bar{\Phi} e^{2V})$$

$$N(\Phi, \bar{\Phi} e^{2V}) = N(\Phi, \bar{\Phi}) + 2\gamma N_I (\bar{\Phi} V)^I + 2\gamma N_{IJ} (\bar{\Phi} V)^I (\Phi V)^J$$

In the Wess-Zumino Gauge s.t. $V^3 = 0$

$$\rightarrow \gamma(\frac{1}{2}N) = -\frac{1}{2}i P_L N_I \Omega^I + \frac{1}{2}i P_R N_I \Omega^{\bar{I}} + \dots$$

$$\therefore \frac{1}{8} (i \epsilon^{\mu\nu\rho\sigma} \bar{\Psi}_\mu \gamma_\nu \gamma_\rho \gamma_\sigma) (N_I D_\sigma X^I + \frac{1}{2} N_{IJ} \bar{\Omega}^{\bar{I}} \gamma_\sigma \Omega^{\bar{J}}) + \text{h.c.}$$

Putting this all together we obtain the kinetic term for $N=1$ conformal supergravity coupled to $(n+1)$ -chiral multiplets. That is:

$$\begin{aligned} & \int d^4x \int d^4\theta N(\Phi, \bar{\Phi} e^{2V}) \\ &= \int d^4x \left\{ N_{I\bar{J}} \left(-D_\mu X^I D^\mu \bar{X}^{\bar{J}} - \frac{1}{2} \bar{\Omega}^{\bar{I}} \not{D} \Omega^{\bar{J}} - \frac{1}{2} \bar{\Omega}^{\bar{J}} \not{D} \Omega^{\bar{I}} + \right. \right. \\ & \quad \left. \left. + \frac{1}{2} N_{I\bar{J}} \bar{\Omega}^{\bar{I}} \Omega^{\bar{J}} F^{\mu\nu} F_{\mu\nu} + \bar{\Omega}^{\bar{I}} (\not{D} X^J) \Omega^{\bar{J}} \right) + \text{h.c.} \right. \\ & \quad \left. + \frac{1}{4} N_{I\bar{J}} \bar{\Omega}^{\bar{I}} \Omega^{\bar{J}} \bar{\Omega}^{\bar{K}} \Omega^K \right. \\ & \quad \left. - i N_I K_A^I D^A - \sqrt{2} \bar{\lambda}^A N_{I\bar{J}} (\Omega^{\bar{I}} K_A^{\bar{J}} + \Omega^{\bar{J}} K_A^{\bar{I}}) \right. \\ & \quad \left. + \left[\frac{1}{2} \bar{\Psi} \cdot \gamma (N_{I\bar{J}} \bar{\Omega}^{\bar{I}} \Omega^{\bar{J}} - N_{I\bar{J}} \not{D} \bar{X}^{\bar{J}} - \frac{1}{2} N_{I\bar{J}} \Omega^{\bar{K}} \bar{\Omega}^{\bar{L}} \Omega^{\bar{J}} \right. \right. \\ & \quad \left. \left. + N_{I\bar{J}} P_R \lambda^A K_A^{\bar{J}} \right] \right. \\ & \quad \left. + \frac{1}{8} i \epsilon^{\mu\nu\rho\sigma} \bar{\Psi}_\mu \gamma_\nu \gamma_\rho \gamma_\sigma (N_I D_\sigma X^I + \frac{1}{2} N_{I\bar{J}} \Omega^{\bar{I}} \gamma_\sigma \Omega^{\bar{J}} + \frac{1}{2} N_{I\bar{J}} \bar{\Omega}^{\bar{I}} \gamma_\sigma \Omega^{\bar{J}} \right. \\ & \quad \left. + \frac{1}{6} N(R(\omega)) + \frac{1}{2} \bar{\Psi}_\mu \gamma^{\mu\nu\rho} R'_{\nu\rho}(A) - \frac{1}{6} \sqrt{2} (N_I \bar{\Omega}^{\bar{I}} + N_D \bar{\Omega}^{\bar{D}}) \not{D}_{\mu\nu} R'_{\mu\nu}(A) \right\} \end{aligned}$$

* Now, we must expand the Gauge term as well as

$$\int d^4x \text{Re} \left\{ \int d^2\theta W(\mathcal{Z}) \right\} =$$

$$\int d^4x e \left(W_I F^I - \frac{1}{2} W_{\bar{I}\bar{J}} \bar{\Omega}^{\bar{I}} \Omega^{\bar{J}} + W_I \bar{\Psi} \cdot \gamma \Omega^I \right.$$

$$\left. + \frac{1}{2} W \bar{\Psi}_m \not{P}_R \gamma^{\mu\nu} \psi_{\nu} + \text{h.c.} \right)$$

(Superpotential)

* The gauge term will contain gauge-field strength $\hat{F}_{\mu\nu}$ and not $F_{\mu\nu}$.

$$\hat{F}_{\mu\nu}^A := e_a^\mu e_b^\nu (2 \partial_{[\mu} A_{\nu]}^A + f_{BC}^A A_\mu^B A_\nu^C + \bar{\Psi}_{\mu\alpha} \gamma_{\nu\beta} \lambda^A)$$

The covariant der. of the chiral multiplet fields are given by

$$D_\mu X^I := (\partial_\mu - b_\mu - i A_\mu) X^I - \frac{1}{2} \bar{\Psi}_\mu \Omega^I - A_\mu^A K_A^I$$

Here A_μ and A_μ^A are two different Gauge fields

A_μ : $U(1)$ -R-symmetry

A_μ^A : G-internal symmetry. Often $G \cong SU(N)$

• This term is the supersymmetry term, i.e.

$$\delta \mathcal{O} X^I = \frac{1}{2} \bar{\epsilon} \Omega^I \rightarrow \text{Replace } \bar{\epsilon} \text{ w/ gauge field } \bar{\Psi}_\mu$$

For the fermion:

$$D_\mu \Omega^I = \left(\partial_\mu - \frac{3}{2} b_\mu + \frac{1}{4} \omega_{\mu}^{ab} \gamma_{ab} + \frac{1}{2} i A_\mu \right) \Omega^I$$

$$- \frac{1}{2} \not{P}_L (\not{D} X^I + F^I) \Psi_\mu - \sqrt{2} X^I \not{P}_L \phi_\mu - A_\mu^A \Omega^J \not{D}_J K_A^I$$

Elimination of the Auxiliary fields

→ Find E.O.M for aux. fields ... brch.

E.O.M for F^I : (find for $\bar{F}^{\bar{I}}$ instead)

$$N_{\bar{I}\bar{J}} \bar{F}^{\bar{J}} - \frac{1}{2} N_{\bar{I}\bar{K}} \bar{\Omega}^{\bar{K}} \Omega^{\bar{J}} + W_{\bar{I}} + \frac{1}{2\sqrt{2}} \bar{\Psi} \cdot \gamma N_{\bar{I}\bar{J}} \Omega^{\bar{J}}$$

$$- \frac{1}{4} f_{AB\bar{I}} \bar{\lambda}^A \not{P}_L \lambda^B = 0$$

$$- \bar{F}^J N_{I\bar{J}} = W_I - \frac{1}{2} N_{I\bar{J}} \bar{\Omega}^{\bar{J}} \Omega^I - \frac{1}{4} f_{AB I} \bar{\chi}^A \chi^B$$

→ E.O.M For D^A :

$$\text{Re}(f_{AB}) D^A =: P_A + P_A^F$$

for which, the D^A terms gives rise to the moment map plus its Gauge extension.

$$P_A = i N_{I\bar{J}} K_A^{\bar{J}}$$

$$P_A^F := -i \frac{1}{2\sqrt{2}} f_{AB I} \bar{\Omega}^{\bar{I}} \lambda^B + i \frac{1}{2\sqrt{2}} f_{AB I} \bar{\Omega}^{\bar{I}} \lambda^B$$

* The A_m is also an Auxillary field

$$A_m = \mathcal{A}_m + A_m^F$$

$$\mathcal{A}_m := i \frac{1}{2N} [N_{\bar{I}} (\partial_m x^{\bar{I}} - A_m^{\bar{I}} K_A^{\bar{I}}) - N_I (\partial_m x^I + A_m^I K_A^I)]$$

$$= i \frac{1}{2N} [N_{\bar{I}} \partial_m x^{\bar{I}} - N_I \partial_m x^I] + \frac{1}{N} A_m^A P_A$$

$$A_m^F := i \frac{1}{4N} [\sqrt{2} \bar{\Psi}_m (N_I \Omega^I - N_{\bar{I}} \bar{\Omega}^{\bar{I}}) + N_{I\bar{J}} \bar{\Omega}^{\bar{I}} \chi_m \Omega^{\bar{J}} + \frac{2}{3} (\text{Re } f_{AB})]$$

* Note: A_m and D^A are split into Bosonic & Fermionic parts.

The moment map P_A of the Kähler manifold appears as the E.O.M for the auxillary D^A term along with a fermionic gen. which comes from the Gauge terms.

→ We now Insert the Auxillary fields back to obtain $e^{-1\mathcal{L}} = () \leftarrow$ see paper for full expansion.

$\mathcal{N}^{(0)}$ is defined as $\mathcal{N}$ w/o the fermions

Define: $R^\mu := \gamma^\mu \rho^\alpha (\partial_\mu + \frac{1}{2} b_\mu + \frac{1}{4} \omega_\mu{}^{ab} e_{ab} - \frac{2}{3} i A_\mu \gamma_5) \psi_\alpha$

Note $N_I D_\mu X^I = N_I D_\mu X^{\bar{I}} =$

Typo:
17.26 $N_I D_\mu X^I + N_{\bar{I}} D_\mu X^{\bar{I}}$
text has =

$N_I (\partial_\mu X^I - b_\mu X^I - A_\mu^A K_A^I - i A_\mu X^I)$

$= \partial_\mu X^I N_I - b_\mu N - A_\mu^A K_A^I N_I - i A_\mu N$

$N_{\bar{I}} D_\mu X^{\bar{I}} = N_{\bar{I}} (\partial_\mu X^{\bar{I}} - b_\mu X^{\bar{I}} - A_\mu^A K_A^{\bar{I}} + i A_\mu X^{\bar{I}})$

$\therefore$

$N_I D_\mu X^I + N_{\bar{I}} D_\mu X^{\bar{I}}$

$= (\partial_\mu X^I N_I - 2b_\mu N - A_\mu^A (K_A^I N_I + K_A^{\bar{I}} N_{\bar{I}}) + \partial_\mu X^I N_I)$

$\Rightarrow \partial_\mu N = \partial_\mu (X^I N_I) = \partial_\mu X^I N_I + \partial_\mu X^{\bar{I}} X^I N_{\bar{I}} + \partial_\mu X^{\bar{I}} X^{\bar{I}} N_{\bar{I}}$
 $= \partial_\mu X^I N_I + \partial_\mu X^{\bar{I}} N_{\bar{I}}$

$\Rightarrow = \frac{1}{2} (\partial_\mu - b_\mu) N$

* Many of the terms in $e^{-1} \mathcal{L}$ can be written as Kähler covariant quantities. $\bar{\Omega}^{\bar{P}} \gamma^\mu \Omega^{\bar{K}} D_\mu X^{\bar{I}}$ can be included in the kinetic term of Ω^I using a modified covariant der.

$\hat{D}_\mu^{(0)} \Omega^{\bar{P}} = D_\mu^{(0)} \Omega^{\bar{P}} + \Gamma_{S\bar{K}}^{\bar{P}} \Omega^{\bar{K}} D_\mu X^{\bar{I}}$

* The $F^{\bar{P}}$ terms can be extended w/ these Kähler connection components.

$\Gamma_{S\bar{K}}^{\bar{P}} = N^{\bar{I}\bar{I}} \partial_{\bar{S}} N_{\bar{I}\bar{I}}$

$-\bar{F}^{\bar{P}} F_{\bar{S}} = W_{\bar{P}} - \frac{1}{2} N_{\bar{I}\bar{I}} \bar{K}^{\bar{I}} \bar{\Omega}^{\bar{J}} \Omega^{\bar{K}} - \frac{1}{4} f_{AB} \bar{\lambda}^A \rho_{AB}$

$$-\bar{F}^{\bar{I}} N_{I\bar{J}} N^{I\bar{K}} = N^{I\bar{K}} W_I - \frac{1}{2} N_{I\bar{J}} \bar{K} N^{I\bar{K}} \bar{\Omega}^{\bar{J}} \Omega^{\bar{I}} \\ - \frac{1}{4} f_{AB I} N^{I\bar{K}} \bar{\lambda}^A P_L \lambda^B$$

$$\bar{F}^{\bar{I}} = -N^{J\bar{I}} W_J + \frac{1}{2} \Gamma_{\bar{J}\bar{K}}^{\bar{I}} \bar{\Omega}^{\bar{J}} \Omega^{\bar{K}}$$

$$\bar{F}^{\bar{I}} = -N^{J\bar{I}} W_J + \frac{1}{2} \Gamma_{\bar{J}\bar{K}}^{\bar{I}} \bar{\Omega}^{\bar{J}} \Omega^{\bar{K}} - \frac{1}{4} f_{AB I} N^{I\bar{K}} \bar{\lambda}^A P_L \lambda^B$$

$\therefore F^I F^{\bar{J}}$ can be expanded as

$$\begin{pmatrix} N_{I\bar{I}} \\ -N^{J\bar{I}} W_J + \frac{1}{2} \Gamma_{\bar{J}\bar{K}}^{\bar{I}} \bar{\Omega}^{\bar{J}} \Omega^{\bar{K}} - \frac{1}{4} f_{AB I} N^{I\bar{K}} \bar{\lambda}^A P_L \lambda^B \\ (-N^{\bar{J}I} W_{\bar{J}} + \frac{1}{2} \Gamma_{\bar{J}\bar{K}}^I \Omega^{\bar{J}} \Omega^{\bar{K}} - \frac{1}{2} f_{AB I} \bar{\lambda}^A P_R \lambda^B) \end{pmatrix}$$

$$= N_{I\bar{I}} N^{J\bar{I}} N^{\bar{J}I} W_{\bar{J}} W_J + N_{I\bar{I}} W^{\bar{J}I} W_{\bar{J}} \left(\frac{1}{2} \Gamma_{\bar{J}\bar{K}}^I \Omega^{\bar{J}} \Omega^{\bar{K}} \right) \\ + \frac{1}{2} N_{I\bar{I}} N^{J\bar{I}} W_J f_{AB I} \bar{\lambda}^A P_R \lambda^B$$

$\rightarrow$ Need a $\nabla_I W_{\bar{J}} \bar{\Omega}^{\bar{I}} \Omega^{\bar{J}}$ term

$$\rightarrow \nabla_I W_{\bar{J}} = \frac{\partial}{\partial x^I} W_{\bar{J}} + \Gamma_{I\bar{J}}^{\bar{K}} W_{\bar{K}}$$

$$\rightarrow -\delta_I^{\bar{J}} W_{\bar{J}} \frac{1}{2} \Gamma_{\bar{J}\bar{K}}^{\bar{I}} \bar{\Omega}^{\bar{J}} \Omega^{\bar{K}} \\ = \frac{1}{2} W_I \Gamma_{\bar{J}\bar{K}}^{\bar{I}} \bar{\Omega}^{\bar{J}} \Omega^{\bar{K}}$$

add this to $\psi \psi^c$ term: $-\frac{1}{2} W_I \bar{\Omega}^{\bar{I}} \Omega^{\bar{I}}$

* we can now split the superconformal action into several parts

$$e^{\pm 1} \mathcal{L} = \frac{1}{8} N [-R(e, b) + \bar{\psi}_m \not{D}^m \psi + e^{\pm 1} \partial_\mu (e \bar{\psi} \cdot \gamma \psi^\mu)]$$

$$+ \mathcal{L}_0 + \mathcal{L}_{1/2} + \mathcal{L}_2 = V + \mathcal{L}_m + \mathcal{L}_{mix} + \mathcal{L}_d + \mathcal{L}_{4f}$$

$$\mathcal{L}_0 = -G_{I\bar{J}} D_\mu X^I D^\mu X^{\bar{J}}$$

$$\mathcal{L}_{1/2} := -\frac{1}{2} G_{I\bar{J}} [\bar{\Omega}^{\bar{I}} \not{D}^{(0)} \Omega^{\bar{J}} + \bar{\Omega}^{\bar{J}} \not{D}^{(0)} \Omega^{\bar{I}}]$$

$$\mathcal{L}_2 := \text{Re}\{f_{AB}\} \left[-\frac{1}{4} F_{\mu\nu}^A F^{\mu\nu B} - \frac{1}{2} \bar{\lambda}^A \not{D}^{(0)} \lambda^B \right] \\ + \frac{1}{4} i \left[\text{Im}(f_{AB}) F_{\mu\nu}^A F^{\mu\nu B} + (\not{D}_\mu \epsilon \wedge f_{AB}) \bar{\lambda}^A \not{D}_\mu \lambda^B \right]$$

$$\begin{aligned}
 V &= V_F + V_D = G^{IJ} W_I W_J + \frac{1}{2} (\text{Ref})^{AB} P_A P_B \\
 \mathcal{L}_m &= \frac{1}{2} W \bar{\Psi}_m P_R \gamma^{\mu\nu} \Psi_\nu - \frac{1}{2} \nabla_I W_J \bar{\Omega}^I \Omega^J + \frac{1}{4} G^{IJ} \bar{W}_I W_J \quad (1) \\
 &\quad + \sqrt{2} i \left[-\partial_I P_A + \frac{1}{4} f_{AB I} (\text{Ref})^{-1/2} P_C \right] \bar{\chi}^A \Omega^I + \text{h.c.} \\
 (\quad) &= f_{AB I} \bar{\chi}^A P_L \chi^B \\
 \mathcal{L}_{m\chi} &= (\bar{\Psi} \cdot \chi) P_L \left(\frac{1}{2} i \not{D} \chi^A + \frac{1}{\sqrt{2}} W_I \Omega^I \right) + \text{h.c.} \\
 \mathcal{L}_d &= \frac{1}{8} \text{Re}(f_{AB}) F_{\mu\nu} \gamma^{\alpha\beta} (F_{\alpha\beta}^A + F_{\alpha\beta}^{\bar{A}}) \gamma^\mu \chi^\nu + \frac{1}{2} \{ G_{IJ} \bar{\Psi}_m \not{D} \chi^J \gamma^\mu \\
 &\quad - \frac{1}{4} f_{AB I} \bar{\Omega}^I \gamma^{\alpha\beta} F_{\alpha\beta}^A \chi^B - \frac{e}{2} N F \bar{\Omega}^I \gamma^{\mu\nu} D_\mu \Psi_\nu + \text{h.c.} \}
 \end{aligned}$$

$$\mathcal{L}_{gf} = \dots$$

Notes:

(i) $\frac{1}{8} N \left[-R(e, b) + \bar{\Psi}_m \not{D} \chi^m + e^{-1} \partial_\mu (e \bar{\Psi} \cdot \gamma \Psi^\mu) \right]$

Is N times a ^{pure} Supergravity action where the χ -fermion term has been included.

(ii) $\mathcal{L}_0$ & $\mathcal{L}_{1/2}$: Kinetic terms for the chiral multiplets

(iii) $\mathcal{L}_1$: Kinetic terms for the Gauged WZ multiplet.

(iv) V : Scalar Potential

$$V \sim (\delta_{\text{fermionic}})(\text{metric})(\delta_{\text{fermionic}})$$

$\Rightarrow$ view $(\text{Re}(D))^{-2AB}$ as the metric for the

(v) $\mathcal{L}_m$: Fermionic mass terms.

(vi) $\mathcal{L}_{mix}$: Goldstino terms

(vii) $\mathcal{L}_d$: $\bar{\Psi}_m$ is contracted w/ the Noether Supercurrent for both gauge & chiral Mult. s.
* Other interactions

(viii) $\mathcal{L}_{gf}$: χ -ferm. interactions

* We must now gauge fix these fields/sym which

are not part of the Super Poincaré alg.
 $\rightarrow$ Fix: K, D, S, T_R

Claim: $V = G^{I\bar{J}} W_I \bar{W}_{\bar{J}} + \frac{1}{2} (R_{ef})^{\alpha\beta} P_\alpha P_\beta$
 can be obtained via a sum over the
 Super symmetric Bosonic terms for each type
 of Multiplet in the theory:

Proof: $\delta_\alpha \Omega^I = \frac{1}{\sqrt{2}} P_L (\not{D} X^I + F^I) \epsilon$

$$\therefore \delta_S \Omega^{\pm} = \frac{1}{\sqrt{2}} F^{\pm}$$

$$\bar{F}^{\bar{I}} = -N^{\bar{I}\bar{J}} W_{\bar{J}} + \frac{1}{2} N^{\bar{I}\bar{J}} N_{\bar{I}\bar{K}} \bar{\epsilon}^{\bar{K}} \Omega^{\bar{I}} \\ + \frac{1}{4} f_{AB\bar{I}} N^{\bar{I}\bar{J}} \bar{\chi}^A P_{\bar{J}} \chi^B$$

$\therefore$ If we take $(\bar{F}^{\bar{J}})_B = -W^{\bar{J}}$

then $F^I G_{I\bar{J}} \bar{F}^{\bar{J}}$

$$F^I G_{I\bar{J}} \bar{F}^{\bar{J}} = W^I G_{I\bar{J}} \bar{W}^{\bar{J}}$$

• For D^A : $Re(f_{AB}) D^B = P_A + P_A^F$

$$[Re(f_{AB}) D^B]_{\beta_0} = P_A$$

$$\therefore V = W^I G_{I\bar{J}} \bar{W}^{\bar{J}} + (g^{-1})^{\alpha\beta} P_\alpha P_\beta \quad \square$$

Partial Gauge fixing:

(i) Choose the $\mathcal{H}$ -Gauge condition $b_{\mu\nu} = 0$

(ii) Dilations: In the global case we simply set
 $\phi = \frac{\sqrt{2}}{R}$ to obtain the Einstein-Hilb.
 Action. Since

$$e^{-2\phi} \sim \frac{1}{6} N R + \dots$$

it is natural to choose the $\mathcal{D}$ -Gauge

$$N = -\frac{3}{R^2} \quad (\mathcal{D}\text{-Gauge})$$

$\rightarrow$ Also note that this Gauge eliminates mixed kinetic

graviton & scalars.

→ This choice also defines a submanifold of $(n+1)$ complex-dim space spanned by X^I .

(iii) S-supersymmetry: Chosen to eliminate kinetic mixing terms between the gravitino and fermions.

• These are: $\mathcal{L}_d$

$$\sim -\frac{2}{3} N^I \bar{\Omega}_I \gamma^{\mu\nu} \hat{D}_\mu \psi_{\nu L} + \frac{1}{2} \bar{\psi}_R \cdot \gamma (N_I \bar{\psi}_L^I \lambda^\alpha + 2 \bar{\psi} \Omega^\alpha)$$

$$\boxed{N_I \Omega^I = 0} \quad (\text{S-gauge})$$

$$\boxed{N_I \Omega^I = 0}$$

* Note: this is a good Gauge choice since its S-transf. is proportional to $N_I X^I = N$.

$$\delta_S \Omega^I = \sqrt{2} \epsilon X^I P_L \Omega$$

$$\Rightarrow \delta_S (N_I \Omega^I) = \delta_S X^I \cancel{N_I} \Omega^I + \delta_S \bar{X}^I \cancel{N_I} \bar{\Omega}^I + N_I (\sqrt{2} \epsilon X^I P_L \Omega)$$

$$\delta_S (N_I \Omega^I) = 0 = \epsilon N \sqrt{2} P_L \Omega = 0$$

* The Q-supersymmetry on these Gauge conditions tell us $\lambda_D = 0$

• Now need to understand X^I : Select n-physical scalars Z^α from a larger set $\bar{\mathcal{Q}}$.

Projective Kähler Manifolds:

• X^I were initially the coords on a target Kähler mfd. D & T_R act as symm. of this manifold.

by identifying pts which lie on the orbit of the vector field that generate a symm.

~ Identified pts on a symm orbit is essentially eq. to gauge fixing the symm.

$$e^{i\alpha} \phi^i = \phi'^i$$

$$\Rightarrow \delta \phi^i = \mathcal{L}_\eta \phi^i$$

$$\mathcal{M} = \sqcup_i [\phi_i]$$

$$C^0 \mathcal{M}^D \equiv \{[\phi_i], \dots\}$$

$[\phi_i]$ define the coordinates which we are using on the manifold. We



~ Idea: Will find/Require for $\mathcal{H}$ to be projective the Kähler potential N must admit the homogeneity property (of $U(1)$)
 $N(\lambda X, \bar{\lambda} \bar{X}) = |\lambda|^2 N(X, \bar{X})$, $\lambda \in U(1)$

$\mathbb{CP}^n$: (Recap of previous discussion)

$\mathbb{CP}^n \subset \mathbb{C}^{n+1}$. Let $(x^1, \dots, x^{n+1}) \in \mathbb{C}^{n+1}$

as well as $(x'^1, \dots, x'^{n+1})$

Then $(x^i) \sim (x'^i) \Leftrightarrow \exists \rho \in \mathbb{C}$

$$(x'^i) = \rho (x^i)$$

(ex $\mathbb{CP}^2$: $\rho(z, w) = (\rho z, \rho w)$
 $w = a z$
 is a line in $\mathbb{C}^2$ through the origin.)

$$\left\{ \rho w = a \rho z \Leftrightarrow w = a z \right\}$$

• Note that this is simply a U(1) symm. similar in all coordinates.

• $x^{n+1} \neq 0 \Rightarrow z^\alpha := x^\alpha / x^{n+1}$
all lines where $x^{n+1} \neq 0$

$$z_{\bar{I}}^\alpha = \frac{x^\alpha}{x^I}, \quad I = 1, \dots, n+1$$

$$x^I \neq 0$$



Impose $x^I \bar{x}_I = 1, \quad I = 1, \dots, n+1$

$$ds^2 = dx^I G_{I\bar{J}} d\bar{x}^{\bar{J}}$$

Since we take $x^I \bar{x}_I = 1 \Leftrightarrow G_{I\bar{J}} = \delta_{I\bar{J}}$

But since N has the property $N = x^I x^{\bar{J}} N_{I\bar{J}}$
then $N = x^I \delta_{I\bar{J}} x^{\bar{J}}$ (Kähler Potential)

Scaling: $x^I = \rho x'^I$:

$$ds^2 = |\rho|^2 dx'^I \delta_{I\bar{J}} dx'^{\bar{J}} = dx'^I \gamma'_{I\bar{J}} dx'^{\bar{J}}$$

$$\gamma'_{I\bar{J}} \equiv |\rho|^2 \delta_{I\bar{J}}$$

Thus, this scaling is simply a conformal symmetry.

not too surprising if suppose:

* Let us look at the U(1) generators of ρ :

$$\rho = \exp(-\lambda_0 - \epsilon \lambda_+)$$

Q: Why is $h=1$ here? I guess it makes sense that $x^\pm$ has conf. weight 1.

$$\delta x^I = x'^I - x^I = \rho^{-1} x^I - x^I = \lambda_0 K_0^I + \lambda_+ K_+^I$$

• Note, as claimed, K_T, K_D are holomorphic.

(Recall $\phi^i \rightarrow \phi^i + \theta^A K_A^i(\phi)$)

$$\therefore \delta G_{I\bar{J}} = \delta_{I\bar{J}} - G_{I\bar{J}} = -\theta^A \mathcal{L}_{K_A} G_{I\bar{J}}$$

$$\delta G_{I\bar{J}} = -(\lambda_D \mathcal{L}_{K_D} + \lambda_T \mathcal{L}_{K_T}) G_{I\bar{J}}$$

: For "general" K

$$\begin{aligned} (\mathcal{L}_K G)_{I\bar{J}} &= K^\mu \partial_\mu G_{I\bar{J}} + K^{\bar{\mu}} \partial_{\bar{\mu}} G_{I\bar{J}} \\ &\quad + \partial_I K^\mu G_{\mu\bar{J}} + \partial_{\bar{I}} K^{\bar{\mu}} G_{\bar{\mu}\bar{J}} \\ &\quad + \partial_{\bar{J}} K^\mu G_{I\mu} + \partial_J K^{\bar{\mu}} G_{I\bar{\mu}} \end{aligned}$$

$$\mathcal{L}_K G_{I\bar{J}} = K^\mu \partial_\mu G_{I\bar{J}} + K^{\bar{\mu}} \partial_{\bar{\mu}} G_{I\bar{J}} + \partial_I K^\mu G_{\mu\bar{J}} + \partial_{\bar{I}} K^{\bar{\mu}} G_{\bar{\mu}\bar{J}}$$

(i) Dilations: $K_D^\mu = x^\mu$

$$\begin{aligned} (\mathcal{L}_{K_D} G)_{I\bar{J}} &= x^\mu \partial_\mu G_{I\bar{J}} + x^{\bar{\mu}} \partial_{\bar{\mu}} G_{I\bar{J}} \\ &\quad + \delta_I^\mu G_{\mu\bar{J}} + G_{I\bar{J}} \\ &= 2 G_{I\bar{J}} \end{aligned}$$

As expected, K_D is not Killing but rather it is a homothetic.

$$\mathcal{L}_K G_{i\bar{j}} = 2K g_{i\bar{j}}$$

(ii) R-symmetry:

$$(\mathcal{L}_{K_T} G)_{I\bar{J}} = 0 \quad K_T: \text{Killing.}$$

Dilations is Holomorphic Homothetic Killing Vectors:

→ Note: $\delta(D_\mu \phi^i) = \lambda_D (D_\mu \phi^i + \partial_j \phi^i D_\mu \phi^j) - 2\lambda_D \phi^i D_\mu \phi^i$

so Dilations induces SCT of covariant der.'s.

For a holomorphic Killing vector K^μ , the Lie derivative $\mathcal{L}_K$ is

Killing Vector.

$$\nabla_i K_j^i = \delta_i^i \text{ (closed Homothetic)}$$

$$\Rightarrow \text{locally } K_{pi} = \partial_i \tilde{u}$$

$$\text{Assign weyl weight } h=1 = \frac{D-2}{2} \mid_{D=4}$$

* Given a closed Homothetic vector K_0^i , we can use the complex structure J_i^j to construct the Killing vector K_T

$$(K_T)^i := K_0^j J_j^i$$

why? Well, we know in CP^n $K_D^I = \epsilon K_T^I$ and so we can generalize this on an arbitrary M^D manifold. Indeed.

If M is Ricci flat, then

$$\nabla_k J_i^j = 0, \quad \nabla_i K_j = h g_{ij} \text{ (homothetic/closed)}$$

w/ these two properties:

$$\nabla_i (K_0^j J_{kj}^i) = \nabla_i K_T^i = g_{ik} h J^k_i = J_i^i$$

Since J_{ij} is symmetric, K_T is a Killing vector(?)

* Now on $M^D: (x^I, \bar{x}^{\bar{I}})$ w/ Herm. metric

$$G_{I\bar{J}} \text{ i.e. } G(JX, JY) = G(X, Y)$$

Def'n: A holomorphic homothetic Killing vector

$$h := h^I(x) \frac{\partial}{\partial x^I}$$

such that

$$(L_h G)_{I\bar{J}} = G_{K\bar{J}} \nabla_I h^K = 2 G_{I\bar{J}}$$

Note:

$$K_0 = \frac{1}{2} (h + \bar{h})$$

$$K_0^I = \frac{1}{2} h^I$$

$$K_0^{\bar{I}} = \frac{1}{2} \bar{h}^{\bar{I}}$$

$$K_{-I} = \frac{1}{2} (h - \bar{h})$$

$$K_{-I}^I = \frac{1}{2} h^I$$

$$K_{-I}^{\bar{I}} = -\frac{1}{2} \bar{h}^{\bar{I}}$$

$$\therefore (2\bar{g})_{I\bar{J}} = G_{KJ} \nabla_{\bar{I}} \bar{h}^{\bar{K}} = 2G_{I\bar{J}}$$

$$\therefore (2\kappa_D)_{I\bar{J}} = \frac{1}{2} (G_{I\bar{K}} \nabla_{\bar{J}} \bar{h}^{\bar{K}} + G_{K\bar{J}} \nabla_I h^K)$$

→ Note: $\kappa_D = \kappa_D^{\bar{I}} \partial_{\bar{I}} + \kappa_D^{\bar{I}} \partial_{\bar{I}}$
 $\kappa_{\pm} = \kappa_{\pm}^{\bar{I}} \partial_I + \kappa_{\pm}^{\bar{I}} \partial_{\bar{I}}$

* The Kähler potential $N = N_I X^I = N_{\bar{I}} \bar{X}^{\bar{I}} = X^I \bar{X}^{\bar{J}} N_{I\bar{J}}$

Then $\nabla_i \partial_{\bar{j}} \tilde{\chi} = g_{i\bar{j}}$ is satisfied for

$$\tilde{\chi} = N$$

$$\nabla_{\bar{I}} \partial_{\bar{J}} N = \nabla_{\bar{I}} N_{\bar{J}} = N_{I\bar{J}} + \Gamma_{I\bar{J}}^K N_K$$

$$\begin{aligned} \Gamma_{I\bar{J}}^K N_K &= g^{K\bar{L}} \partial_I g_{\bar{L}\bar{J}} N_K \\ &= N^{K\bar{L}} N_{I\bar{L}} N_{K\bar{J}} X^{\bar{L}} = N_{I\bar{J}} X^{\bar{L}} \\ &= N_{I\bar{J}} \end{aligned}$$

$$\text{LHS} = 2N_{I\bar{J}} = N_{I\bar{J}} \Leftrightarrow N_{I\bar{J}} = 0$$

$$\nabla_I \partial_I N = 0$$

and, trivially $\partial_I \partial_{\bar{J}} N = N_{I\bar{J}}$ ✓

The Projective Parametrization:

- The homothetic Killing vector $h(X)$ defines a holomorphic map. of the embedding mfd to itself.

(target mfd instead?)

- This map defines the action of the Lie group of combined dilations & R-sym.

$$h = \kappa_D - i\kappa_T$$

$$\Rightarrow N = \exp(\theta h)$$

• The group acts freely, $\sigma(G, P) = P$, then $g \neq e$
 $\Rightarrow h$ defines a "holonomy" of the embedding mfd.

By identifying all points related by a group tras. we obtain a Kähler mfd.

* By the Frobenius theorem: allows us to choose holomorphic coords y and z^α s.t. $\partial/\partial y$ points in the direction of h . Thus

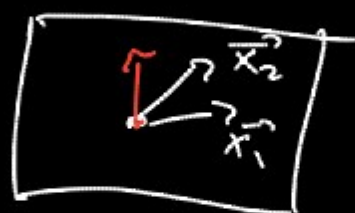
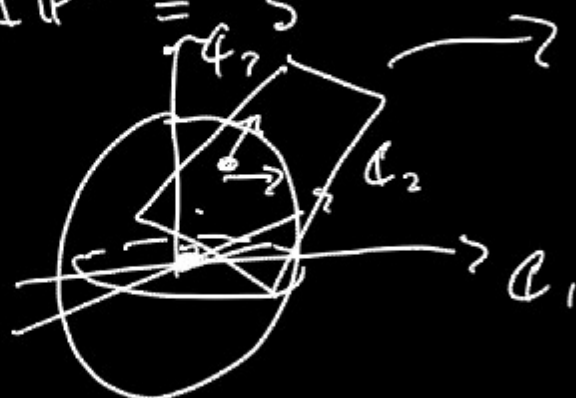
$$h = 2y \frac{\partial}{\partial y}$$

→ Frobenius theorem: Recall that in GR we would like to understand null Hypersurfaces. In particular we can always find a set of tangent space elements such that

$$T_p K = H \oplus H^\perp$$

Here H is the surface in $\mathbb{P}^{n+1}$ such that z^α are tangent to $K^D \subset \mathbb{C}^{n+1}$ while y has its associated KM vector which is orthogonal to K .

ex: Take $\mathbb{CP}^2 \subset \mathbb{C}^3$. Since $\mathbb{CP}^2 \cong S^{2^2}$



$$\vec{h} = y \frac{\partial}{\partial y}$$

* In the simple case of $\mathbb{CP}^n$ we see

that MP^n admits a global orthogonal vector field (i.e. section).

→ This induces a (trivial) Bundle $\mathbb{R}$

Now require $(n+1)$ x^I be a set of homogeneous coords. Thus, $\exists z^I$ and $y \in \mathbb{C}$ s.t.

$$x^I = y z^I(z), \quad z = z^\alpha, \quad \alpha = 1, \dots, n$$

$$I = 0, \dots, n$$

Recall, we defined,

$$(x^0, \dots, x^n) = \frac{1}{\sqrt{1+z^\alpha z_\alpha}} (z^1, \dots, z^n, 1)$$

$$\text{Here } y = \frac{1}{\sqrt{1+z^\alpha z_\alpha}}$$

$$z^\alpha = x^\alpha / x^0$$

So, the choice for z is such that we obtain the form above and thus:

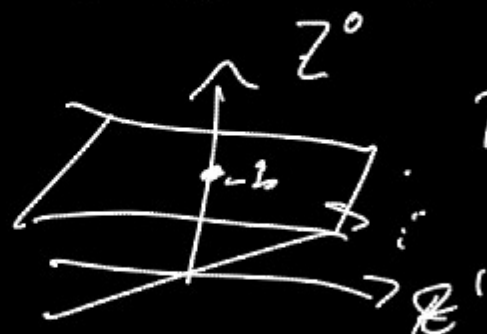
$$z^0 = 1, \quad z^\alpha = z^\alpha$$

x^I : Embedding scalars

z^α : Physical scalar fields

⇒ n -chiral mults: $\alpha = 1, \dots, n, \quad I = 0, \dots, n$

With these values: Basically working w/



Thus

$$x^0 = y$$

$$x^\alpha = y z^\alpha$$

$$\therefore h = 2y \frac{\partial}{\partial y} = 2y \frac{\partial x^I}{\partial y} \frac{\partial}{\partial x^I} =$$

$$= 2\gamma \frac{\partial}{\partial x^0} + 2\gamma z \frac{\partial}{\partial x^\alpha} - 2X^\alpha \frac{\partial}{\partial y^0} + X^\alpha \frac{\partial}{\partial x^\alpha}$$

$$\therefore \boxed{h = 2x^2 \frac{\partial}{\partial x^I}} \quad h^I = 2x^I$$

Now, h is holomorphic:

$$G_{K\bar{J}} \nabla_I h^K = 2G_{I\bar{J}}$$

$$\rightarrow G_{K\bar{J}} (2\delta^K_I + 2\Gamma_{I\bar{m}}^K X^{\bar{m}}) = 2G_{I\bar{J}}$$

$$\left(G_{I\bar{J}} + g^{K\bar{m}} \partial_I g_{m\bar{n}} X^{\bar{n}} g_{K\bar{J}} \right)$$

$$\rightarrow X^{\bar{m}} \frac{\partial}{\partial x^m} (G_{I\bar{J}}) = 0$$

$$\boxed{h G_{I\bar{J}} = 0}$$

• Call the Kähler potential N

$$G_{I\bar{J}} = \frac{\partial}{\partial x^I} \frac{\partial}{\partial \bar{x}^{\bar{J}}} N$$

The above $h G_{I\bar{J}} = 0$ implies that N is not uniquely defined. i.e., note,

$$\frac{\partial}{\partial x^I} \frac{\partial}{\partial \bar{x}^{\bar{J}}} \left(X^K \frac{\partial}{\partial x^K} N = N' \right) = \partial_I (X^K N_{K\bar{J}}) = N_{I\bar{J}} = 0.$$

Then by a Kähler trans. and the complex conj.

of the above we can define N s.t.

$$x^I \frac{\partial}{\partial x^I} N \quad \& \quad \bar{x}^{\bar{I}} \frac{\partial}{\partial \bar{x}^{\bar{I}}} N = N$$

$$[\partial_I, h] N = N_{I\bar{J}} ?$$

$$\partial_I \partial_{\bar{J}} (X^{\bar{I}} N + h f + h \bar{f} - N - f - \bar{f})$$

$$= \partial_I \partial_{\bar{J}} (h f - f) = 0$$

• Thus, N is homogeneous in x^I & $\bar{x}^{\bar{J}}$

ex: $f(x) = x \rightarrow x \frac{\partial}{\partial x} f = f(x)$

* For our Super theory in gauges k_D is k_T
 $\delta X^I = \lambda_D X^I + i \lambda_T X^I \quad \left| \quad \delta \bar{X}^I = \lambda_D \bar{X}^I - i \lambda_T \bar{X}^I \right.$
 $k_D = \frac{1}{2}(\psi + \bar{\psi}) \quad k_T = \frac{1}{2}i(\psi - \bar{\psi})$
 $k_D = (\gamma \partial_\gamma + \bar{\gamma} \partial_{\bar{\gamma}}) \quad , \quad k_T = i(\gamma \partial_\gamma - \bar{\gamma} \partial_{\bar{\gamma}})$

$$\delta \psi = (\lambda_D + i \lambda_T) \psi \quad \delta \bar{\psi} = \delta \bar{\psi} = 0$$

$$\delta \bar{\psi} = (\lambda_D - i \lambda_T) \bar{\psi}$$

* We now want to write an action for the scalar fields X^I coupled to a general spacetime metric $g_{\mu\nu}(x)$ that is invariant under local versions of these transformations: $\lambda_D(x), \lambda_T(x)$

$\lambda_T(x)$: Is a $U(1)$ symmetry and implemented through an Auxiliary gauge field $A_\mu(x)$ (we obtain the scalar kin. & grav. term of e^{-2} & ω $b_\mu = 0$)

$$S = \int d^4x \mathcal{L} = - \int d^4x e \left[\frac{1}{16} R(e) + G_{I\bar{J}} g^{\mu\nu} (\partial_\mu - i A_\mu) X^I (\partial_\nu + i A_\nu) \bar{X}^{\bar{J}} \right]$$

$$\delta A_\mu = \partial_\mu \lambda_T$$

Claim: S is scale invariant. $\delta_D S = 0$

Proof: We must write the

$$\delta X^I = \lambda_D X^I \quad , \quad \delta e_\mu^a = -\lambda_D e_\mu^a$$

$$\delta \omega_\mu^{ab} = 0$$

$$\delta S = \int d^4x -4e \lambda_D \left[\frac{1}{16} R(e) + G_{I\bar{J}} g^{\mu\nu} (\partial_\mu - i A_\mu) X^I (\partial_\nu + i A_\nu) \bar{X}^{\bar{J}} \right]$$

$$= \int d^4x (-u \cdot e) \lambda_D [3 + \int d^4x \psi \lambda_D e^{NR(e)}]$$

trivially, $\partial_\mu (G_{\pm 5} g^{\mu\nu} G) (\partial_\mu X^\pm \partial_\nu \bar{X}^\mp)$

* Recall $A_\mu = A_\mu^I + A_\mu^F$

The E.O.M: A_μ :

$$\begin{aligned} (G_{\pm 5}) &= i X^E (\partial_\mu \bar{X}^\mp) g^{\mu\nu} + G_{\pm 5} g^{\mu\nu} (\partial_\mu X^E) i \bar{X}^\mp = 0 \\ g^{\mu\nu} G_{\pm 5} (X^E (\partial_\mu + i A_\mu) X^\mp - \bar{X}^\mp (\partial_\mu - i A_\mu) X^E) &= 0 \\ N_\mp (\partial_\mu X^\mp + i A_\mu X^\mp) - N_\pm \partial_\mu X^E + i N_\pm X^E A_\mu &= 0 \\ 2 i N A_\mu &= (N_\pm \partial_\mu X^E - N_\mp \partial_\mu \bar{X}^\mp) \end{aligned}$$

$$\therefore \boxed{2 i A_\mu = -\frac{1}{N} (\partial_\mu \bar{X}^\mp N_\mp - \partial_\mu X^E N_\pm)}$$

This agrees with $A_\mu = A_\mu^I + A_\mu^F$ if we ignore fermionic terms.

Plugging this into the action:

$$\begin{aligned} (\partial_\mu - i A_\mu) X^E &= \partial_\mu X^E + \frac{1}{2N} X^E (\partial_\mu \bar{X}^\mp N_\mp - \partial_\mu X^\mp N_\pm) \\ &= \partial_\mu X^E + \frac{1}{2N} X^E \partial_\mu \bar{X}^\mp N_\mp - \frac{1}{2N} X^\mp \partial_\mu X^\mp N_\pm \end{aligned}$$

Note $\frac{\partial}{\partial X^u} \ln(N) = \frac{1}{N} N_u$

$$\partial_\mu \ln(N) = \frac{1}{N} \partial_\mu X^\pm N_\pm + \partial_\mu X^\mp N_\mp$$

$$\boxed{(\partial_\mu - i A_\mu) X^E = \partial_\mu X^\pm - X^\mp \partial_\mu X^u \frac{\partial}{\partial X^u} \ln(N) + \frac{1}{2} X^\mp \partial_\mu \ln(N)}$$

$$\begin{aligned} \therefore g^{\mu\nu} G_{\pm 5} &(\partial_\mu X^E - X^\mp \partial_\mu X^u \frac{\partial}{\partial X^u} \ln(N) + \frac{1}{2} X^\mp \partial_\mu \ln(N)) \\ &\times (\partial_\nu X^\mp + \bar{X}^\mp \frac{\partial}{\partial X^u} () - \frac{1}{2} \bar{X}^\mp \partial_\nu \ln(N)) \\ \Rightarrow g^{\mu\nu} G_{\pm 5} \partial_\mu X^E \partial_\nu X^\mp &- g^{\mu\nu} G_{\pm 5} \partial_\nu X^\mp X^E \partial_\mu X^u \frac{\partial}{\partial X^u} \ln(N) \\ &+ \frac{1}{2} g^{\mu\nu} G_{\pm 5} X^E \partial_\mu \ln(N) \partial_\nu X^\mp \\ g^{\mu\nu} G_{\pm 5} \partial_\mu X^E \bar{X}^\mp \partial_\nu X^u \frac{\partial}{\partial X^u} \ln(N) &- X^\mp \partial_\mu X^u \frac{\partial}{\partial X^u} \ln(N) \end{aligned}$$

$$+ 5^{\mu\nu} G_{\pm\bar{5}} \left(\partial_\mu X^\pm - \dot{X}^\pm \partial_\mu X^\mu \frac{\partial}{\partial X^\mu} l_1(N) + \frac{1}{2} X^\pm \partial_\mu l_1(N) \right) \\ \times \left(-\frac{1}{2} \bar{X}^\mp \partial_\mu l_1(N) \right)$$

Note: $N_{\pm} \left(\partial_\mu X^\pm - X^\pm \partial_\mu X^\mu \frac{\partial}{\partial X^\mu} l_1(N) \right)$
 $= N_{\pm} \partial_\mu X^\pm - X^\pm \partial_\mu X^\mu \frac{1}{N} N_K$
 $= N_{\pm} \partial_\mu X^\pm - \partial_\mu X^\pm N_{\pm} = 0$

$$G_{\pm\bar{5}} g^{\mu\nu} (\partial_\mu - i A_\mu) X^\pm (\partial_\nu + i A_\nu) \bar{X}^\mp \\ (\partial_\mu - i A_\mu) X^\pm (\partial^\mu \bar{X}^\mp G_{\pm\bar{5}} + i A^\mu N_{\pm}) \\ = (\partial_\mu X^\pm - i A_\mu X^\pm) \partial^\mu \bar{X}^\mp G_{\pm\bar{5}} + i A^\mu \left(N_{\pm} \frac{1}{2} X^\pm \partial_\mu l_1(N) \right) \\ = \partial_\mu X^\pm \partial^\mu \bar{X}^\mp G_{\pm\bar{5}} + \frac{1}{2N} \partial_\mu N \partial^\mu \bar{X}^\mp G_{\pm\bar{5}} X^\pm \\ - i \partial_\mu X^\pm N_K X^\pm \partial^\mu \bar{X}^\mp G_{\pm\bar{5}} + \dots \\ = (\partial X)^2 + \frac{1}{2N} \partial_\mu N \partial^\mu \bar{X}^\mp N_{\pm} - i \partial_\mu X^\pm N_K \partial^\mu \bar{X}^\mp N_{\pm} \\ + \dots$$

$$\star \partial_{\pm} \partial_{\bar{5}} l_1(N) = \partial_{\pm} \left(\frac{1}{N} N_{\bar{5}} \right) \\ = -\frac{1}{N^2} N_{\pm} N_{\bar{5}} + \frac{1}{N} N_{\pm\bar{5}}$$

$$\therefore N (\partial_\mu X^\pm) (\partial^\mu X^{\bar{5}}) \left(\frac{1}{N} N_{\pm\bar{5}} - \frac{1}{N^2} N_{\pm} N_{\bar{5}} \right) \\ = G_{\pm\bar{5}} \partial_\mu X^\pm \partial^\mu \bar{X}^{\bar{5}} - \frac{1}{N} (N_{\pm} \partial_\mu X^\pm) (\partial^\mu \bar{X}^{\bar{5}} N_{\bar{5}})$$

$$\star \partial_\mu N \partial^\mu N = (\partial_\mu X^\pm N_{\pm} + \partial_\mu X^{\bar{5}} N_{\bar{5}}) \left(\dots \right)$$

$$\Rightarrow i A_\mu = -\frac{1}{2} \partial_\mu l_1(N) + \partial_\mu X^\pm N_{\pm}$$

$$(\partial_\mu X^\pm - i A_\mu X^\pm) \partial^\mu \bar{X}^{\bar{5}} G_{\pm\bar{5}} + i A^\mu \left(N_{\pm} \frac{1}{2} \partial_\mu l_1(N) \right) \\ = \partial_\mu X^\pm \partial^\mu \bar{X}^{\bar{5}} G_{\pm\bar{5}} - i A_\mu \partial^\mu \bar{X}^{\bar{5}} N_{\bar{5}} + i A^\mu \frac{1}{2} N \partial_\mu l_1(N) \\ = \partial_\mu X^\pm \partial^\mu \bar{X}^{\bar{5}} G_{\pm\bar{5}} - i \left(\frac{1}{2} \partial_\mu l_1(N) \partial^\mu \bar{X}^{\bar{5}} N_{\bar{5}} \right) \\ - i \partial_\mu X^\pm N_{\pm} \partial^\mu \bar{X}^{\bar{5}} N_{\bar{5}}$$

$\Leftrightarrow$

$$\mathcal{L}_r = -\frac{1}{4N} \partial_\mu N \partial^\mu N - N (\partial_\mu x^I) (\partial^\mu \bar{x}^{\bar{J}}) \partial_\mu \partial_{\bar{J}} (\ln(N))$$

* Ignore $N R(e)$ term for now, to understand geometry.

Ex: Consider $\mathcal{L}_r$ in the flat space limit:

$g_{\mu\nu} \mapsto \eta_{\mu\nu}$

Claim: the flat space-limit is given by a $U(1)$ gauge theory on $\mathbb{CP}^n$ subject to the constraint $x^I \bar{x}_{\bar{I}} = 1$ with a flat Kähler target space. $N \rightarrow x^I \partial_{\bar{I}} \bar{x}^{\bar{J}} x_{\bar{J}}$.

Proof:

$$\begin{aligned} \partial_\mu N &= \partial_\mu x^I \bar{x}_{\bar{I}} + \partial_\mu \bar{x}^{\bar{J}} x_{\bar{J}} \\ -\frac{1}{4} (\partial_\mu \partial^\mu N) &= (\partial_\mu x^I \bar{x}_{\bar{I}} + \partial_\mu \bar{x}^{\bar{J}} x_{\bar{J}}) (\partial^\mu x^K \bar{x}_{\bar{K}} + \partial^\mu \bar{x}^{\bar{L}} x_{\bar{L}}) \\ &= -\frac{1}{4} (\partial_\mu x^I \bar{x}_{\bar{I}} \partial^\mu x^K \bar{x}_{\bar{K}} + \partial_\mu x^I \bar{x}_{\bar{I}} \partial^\mu \bar{x}^{\bar{L}} x_{\bar{L}} \\ &\quad + \partial_\mu \bar{x}^{\bar{J}} x_{\bar{J}} \partial^\mu x^K \bar{x}_{\bar{K}} + \partial_\mu \bar{x}^{\bar{J}} x_{\bar{J}} \partial^\mu \bar{x}^{\bar{L}} x_{\bar{L}}) \\ &= \partial_\mu x^I \partial^\mu \bar{x}_{\bar{I}} \end{aligned}$$

Note $\partial_\mu (x^I \bar{x}_{\bar{I}}) = 0 \quad \partial_\mu x^I \bar{x}_{\bar{I}} = -x^{\bar{I}} \partial_\mu \bar{x}_{\bar{I}}$

$$\begin{aligned} \Leftrightarrow \partial_\mu x^I \bar{x}_{\bar{I}} &= -\partial_\mu \bar{x}^{\bar{I}} x_{\bar{I}} \\ -\partial x^I \bar{x}_{\bar{I}} &= (\partial_\mu x^I) \bar{x}_{\bar{I}} - x_{\bar{I}} \partial_\mu \bar{x}^{\bar{I}} \\ &= -\partial x^I - \bar{x}_{\bar{I}} \partial x^I x_{\bar{J}} \partial^\mu \bar{x}^{\bar{J}} \end{aligned}$$

• Can also simplify. Note $2iA_\mu = -\frac{1}{2i} (\partial_\mu \bar{x}^{\bar{I}} x_{\bar{I}} - \partial_\mu x^I \bar{x}_{\bar{I}})$

$$\begin{aligned} A_\mu &= \frac{1}{2i} (\partial_\mu x^I \bar{x}_{\bar{I}} - x_{\bar{I}} \partial_\mu \bar{x}^{\bar{I}}) \\ &= -\frac{1}{2i} \bar{x}_{\bar{I}} \partial_\mu x^I \end{aligned} \quad \square$$

$$\Rightarrow \psi \quad \psi = \psi^I \partial_I \quad (\psi^I \leftrightarrow \bar{\psi}^{\bar{I}}) \quad (\psi^I \leftrightarrow \bar{\psi}^{\bar{I}})$$

subject to $\bar{X}_I X^I = 1$
 $g_{\alpha\bar{\beta}} \mapsto g_{\alpha\bar{\beta}}, \quad N \mapsto X^I \delta_{I\bar{J}} \bar{X}^{\bar{J}}$

The Kähler cone:

* The target space has the topology of a cone:

In $\mathbb{R}^3$, the cone has induced metric

$$ds^2 = dr^2 + r^2(dx^2 + dy^2)$$

* In the basis $\{y, z^a\}$ of $\mathbb{CP}^n$, X^I (and hence) $y \bar{z}^I = X^I$ are zero modes of

$$\partial_I \partial_{\bar{J}} \ln(N) = \frac{G_{I\bar{J}}}{N} - (G_{I\bar{K}} \bar{X}^{\bar{K}} G_{\bar{J}L} X^L) / N^2$$

Since $(\partial_I \partial_{\bar{J}} \ln(N)) X^I = 0$

so, the spacetime gradient $\partial_\mu Y$ does not survive the second term:

$$\ln(y^2 z^I G_{I\bar{J}} \bar{z}^{\bar{J}}) = 2\ln(y) + \ln(z^I G_{I\bar{J}} \bar{z}^{\bar{J}})$$

$$X^I = y z^I \Rightarrow I = \frac{\partial y}{\partial X^I} z^I + y \frac{\partial z^I}{\partial X^I}$$

$$\Rightarrow \boxed{\frac{\partial y}{\partial X^I} = 0}$$

$$\therefore \frac{\partial}{\partial X^I} = \frac{1}{y} \frac{\partial}{\partial z^I}$$

The second term of the $\mathcal{L}$: $\partial_\mu z^a = 0$

$$\mathcal{L} = -\frac{1}{4N} (\partial_\mu N) (\partial^\mu N) - N (\partial_a z^\alpha) \partial^a \bar{z}^{\bar{\beta}} \frac{\partial}{\partial z^\alpha} \frac{\partial}{\partial \bar{z}^{\bar{\beta}}} (\ln [z^I G_{I\bar{J}} \bar{z}^{\bar{J}}])$$

→ If the spacetime has positive signature, such as $\mathbb{R}P^n$, then

$$\boxed{N := r^2} \quad \partial_\mu N = 2r \partial_\mu r$$

$$\mathcal{L} = -\partial_\mu r \partial^\mu r - r^2 \partial_\mu z^\alpha \partial^\mu \bar{z}^{\bar{\beta}} g_{\alpha\bar{\beta}}(z, \bar{z})$$

Note this 'looks' like the metric of a cone.

whose 'Cross-section' is an n -dim Kähler

manifold w/ metric $g_{\alpha\bar{\beta}}(z, \bar{z})$

defined by

$$\boxed{g_{\alpha\bar{\beta}}(z, \bar{z}) := \frac{\partial}{\partial z^\alpha} \frac{\partial}{\partial \bar{z}^{\bar{\beta}}} \ln(z^\pm G_{\pm\bar{z}} \bar{z}^{\bar{\beta}})}$$

$$z^0 \equiv 1, \quad z^\alpha = z^\alpha$$

* For Lorentzian embeddings $N = -r^2$

$$\boxed{\mathcal{L} = \partial_\mu r \partial^\mu r - r^2 g_{\alpha\bar{\beta}}(z, \bar{z}) \partial_\mu z^\alpha \partial^\mu \bar{z}^{\bar{\beta}}}$$

we will define $g_{\alpha\bar{\beta}}$ below.

A Kähler mfd w/ a homothetic Killing vec. is called a Sasakian Manifold. The Killing vec.

generating the $U(1)$ symm, U_1 , is called the Reeb vec. field.

The projection:

* The embedding Kähler has dim $(n+2)$, the projective Kähler mfd dim n , scaling factor

$$\sim \frac{1}{r}$$

$$\boxed{\mathbb{R}P^{n+1} \simeq \mathbb{R}P^1 \times \mathbb{R}P^n}$$

$$\{x^\pm\} \quad \{z^\alpha\} \quad \{y\}$$

* We can pick a rep of each eq. class by imposing a constraint.

$$R = e^a e^b R_{ab}, \quad \delta_b e^a = +1 \lambda_b e^a$$

$$\delta R = 2 \lambda_b R \quad \text{so we must have; from}$$

$$\int d^4x \delta N R \Rightarrow \boxed{\delta N = 2 \lambda_b N}$$

• Thus, we impose the constraint on N directly. As before we impose:

$$\boxed{N(x, \bar{x}) = -a} \quad (\text{Dilatation Gauge})$$

The value of a depends on the theory. For $\mathcal{N}^4$ we simply choose $a = 1 \Rightarrow x^\pm \bar{x}^\pm$

$$\boxed{\begin{array}{l} \underline{N=1}: \quad a = 3\kappa^{-2} \\ \underline{N=2}: \quad a = \kappa^{-2} \end{array}}$$

$$N = -a = y \bar{y} (z^\pm G_{\pm\bar{\pm}} \bar{z}^\pm)$$

$$\therefore \boxed{y \bar{y} = -a [z^\pm G_{\pm\bar{\pm}} \bar{z}^\pm]^{-2}}$$

In this fixed gauge: $\partial_\mu N \partial^\mu N = 0$

$$\mathcal{L} = a \partial_\mu z^\alpha \partial^\mu \bar{z}^{\bar{\alpha}} \frac{\partial}{\partial z^\alpha} \frac{\partial}{\partial \bar{z}^{\bar{\alpha}}} \ln(z^\pm G_{\pm\bar{\pm}} \bar{z}^\pm)$$

$$\ln(y \bar{y}) = \ln(-a [z^\pm G_{\pm\bar{\pm}} \bar{z}^\pm]^{-2})$$

$$\ln(y \bar{y}) = \ln(-a) - \ln(z^\pm G_{\pm\bar{\pm}} \bar{z}^\pm)$$

$$= -\ln(-a^{-1}) - \ln(z^\pm G_{\pm\bar{\pm}} \bar{z}^\pm)$$

$$= -\ln(-a^{-1} z^I G_{I\bar{J}} \bar{z}^{\bar{J}})$$

$$\Rightarrow \ln(z^I G_{I\bar{J}} \bar{z}^{\bar{J}}) = \ln\left(\frac{a}{a}\right)$$

$$\therefore \mathcal{L} = a \partial_\alpha z^\alpha \partial^{\bar{\alpha}} \bar{z}^{\bar{\alpha}} \frac{\partial}{\partial z^\alpha} \frac{\partial}{\partial \bar{z}^{\bar{\alpha}}} \ln(-a^{-1} z^I G_{I\bar{J}} \bar{z}^{\bar{J}})$$

$\therefore$ The Kähler metric for the embedding space is

$$K(z, \bar{z}) \equiv -a \ln[-a^{-1} z^I G_{I\bar{J}} \bar{z}^{\bar{J}}] = a \ln(\gamma \bar{\gamma})$$

After Gauge fixing the Action becomes a nonlinear σ -model on an n -dim Kähler manifold:

$$\mathcal{L} = -g_{\alpha\bar{\beta}} \partial_\alpha z^\alpha \partial^{\bar{\alpha}} \bar{z}^{\bar{\alpha}} = -\partial_\alpha \partial_{\bar{\beta}} K(z, \bar{z}) \partial_\alpha z^\alpha \partial^{\bar{\alpha}} \bar{z}^{\bar{\alpha}}$$

Thus, the gauge fixing process has simply chosen a slice of the Kähler cone to work on. Since the manifold/Bundle structure of $C(\Sigma)$ is simply connected we have no problems choosing different Gauges. However, different choices amount to diff. Geometries.

$$\partial_\alpha: \text{On } PK^n$$

Claim: The Homogeneity conditions on N :

$$N = N_I X^I = N_{\bar{I}} \bar{X}^{\bar{I}} = X^I N_{I\bar{J}} \bar{X}^{\bar{J}}$$

imply that

$$\text{Lid } \partial_\alpha K = -a \frac{1}{N} N_I \partial_\alpha z^I$$

$$(\therefore) \text{ it}_\alpha = \frac{1}{2} a^{-1} (\partial_\alpha \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K - \partial_\alpha z^\alpha \partial_\alpha K) + \frac{1}{2} \partial_\alpha (\ln(\gamma \bar{\gamma}))$$

→ Yorc Gauge

$$\text{Proof: } (i) \partial_\alpha K = (-a)^2 \left(\frac{1}{2\sqrt{G_{I\bar{J}}} \bar{z}^{\bar{J}}} \right) \left(\partial_\alpha (-a^{-1}) z^I G_{I\bar{J}} \bar{z}^{\bar{J}} + (-a^{-1}) \partial_\alpha z^I N_I \right)$$

$$\partial_\alpha K = (-a)^2 (z^I G_{I\bar{J}} \bar{z}^{\bar{J}})^{-1} (-a^{-1} \partial_\alpha z^I G_{I\bar{J}} \bar{z}^{\bar{J}}(\bar{z}))$$

$$\partial_\alpha K = \left(-a \quad \gamma \bar{\gamma} / N \right) (G_{I\bar{J}} \bar{z}^{\bar{J}})$$

$$\partial_\alpha K = -a \frac{\gamma}{N} N_I \partial_\alpha z^I \quad \square$$

$$(ii) \text{Zitun} = -\frac{1}{N} (\partial_m \bar{X}^{\bar{J}} N_{\bar{J}}^{\bar{I}} - \partial_m X^I N_I)$$

Work w/ first term

$$a^{-1} (\partial_m (\gamma z^{\bar{J}})) N_{\bar{J}}^{\bar{I}} = (\partial_m \gamma) \bar{z}^{\bar{J}} N_{\bar{J}}^{\bar{I}} X^I + \gamma \partial_m \bar{z}^{\bar{J}} N_{\bar{J}}^{\bar{I}}$$

$$\underbrace{\partial_m (\gamma) \frac{1}{\gamma} N}_{\text{part of gauge}} + \underbrace{\gamma \partial_m \bar{z}^{\bar{J}} N_{\bar{J}}^{\bar{I}}}$$

$$\rightarrow \bar{\gamma} \partial_m \bar{z}^{\bar{J}} : \quad \partial_{\bar{\alpha}} K = -\frac{a}{N} \bar{\gamma} N_I \partial_{\bar{\alpha}} z^I$$

$$\bar{\gamma} \partial_m \bar{z}^{\bar{J}} N_{\bar{J}}^{\bar{I}} \quad \partial_m \bar{z}^{\bar{J}} = \partial_m \bar{z}^{\bar{J}} \partial_\alpha z^{\bar{I}} N_I$$

$$\bar{\gamma} \partial_m \bar{z}^{\bar{J}} \partial_{\bar{\alpha}} z^{\bar{I}} N_I$$

→ $\partial_{\bar{\alpha}} K$

$$\text{Zitun} = a^{-1} (\partial_m \bar{z}^{\bar{J}} \partial_{\bar{\alpha}} K - \partial_m z^I \partial_\alpha K) + \partial_m \ln\left(\frac{\gamma}{\bar{\gamma}}\right)$$

Kähler transformations:

* Note N is not preserved under Kähler transf. However $K(z, \bar{z})$ is. The variable γ is not completely fixed.

$$\text{Redefine: } \gamma' := \gamma e^{f(z)/a}, \quad z'^I := z^I e^{-f(\bar{z})/a}$$

$$K(z, \bar{z}) = -a \ln \left(-a e^{f(z)/a} z^I G_{I\bar{J}} \bar{z}'^{\bar{J}} e^{-f(\bar{z})/a} \right) \\ = -a \ln \left(\exp(f(z)/a - f(\bar{z})/a) \right) +$$

$$K(z, \bar{z}) = K(z, z) + f(z) + \bar{f}(\bar{z})$$

* If a quantity transforms under Kähler trans. it is useful to include a new term, the Kähler connection

• Kähler covariant derivative:

For $Z^I(z)$ & $\bar{Z}^{\bar{I}}(\bar{z})$

$$\begin{array}{l|l} \nabla_\alpha Z^I := \partial_\alpha Z^I + a^{-1}(\partial_\alpha K) Z^I & \bar{\nabla}_{\bar{\alpha}} Z^I = \partial_{\bar{\alpha}} Z^I = 0 \\ \bar{\nabla}_{\bar{\alpha}} \bar{Z}^{\bar{I}} := \partial_{\bar{\alpha}} \bar{Z}^{\bar{I}} + a^{-1}(\partial_{\bar{\alpha}} K) \bar{Z}^{\bar{I}} & \nabla_\alpha \bar{Z}^{\bar{I}} = \partial_\alpha \bar{Z}^{\bar{I}} = 0 \end{array}$$

* Here we view $f(z)$ and $\bar{f}(\bar{z})$ as independent Gauge parameters: This leaves ∇Z & $\bar{\nabla} \bar{Z} \rightarrow e^{f/a} \nabla Z$

proof:

$$\begin{aligned} \nabla_\alpha Z^I &= \partial_\alpha (e^{-f(z)/a} Z^I) + (\partial_\alpha K/a) e^{-f(z)/a} Z^I \\ &= \partial_\alpha f/a e^{-f(z)/a} Z^I + (\partial_\alpha K + \partial_\alpha f) e^{-f(z)/a} Z^I \\ &= e^{-f(z)/a} (\partial_\alpha Z^I + \partial_\alpha K Z^I) \\ &= e^{-f(z)/a} \nabla_\alpha Z^I \end{aligned}$$

$\partial_\alpha K$: Holomorphic $U(1)$ Gauge field

$\partial_{\bar{\alpha}} K$: Anti Holo $\overline{U(1)}$ Gauge field.

$\leadsto$ Basically a new Sym for which we fix the value of y & $\bar{y}$.

When an object $\mathcal{O}_\alpha Z \dots$ transforms under complex coord. trans. on $\mathbb{C}K^n$ we must take $\partial_\alpha \mapsto \nabla_\alpha$

* More general functions: $V(z, \bar{z})$

$$V'(z, \bar{z}) = V(z, \bar{z}) \exp[-a^{-1}(\omega_+ f(z) + \omega_- \bar{f}(\bar{z}))]$$

V is said to have Kähler weight (ω_+, ω_-) .

In general V has $[\text{Weyl, char}] = [h, c]$

$$\omega_{\pm} := \frac{1}{2}h \mp \frac{3}{2}c$$

$$Z^{\pm} : (1, 0) \quad , \quad \bar{Z}^{\pm} : (0, 1)$$

$$\chi : (-1, 0)$$

- Functions on $\mathbb{R}^{n+1}$ do not transform under this type of Kähler trans. For $V(x, \bar{x})$ which transforms as

$$\delta V = \omega_+(\lambda_D + i\lambda_T)V + \omega_-(\lambda_D - i\lambda_T)V$$

$$V(x, \bar{x}) = y^{\omega_+} \bar{y}^{\omega_-} V(z, \bar{z})$$

$V(x, \bar{x})$ transforms as $V'(z, \bar{z})$ so we must include the ω_+ and ω_- terms:

$$\nabla_{\alpha} V(z, \bar{z}) = \partial_{\alpha} V(z, \bar{z}) + \omega_+ a^{-1}(\partial_{\alpha} K) V(z, \bar{z})$$

$$\bar{\nabla}_{\bar{\alpha}} V(z, \bar{z}) = \partial_{\bar{\alpha}} V(z, \bar{z}) + \omega_- a^{-1}(\partial_{\bar{\alpha}} K) V(z, \bar{z})$$

Proof of transf.: $\nabla_{\alpha} V$ only, $\bar{\nabla}_{\bar{\alpha}} V$ follows identically.

$$\begin{aligned} \nabla_{\alpha} V^c &= \partial_{\alpha} (e^{-a^{-1}(\omega_+ f + \omega_- \bar{f})} V) + \omega_+ a^{-1} \partial_{\alpha} K^* V \\ &= -a^{-1} \omega_+ \partial_{\alpha} f e^{c_1} V + \omega_+ a^{-1} \partial_{\alpha} K V + \omega_+ a^{-1} \partial_{\alpha} f^c V \\ &= e^{-a^{-1}(\omega_+ f + \omega_- \bar{f})} \nabla_{\alpha} V(z, \bar{z}) \quad \square \end{aligned}$$

Note $\boxed{\nabla_{\alpha} = \partial_{\alpha} + \omega_+ a^{-1}(\partial_{\alpha} K)}$

$$\begin{aligned}
[\nabla_\alpha, \nabla_\beta] &= [\partial_\alpha + \omega_+ a^{-1} \partial_{\alpha K}, \partial_\beta + \omega_- a^{-1} \partial_{\beta K}] \\
&= [\partial_\alpha, \partial_\beta] + \omega_+ a^{-1} [\partial_\alpha K, \partial_\beta] + [\partial_\alpha, \omega_- a^{-1} \partial_{\beta K}] \\
&\quad + \omega_+^2 a^{-2} [\partial_\alpha K, \partial_{\beta K}] \\
&= \omega_+ a^{-1} (\partial_\alpha K \partial_\beta - (\partial_\beta \partial_\alpha K + \partial_\alpha K \partial_\beta)) \quad \textcircled{2} \\
&= \omega_+ a^{-1} (\partial_\beta K \partial_\alpha - (\partial_\alpha \partial_\beta K + \partial_\beta \partial_\alpha)) \\
&= \omega_+ a^{-1} (-\partial_\beta \partial_\alpha K + \partial_\alpha \partial_\beta K) = 0
\end{aligned}$$

$$\therefore \boxed{[\nabla_\alpha, \nabla_\beta] = 0}$$

For scalar $\mathcal{L}^{NS}$ $V(z, \bar{z})$, we can find $[\nabla_\alpha, \tilde{\nabla}_{\bar{\alpha}}]$ by noting above in the second line $\textcircled{2}$ $\omega_+ \rightarrow \omega_-$

$$\boxed{[\nabla_\alpha, \tilde{\nabla}_{\bar{\alpha}}]V = a^{-1} (\omega_+ - \omega_-) g_{\alpha\bar{\alpha}} V(z, \bar{z})}$$

Note for Higher order Tensor terms $V^{\alpha\bar{\alpha}}_{\beta\bar{\beta}} \dots$ we must include $(Riem)_{\alpha\bar{\alpha}} \dots$

How is $R_{\mu\nu}^A$ defined? Recall for a general Gauge theory w/ Gauge Parameter B_μ^A

$$R = -2i g_{\alpha\bar{\beta}} d\bar{z}^\alpha \wedge dz^\beta \quad (\text{fundamental 2-form})$$

*Note, ∇_α is on $\mathbb{C}K^n$ so we can define the covariant derivative in $M^{n+1}(x^n)$

$$\boxed{\nabla_\mu = \partial_\mu z^\alpha \nabla_\alpha + \partial_\mu \bar{z}^{\bar{\alpha}} \tilde{\nabla}_{\bar{\alpha}}}$$

• After complete gauge fixing $\gamma = \gamma(z, \bar{z})$. We have fixed its modulus but it remains to fix its phase. This is where the T-gauge fixing comes into play.

$y: \text{Kähler}(-1,0)$, $C = \mathbb{I}$
add the T-gauge field:

$$\nabla_m y \equiv (\partial_m - i\tau_m - a^{-1}(\partial_m z^\alpha) \partial_\alpha K) y = 0$$

Proof: Recall, in the depths of all of this, τ_m is the $U(1)_R$ gauge field, and

$$i\tau_m = \frac{1}{2} a^{-1} (\partial_m \bar{z}^\alpha \partial_\alpha K - \partial_m z^\alpha \partial_\alpha K) + \frac{1}{2} \partial_m \ln\left(\frac{K}{y}\right)$$

$$i\tau_m - a^{-1} (\partial_m z^\alpha \partial_\alpha K) = \frac{1}{2} a^{-1} (\partial_m K) - \frac{1}{2} \partial_m \ln\left(\frac{K}{y}\right)$$

$$\begin{aligned} \therefore \nabla_m y &= \left(\frac{1}{2} a^{-1} \partial_m K - \frac{1}{2} \partial_m \ln\left(\frac{K}{y}\right) \right) y \\ &= \partial_m y - \frac{1}{2} (\partial_m \ln(y \bar{y}) + \ln\left(\frac{K}{y}\right)) y \\ &= \partial_m y - \frac{1}{2} \partial_m \ln(y^2) y = \partial_m y - \partial_m y = 0 \quad \square \end{aligned}$$

• After any T-Gauge transformation $y = y(z, \bar{z})$
 $\rightarrow$ split τ_m into $\partial_m z^\alpha$ and $\partial_m \bar{z}^\alpha$.

$\nabla_m y$ splits as

$$\nabla_m y = \partial_m z^\alpha \nabla_\alpha y + \partial_m \bar{z}^\alpha \nabla_{\bar{\alpha}} y = 0$$

so $\boxed{\nabla_\alpha y = \nabla_{\bar{\alpha}} y = 0}$

Form of ∇_α is not determined until we fix Gauge.

Claim: Since X does not transform under Kähler transf., then

$$(\partial_m - i\tau_m) X^\pm = \nabla_m X^\pm = \nabla_m y z^\pm + y \nabla_m z^\pm = y (\partial_m z^\alpha) \nabla_\alpha z^\pm$$

Proof:

Since $X^\pm$ has Kähler weight $(0,0)$, it

$$s_0 \quad \nabla_m = \partial_m - iA_m$$

and so

$$\begin{aligned} \nabla_m X^I &= \nabla_m \gamma Z^I + \gamma \nabla_m Z^I \\ &= \gamma (\partial_m Z^\alpha \nabla_\alpha + \partial_m \bar{Z}^{\bar{\alpha}} \bar{\nabla}_{\bar{\alpha}}) Z^I \end{aligned}$$

Since Z^I has Kähler weight $(1, 0)$

$$\bar{\nabla}_{\bar{\alpha}} Z^I = 0 \Rightarrow \bar{\nabla}_{\bar{\alpha}} Z^I = 0$$

$$\therefore \boxed{\nabla_m X^I = \gamma \partial_m Z^\alpha \nabla_\alpha Z^I} \quad \square$$

Since N is invariant under chiral $\mathfrak{s}$ Kähler:

$$\begin{aligned} N_I \nabla_\alpha Z^I &= 0, \quad N_I = G_{I\bar{J}} \bar{\gamma} \bar{Z}^{\bar{J}} \\ g_{\alpha\bar{\beta}} &= \gamma \bar{\gamma} \nabla_\alpha Z^I G_{I\bar{J}} \bar{\gamma} \bar{Z}^{\bar{J}} \end{aligned}$$

Proof:

$$\nabla_\alpha N = 0 \Rightarrow \nabla_\alpha (\gamma \bar{\gamma} Z^I G_{I\bar{J}} \bar{Z}^{\bar{J}}) = 0$$

$$\gamma \neq 0 \Rightarrow \boxed{\nabla_\alpha Z^I N_I = 0} \quad \checkmark \quad N_I = G_{I\bar{J}} \bar{\gamma} \bar{Z}^{\bar{J}}$$

$$\begin{aligned} \bar{\nabla}_{\bar{\beta}} \nabla_\alpha N &= \bar{\nabla}_{\bar{\beta}} (\nabla_\alpha Z^I \bar{\gamma} G_{I\bar{J}} \bar{Z}^{\bar{J}}) \\ &= \nabla_\alpha Z^I \bar{\gamma} G_{I\bar{J}} \bar{\nabla}_{\bar{\beta}} \bar{Z}^{\bar{J}} \end{aligned}$$

$$[\nabla_\alpha, \bar{\nabla}_{\bar{\beta}}] N = a^{-1} g_{\alpha\bar{\beta}} N$$

$$\nabla_\alpha Z^I G_{I\bar{J}} \bar{\nabla}_{\bar{\beta}} \bar{Z}^{\bar{J}} \gamma \bar{\gamma} = g_{\alpha\bar{\beta}}$$

Not sure how to the difference occurs and

how $g_{\alpha\bar{\beta}} \neq 0$!

• $\gamma \bar{\gamma} = a^{-1} (Z^I G_{I\bar{J}} \bar{Z}^{\bar{J}})^{-1}$ and the previous result:

$$\text{Claim: } \begin{pmatrix} -a_0 \\ 0 \end{pmatrix} g_{\alpha\bar{\beta}} = \gamma \bar{\gamma} \begin{pmatrix} Z^I \\ \nabla_\alpha Z^I \end{pmatrix} G_{I\bar{J}} \begin{pmatrix} \bar{Z}^{\bar{J}} \\ \bar{\nabla}_{\bar{\beta}} \bar{Z}^{\bar{J}} \end{pmatrix}$$

proof:

$$\begin{pmatrix} Z^I \\ \nabla_\alpha Z^I \end{pmatrix} = \begin{pmatrix} \bar{Z}^{\bar{J}} \\ \bar{\nabla}_{\bar{\beta}} \bar{Z}^{\bar{J}} \end{pmatrix}$$

$$= \gamma \bar{\gamma} G_{\pm \bar{J}} \begin{pmatrix} z^I \bar{z}^{\bar{J}} & z^I \bar{\nabla}_{\bar{J}} \bar{z}^{\bar{J}} \\ \bar{\nabla}_{\bar{J}} z^I \bar{z}^{\bar{J}} & \bar{\nabla}_{\bar{J}} z^I \bar{\nabla}_{\bar{J}} \bar{z}^{\bar{J}} \end{pmatrix}$$

$$= \begin{pmatrix} N & N^{\bar{J}} \bar{\nabla}_{\bar{J}} X^{\bar{J}} \\ N_{\pm} \bar{\nabla}_{\alpha} z^{\pm} & g_{\alpha \bar{\beta}} \end{pmatrix} = \begin{pmatrix} -a & 0 \\ 0 & g_{\alpha \bar{\beta}} \end{pmatrix}$$

* These ^(matrix) matrices must be invertible so that scalar kinetic terms are non-degenerate. This implies the degeneracy of $\begin{pmatrix} z^{\pm} \\ \bar{\nabla}_{\alpha} z^{\pm} \end{pmatrix}$

$$\begin{pmatrix} z^{\pm} \\ \bar{\nabla}_{\alpha} z^{\pm} \end{pmatrix}^{-1} \begin{pmatrix} -a & 0 \\ 0 & g_{\alpha \bar{\beta}} \end{pmatrix} (\bar{z}^{\bar{J}}, \bar{\nabla}_{\bar{J}} \bar{z}^{\bar{J}})^{-1} \begin{pmatrix} 1 \\ \gamma \bar{\gamma} \end{pmatrix} = G_{\pm \bar{J}}$$

$$G_{\pm \bar{J}} = (\bar{z}^{\bar{J}}, \bar{\nabla}_{\bar{J}} \bar{z}^{\bar{J}}) \begin{pmatrix} -\frac{1}{a} & 0 \\ 0 & g_{\alpha \bar{\beta}}^{-1} \end{pmatrix} \begin{pmatrix} z^{\pm} \\ \bar{\nabla}_{\alpha} z^{\pm} \end{pmatrix} \gamma \bar{\gamma}$$

$$G_{\pm \bar{J}} = (\bar{z}^{\bar{J}}, \bar{\nabla}_{\bar{J}} \bar{z}^{\bar{J}}) \begin{pmatrix} -\frac{1}{a} z^{\pm} \\ g^{\alpha \bar{\beta}} \bar{\nabla}_{\alpha} z^{\pm} \end{pmatrix}$$

$$= \left(-\frac{1}{a} z^{\pm} \bar{z}^{\bar{J}} + \bar{\nabla}_{\alpha} z^{\pm} g^{\alpha \bar{\beta}} \bar{\nabla}_{\bar{J}} \bar{z}^{\bar{J}} \right)$$

$$\therefore \boxed{G^{\pm \bar{J}} = \gamma \bar{\gamma} \left(-a^{-1} z^{\pm} \bar{z}^{\bar{J}} + \bar{\nabla}_{\alpha} z^{\pm} g^{\alpha \bar{\beta}} \bar{\nabla}_{\bar{J}} \bar{z}^{\bar{J}} \right)}$$

By applying $\bar{\nabla}_{\bar{\beta}}$ to the second row

$$\bullet \bar{\nabla}_{\bar{\beta}} \bar{\nabla}_{\alpha} z^{\pm} = -\gamma G^{\pm \bar{I}} N_{\bar{J} K \bar{L}} \bar{\nabla}_{\alpha} z^{\pm} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{K}}$$

$$= -\gamma \Gamma_{\bar{J} K}^{\pm \bar{I}} \bar{\nabla}_{\alpha} z^{\pm} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{K}}$$

$$\boxed{[\bar{\nabla}_{\bar{\alpha}}, \bar{\nabla}_{\bar{\beta}}] \bar{\nabla}_{\alpha} z^{\pm} = a^{-1} g_{\beta \bar{\alpha}} \bar{\nabla}_{\alpha} z^{\pm} + R_{\bar{\alpha} \bar{\beta} \alpha}^{\gamma} \bar{\nabla}_{\gamma} z^{\pm}}$$

Proof: $\bar{\nabla}_{\bar{\alpha}} (\bar{\nabla}_{\bar{\beta}} \bar{\nabla}_{\alpha} z^{\pm})$

$$= \bar{\nabla}_{\bar{\alpha}} \left(-\gamma G^{\pm \bar{I}} N_{\bar{J} K \bar{L}} \bar{\nabla}_{\alpha} z^{\pm} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{K}} \right)$$

$$= -\gamma \bar{\nabla}_{\bar{\alpha}} G^{\pm \bar{I}} N_{\bar{J} K \bar{L}} \bar{\nabla}_{\alpha} z^{\pm} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{K}} - \gamma G^{\pm \bar{I}} \bar{\nabla}_{\bar{\alpha}} N_{\bar{J} K \bar{L}} \bar{\nabla}_{\alpha} z^{\pm} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{K}}$$

$$= -\gamma \bar{\nabla}_{\bar{\alpha}} G^{\pm \bar{I}} N_{\bar{J} K \bar{L}} \bar{\nabla}_{\alpha} z^{\pm} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{K}} - \gamma G^{\pm \bar{I}} \bar{\nabla}_{\bar{\alpha}} N_{\bar{J} K \bar{L}} \bar{\nabla}_{\alpha} z^{\pm} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{K}}$$

$$= -\gamma(\bar{\gamma}) (-a^{-1} z^\mu \bar{V}_\alpha \bar{z}^\mu N_{5KL} V_\alpha z^\mu \bar{V}_\beta z^\mu)$$

$$-\gamma(\bar{\gamma}) (N_{5KL} V_\alpha z^\mu \bar{V}_\beta z^\mu g^{\alpha\bar{\beta}} \nabla_\alpha z^\mu \bar{\nabla}_\alpha \bar{V}_\beta z^\mu) \quad (b)$$

$$\begin{aligned} (b) & -\gamma(\bar{\gamma}) N_{5KL} V_\alpha z^\mu \bar{V}_\beta z^\mu g^{\alpha\bar{\beta}} V_\alpha z^\mu (-\bar{\gamma} \bar{G}^{\bar{L}M} N_{MNP} \bar{V}_\alpha \bar{z}^\mu \bar{V}_\beta \bar{z}^\mu) \\ & = (\bar{\gamma})^2 N_{5KL} V_\alpha z^\mu \bar{V}_\beta z^\mu g^{\alpha\bar{\beta}} V_\alpha z^\mu \bar{G}^{\bar{L}M} N_{MNP} V_\alpha \bar{z}^\mu \bar{V}_\beta \bar{z}^\mu \\ & = (\bar{\gamma})^2 N_{5KL} () (-a^{-1} z^\mu \bar{z}^\mu + g^{\alpha\bar{\beta}} V_\alpha z^\mu \bar{V}_\beta \bar{z}^\mu) () \\ & = (\bar{\gamma})^2 N_{5KL} V_\alpha z^\mu \bar{V}_\beta z^\mu g^{\alpha\bar{\beta}} V_\alpha z^\mu g^{\alpha\bar{\beta}} V_\alpha z^\mu \bar{V}_\beta \bar{z}^\mu N_{MNP} V_\alpha \bar{z}^\mu \end{aligned}$$

→ Note: Sure about this

* Apply $\gamma \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} \bar{\nabla}_\alpha$ to $V_\beta V_\alpha z^\mu$:

$$\begin{aligned} & \gamma \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} \bar{\nabla}_\alpha (-\gamma \bar{G}^{\bar{L}M} V_\alpha z^\mu \bar{V}_\beta z^\mu) \\ & = -\gamma^2 \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} \bar{\nabla}_\alpha \bar{G}^{\bar{L}M} V_\alpha z^\mu \bar{V}_\beta z^\mu \\ & = -\gamma^2 \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} \bar{\nabla}_\alpha \bar{z}^\mu \partial_L \bar{G}^{\bar{L}M} V_\alpha z^\mu \bar{V}_\beta z^\mu \\ & = \gamma^2 \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} R_{KL}{}^I{}_K V_\alpha z^\mu \bar{V}_\beta z^\mu \\ & = \gamma^2 \bar{\gamma} \bar{V}_\beta \bar{z}^\mu R_{KL}{}^I{}_K V_\alpha z^\mu \bar{V}_\beta z^\mu \end{aligned}$$

$$\begin{aligned} & \gamma \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} \bar{\nabla}_\alpha V_\beta V_\alpha z^\mu \\ & = \gamma \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} (a^{-1} g_{\bar{\beta}\alpha} \nabla_\alpha z^\mu + R_{\bar{\alpha}\beta\alpha}{}^\gamma V_\gamma z^\mu) \\ & = \gamma \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} a^{-1} g_{\bar{\beta}\alpha} \nabla_\alpha z^\mu + \gamma \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} R_{\bar{\alpha}\beta\alpha}{}^\gamma V_\gamma z^\mu \\ & = g_{\bar{\beta}\alpha} g_{\beta\bar{\alpha}} + \gamma \bar{\gamma} \bar{V}_\beta \bar{z}^\mu \bar{G}^{\bar{L}M} R_{\bar{\alpha}\beta\alpha}{}^\gamma V_\gamma z^\mu \\ & = g_{\beta\alpha} g_{\bar{\beta}\bar{\alpha}} + g_{\bar{\beta}\gamma} R_{\bar{\alpha}\beta\alpha}{}^\gamma \end{aligned}$$

$$\therefore \gamma^2 \bar{\gamma}^2 \bar{V}_\beta \bar{z}^\mu R_{KL}{}^I{}_K V_\alpha z^\mu \bar{V}_\beta z^\mu = g_{\bar{\beta}\alpha} g_{\beta\bar{\alpha}} + g_{\bar{\beta}\gamma} R_{\bar{\alpha}\beta\alpha}{}^\gamma \nabla_\alpha z^\mu$$

$$(\gamma)^2 R_{K\bar{L}\bar{J}K} \nabla_a z^I \nabla_{\bar{b}} \bar{z}^{\bar{K}} \bar{\nabla}_{\bar{a}} \bar{z}^{\bar{L}} \nabla_{\bar{b}} \bar{z}^{\bar{J}} \\ = a^{-1} g_{B\bar{K}} g_{B\bar{a}} + R_{\bar{a}B\bar{a}\bar{B}}$$

$\therefore$

$$R_{\alpha\bar{\alpha}\beta\bar{\beta}} - 2a^{-1} g_{\alpha\beta} g_{\bar{\alpha}\bar{\beta}} = (\gamma)^2 R_{I\bar{I}J\bar{J}} \nabla_{\alpha} z^I \nabla_{\beta} z^{\bar{J}} \bar{\nabla}_{\bar{\alpha}} \bar{z}^{\bar{I}} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{J}}$$

* From the homogeneity conditions it is clear that z^I is a zero eigenvector of the curvature tensor $z^K N_{K\bar{I}\bar{J}} = 0 \rightarrow N_{\bar{I}\bar{J}} + z^K N_{\bar{I}K\bar{J}} = 0$

• The condition for integrability of a closed homothetic Killing vector k^{μ} is $R_{\mu\nu\rho\sigma} k^{\rho} = 0$

so R.H.S we can replace $\nabla_{\alpha} z^I$ w/ $\partial_{\alpha} z^I$.

Physical Fermions

• To obtain physical Bosons we took the scalar fields from the cont. Basis $\{x^{\pm}\} \mapsto \{y, z^{\alpha}\}$.

Modulus of y is fixed by the Dilation Gauge.

$$N(x, \bar{x}) = -a = y\bar{y} (z^I g_{I\bar{J}} z^{\bar{J}})$$

$\rightarrow$ The phase $f(z)$ corresponds to a freedom in the R -symmetry Gauge.

$$\mathbb{R}^{n+1} \cong \mathbb{C} \times \mathbb{P}K^n$$

$\rightarrow$ Fermions: $\{\Omega^{\pm}\} \mapsto \{\chi^0, \chi^{\alpha}\}$

Recall the S-Gauge Condition $N_{\bar{I}} \Omega^{\bar{I}} = 0$

We choose $\{\Omega^{\pm}\}$ s.t. $\chi^0 = 0$ from this

$$\langle \Omega^{\pm} | \Omega^{\pm} \rangle = \frac{1}{2} = \gamma^{\pm}$$

use the notation $P_L \chi^\alpha \equiv \chi^\alpha$
 $P_R \chi^\alpha \equiv \bar{\chi}^\alpha$

- For $\chi^\pm$ we defined $\chi^\pm = \gamma z^\pm$
 i.e. we chose a rep of PK^n and the $U(1)_R$
 bundle over it. Here,

$$\Omega^\pm = \gamma (\chi^0 z^\pm + \chi^\alpha \nabla_\alpha z^\pm) = \gamma (z^\pm \nabla_\alpha z^\pm) \begin{pmatrix} \chi^0 \\ \chi^\alpha \end{pmatrix}$$

$\nabla_\alpha z^\pm$ is required to ensure uniform behavior under
 Kähler transformations

Claim: The bosonic fields $z^\pm$ & $\nabla_\alpha z^\pm$ are orthogonal
 to one another in HK^{n+1}

Proof: Since $N = \gamma z^\pm N_\pm$

$$\begin{cases} \nabla_\alpha N = 0 \Rightarrow \nabla_\alpha z^\pm N_\pm = 0 & \text{But } N_\pm = G_{\pm\bar{j}} \bar{z}^{\bar{j}} \\ \nabla_\alpha z^\pm G_{\pm\bar{j}} \bar{z}^{\bar{j}} = 0 \end{cases}$$

$$G_{\pm\bar{j}} \gamma^2 \begin{pmatrix} -\frac{1}{a} & \\ & g^{-1} \end{pmatrix} = \left(\bar{z}^{\bar{j}} \nabla_{\bar{\beta}} \bar{z}^{\bar{\beta}} \right)^{-1} \left(\nabla_\alpha z^\pm \right)^{-1}$$

$$G_{\pm\bar{j}} \gamma^2 \begin{pmatrix} -\frac{1}{a} & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} z^\pm \\ \nabla_\alpha z^\pm \end{pmatrix} = \left(\bar{z}^{\bar{j}} \nabla_{\bar{\beta}} \bar{z}^{\bar{\beta}} \right)^{-1}$$

Can use this to solve for $\{\chi^0, \chi^\alpha\}$, however
 Not sure where the Euclidean inner product
 of $(z^\pm \nabla_\alpha z^\pm) \begin{pmatrix} \chi^0 \\ \chi^\alpha \end{pmatrix}$ comes from.

$$\chi^\pm \in I^*(TK \otimes (\chi^\pm)^*(Spin(?)))$$

$$\Omega^\pm = \gamma (z^\pm \nabla_\alpha z^\pm) \begin{pmatrix} \chi^0 \\ \chi^\alpha \end{pmatrix}$$

$$\begin{aligned} \therefore \begin{pmatrix} \chi^0 \\ \chi^\alpha \end{pmatrix} &= \bar{\gamma} G_{\pm\bar{j}} \Omega^\pm \begin{pmatrix} -a^{-1} & 0 \\ 0 & g^{\alpha\bar{\beta}} \end{pmatrix} \begin{pmatrix} \bar{z}^{\bar{j}} \\ \nabla_{\bar{\beta}} \bar{z}^{\bar{\beta}} \end{pmatrix} \\ &= \bar{\gamma} G_{\pm\bar{j}} \Omega^\pm \begin{pmatrix} -a^{-1} \bar{z}^{\bar{j}} & 0 \\ 0 & g^{\alpha\bar{\beta}} \nabla_{\bar{\beta}} \bar{z}^{\bar{\beta}} \end{pmatrix} \end{aligned}$$

$$\chi^0 = \bar{y} G_{\pm} \bar{z} \Omega^{\pm} (-a^{-1}) z^{\pm} = -\bar{y} N_{\pm} \Omega^{\pm}$$

$$\chi^{\alpha} = \bar{y} G_{\pm} \bar{z} \Omega^{\pm} g^{\alpha\beta} \bar{\nabla}_{\beta} z^{\pm}$$

→ We see that the δ -gauge condition $N_{\pm} \Omega^{\pm}$ in this parametrization is equivalent to $\chi^0 = 0$.
So there is no bundle structure here.

Claim: $\delta y = \frac{1}{\sqrt{2} a} \bar{y} \bar{e} \chi^{\alpha} \partial_{\alpha} k$, $\delta \bar{y} = \frac{1}{\sqrt{2} a} \bar{y} \bar{e} \chi^{\alpha} \partial_{\alpha} k$

Proof: $\delta \chi^{\pm} = \bar{e} \Omega^{\pm} \frac{1}{\sqrt{2}}$

$$\delta y z^{\pm} + y \delta z^{\pm} = \bar{e} \frac{1}{\sqrt{2}} (y (z^{\pm}, \bar{\nabla}_{\alpha} z^{\pm})) \begin{pmatrix} \chi^0 \\ \chi^{\alpha} \end{pmatrix}$$

Here we assume the δ -gauge $\chi^0 = 0$

$$\delta y z^{\pm} + y \delta z^{\pm} = \bar{e} \frac{1}{\sqrt{2}} y \bar{\nabla}_{\alpha} z^{\pm} \chi^{\alpha}$$

$$\delta y z^{\pm} + y \frac{\partial z^{\pm}}{\partial z^{\alpha}} \delta z^{\alpha} = \frac{1}{\sqrt{2}} \bar{e} y \bar{\nabla}_{\alpha} z^{\pm} \chi^{\alpha}$$

Here we write $\partial_{\alpha} z^{\pm} + a^{-1} \partial_{\alpha} k z^{\pm} = a^{-1} \partial_{\alpha} k z^{\pm}$

$$\delta y z^{\pm} + y (\bar{\nabla}_{\alpha} - a^{-1} \partial_{\alpha} k) z^{\pm} \delta z^{\alpha} = \frac{1}{\sqrt{2}} \bar{e} y \bar{\nabla}_{\alpha} z^{\pm} \chi^{\alpha}$$

Since $z^0 = 1$

$$\delta y + y (-a^{-1} \partial_{\alpha} k) \delta z^{\alpha} = 0$$

$$\delta y = \frac{y}{a} \partial_{\alpha} k \delta z^{\alpha}$$

$$(z^{\pm} \frac{y}{a} \partial_{\alpha} k + y (\bar{\nabla}_{\alpha} - a^{-1} \partial_{\alpha} k) z^{\pm}) \delta z^{\alpha} = \frac{1}{\sqrt{2}} \bar{e} y \bar{\nabla}_{\alpha} z^{\pm}$$

$$= y \bar{\nabla}_{\alpha} z^{\pm} \delta z^{\alpha} = \frac{1}{\sqrt{2}} \bar{e} y \bar{\nabla}_{\alpha} z^{\pm} \chi^{\alpha}$$

$$\therefore \delta z^{\alpha} = \frac{1}{\sqrt{2}} \bar{e} \chi^{\alpha}$$

$$\delta y = \frac{y}{a} \partial_{\alpha} k \frac{1}{\sqrt{2}} \bar{e} \chi^{\alpha} \text{ as claimed.}$$

Symmetries of projective Kähler

Recall, for a Killing $K^a(z, \bar{z})$

$$\mathcal{L}_K g_{I\bar{J}} = 0 \quad \text{or} \quad (\mathcal{L}_K \eta^j_i = 0$$

Recall H^n admits a holomorphic Killing vector $\mathcal{L}_h g_{I\bar{J}} = G_{A\bar{J}} \nabla_I h^A = 2\delta_{I\bar{J}}$

• To preserve this structure, a Killing vector $K_A^I \frac{\partial}{\partial x^I}$ must commute with $h = 2x^I \frac{\partial}{\partial x^I}$

$$[h, K_A^I \frac{\partial}{\partial x^I}] = 0$$

which means that K_A^I has conformal weight $+\underline{I}$

• N is invariant under δ_D and

$$M_I K_A^I + N_{\bar{I}} K_A^{\bar{I}} = 0$$

?

From h : $K_D = \frac{1}{2}(h + \bar{h})$ | $K_T = \frac{1}{2i}(h - \bar{h})$

$$[K_A^I \frac{\partial}{\partial x^I}, 2x^J \frac{\partial}{\partial x^J}] = 0$$

$$[h, K_A^I \frac{\partial}{\partial x^I}] = 0$$

$$2y \frac{\partial}{\partial y} K_A^I \frac{\partial}{\partial x^I} = 0 \rightarrow \frac{\partial}{\partial y} K_A^I = 0$$

So K_A^I is not a f.m. of y !

$$h = 2y \frac{\partial}{\partial y} = 2y \left(\frac{\partial x^I}{\partial y} \right) \frac{\partial}{\partial x^I} = 2x^I \frac{\partial}{\partial x^I}$$

$$h = 2x^I \frac{\partial}{\partial x^I} = 2x^{\bar{I}} \left(\frac{\partial y}{\partial x^{\bar{I}}} \frac{\partial}{\partial y} + \frac{\partial \bar{z}^J}{\partial x^{\bar{I}}} \frac{\partial}{\partial \bar{z}^J} \right)$$

$$= 2y \frac{\partial}{\partial y} + 2x^{\bar{I}} \frac{\partial \bar{z}^J}{\partial x^{\bar{I}}} \frac{\partial}{\partial \bar{z}^J}$$

$$2x^{\bar{I}} \frac{\partial \bar{z}^J}{\partial x^{\bar{I}}} = 2x^{\bar{I}} \bar{z}^J \frac{\partial \bar{z}^J}{\partial x^{\bar{I}}} = 0?$$

We obtain, not sure, however:

$$\delta \bar{z}^\alpha = \theta^A K_A^\alpha$$

$$\delta y = a^{-1} \theta^A y f_A(z)$$

$f_A(z)$: Component of K_A^I in the $\hat{y}$ direction.

$$\text{Killing: } \nabla_I K_{\bar{J}} + \nabla_{\bar{I}} K_I = 0$$

$$\Rightarrow \left[\nabla_\alpha K_{\bar{B}} + \nabla_{\bar{B}} K_\alpha = 0 \right]$$

So K_A^α and $K_{\bar{A}}^{\bar{\alpha}}$ are holo Killing wrt. the metric $g_{\alpha\bar{\beta}}$.

* The Kähler potential is not invariant under Killing:

$$\delta K = \theta^A \mathcal{L}_{K_A} K = \theta^A [r_A(z) + \bar{r}_A(\bar{z})]$$

$\rightarrow$ This is simply a Kähler transt.

$$\begin{aligned} \delta K &= \frac{\partial K}{\partial y} \delta y + \frac{\partial K}{\partial \bar{y}} \delta \bar{y} \\ &= a\left(\frac{1}{y}\right) (a^{-1} \theta^A y r_A) + a\left(\frac{1}{\bar{y}}\right) (a^{-1} \theta^A \bar{y} \bar{r}_A(z)) \\ &= \theta^A [r_A(z) + \bar{r}_A(\bar{z})] \end{aligned}$$

as claimed.

$$K_A^\alpha = -i g^{\alpha\bar{\beta}} \partial_{\bar{\beta}} P_A(z, \bar{z})$$

* The moment map

$$P_A(z, \bar{z}) = i [K_A^\alpha \partial_\alpha K - r_A(z)]$$

$$\begin{aligned} \delta x^I &= \delta y z^I + y \delta z^I = \theta^A K_A \\ &= (a^{-1} \theta^A y r_A z^I + y (\nabla_\alpha z^I - a^{-1} \partial_\alpha K z^I) \delta z^A \\ &= y [a^{-1} r_A z^I + (\nabla_\alpha z^I - a^{-1} \partial_\alpha K z^I) (K_A^\alpha)] \\ &= y [a^{-1} r_A z^I + i P_A a^{-1} z^I - a^{-1} z^I r_A + K_A^\alpha \nabla_\alpha z^I] \end{aligned}$$

$$\therefore K_A^I = y [a^{-1} r_A(z) z^I + K_A^\alpha \nabla_\alpha z^I]$$

Note: $[\delta_A, \delta_B] \gamma$ must satisfy the gauge algebra:

$$\begin{aligned} [\delta_A, \delta_B] \gamma &= \delta_A (a^{-1} \theta^B y r_B(z)) - \\ &= a^{-1} \theta^B [a^{-1} \theta^A y r_A] r_B(z) + a^{-1} \theta^B y \partial_\alpha r_B(z) \delta r^A \\ &= a^{-1} \theta^B y \partial_\alpha r_B(z) \theta^A K_A^\alpha - A^C B \\ &= (a^{-1} y) \theta^B \theta^A \partial_\alpha r_B K_A^\alpha \\ &= \theta^B \theta^A (a^{-1} y) [K_A^\alpha \partial_\alpha r_B - K_B^\alpha \partial_\alpha r_A] \end{aligned}$$

$$\therefore \int_{AB}^C r_c(z) = [K_A^\alpha \partial_\alpha r_D - K_D^\alpha \partial_\alpha r_A]$$

With: $r_A = K_A^\alpha \partial_\alpha K + i P_A$

$$K_A^{\bar{\alpha}} = +i g^{\bar{\alpha}\beta} \partial_\beta P_A$$

$$\int_{AB}^C K_A^\alpha \partial_\alpha K + \int_{AB}^C i P_C$$

$$\begin{aligned} \text{K1-15} &= K_A^\alpha \partial_\alpha K_B^\beta \partial_\beta K + K_A^\alpha K_B^\beta \partial_\alpha \partial_\beta K \\ &\quad + i K_A^\alpha g_{\alpha\bar{\alpha}} K_B^{\bar{\alpha}} \\ &\quad - K_B^\alpha \partial_\alpha K_A^\beta \partial_\beta K - K_A^\alpha K_B^\beta \partial_\alpha \partial_\beta K \\ &\quad - i K_B^\alpha g_{\alpha\bar{\alpha}} K_A^{\bar{\alpha}} \\ &= K_A^\alpha \partial_\alpha K_B^\beta \partial_\beta K - K_B^\alpha \partial_\alpha K_A^\beta \partial_\beta K \\ &\quad + i (K_A^\alpha g_{\alpha\bar{\alpha}} K_B^{\bar{\alpha}} - K_B^\alpha g_{\alpha\bar{\alpha}} K_A^{\bar{\alpha}}) \end{aligned}$$

$$\therefore \int_{AB}^C P_C = K_A^\alpha g_{\alpha\bar{\alpha}} K_B^{\bar{\alpha}} - K_B^\alpha g_{\alpha\bar{\alpha}} K_A^{\bar{\alpha}}$$

T-gauge field $A_m = \frac{1}{2} a^{-1} (\partial_m \bar{z}^\alpha \partial_\alpha K - \partial_m z^\alpha \partial_\alpha K) + \frac{1}{2} \partial_m \ln(\frac{Y}{\bar{Y}}) + \frac{1}{N} A_m^A P_A$

* In the above analysis we had a projective U(1) theory however the P_A term was not present:

$$\begin{aligned} \mathcal{L}_m &= \frac{1}{2} a^{-1} (\partial_m z^\alpha \partial_\alpha K - \partial_m \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K) - \frac{i}{2} \partial_m \ln(\frac{Y}{\bar{Y}}) - a^{-1} A_m^A P_A \\ &= \frac{1}{2} a^{-1} (\partial_m z^\alpha \partial_\alpha K - \partial_m \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K) - \frac{i}{2} \partial_m \ln(\frac{Y}{\bar{Y}}) \\ &\quad - \frac{1}{2} a^{-1} A_m^A (i K_A^\alpha \partial_\alpha K - \bar{r}_A(z)) \\ &\quad - \frac{1}{2} a^{-1} A_m^A (i K_A^{\bar{\alpha}} \partial_{\bar{\alpha}} K - \bar{r}_A(\bar{z})) \end{aligned}$$

$$\mathcal{L}_m = \frac{1}{2} a^{-1} (\hat{\partial}_m z^\alpha \partial_\alpha K - \hat{\partial}_m \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K) - \frac{1}{2} i \hat{\partial}_m \ln(\frac{Y}{\bar{Y}})$$

$$\hat{\partial}_\mu z^\alpha := \partial_\mu z^\alpha - A_\mu^A K_A^\alpha$$

$$\hat{\partial}_\mu y := \partial_\mu y - a^{-1} 4_\mu^A \zeta_A(z) y$$

With the $U(1)_R$ Gauge now involved we can covariantize $\nabla_\mu = \partial_\mu z^\alpha \nabla_\alpha + \partial_\mu \bar{z}^{\bar{\alpha}} \bar{\nabla}_{\bar{\alpha}}$

$$\nabla_\mu = \hat{\partial}_\mu z^\alpha \nabla_\alpha + \hat{\partial}_\mu \bar{z}^{\bar{\alpha}} \bar{\nabla}_{\bar{\alpha}}$$

This is still consistent w/ $\nabla_\mu y = 0$:

Proof:

$$\text{Left side: } \nabla_\mu y = \hat{\partial}_\mu z^\alpha (\partial_\alpha y - a^{-1} \partial_\alpha K y) + \hat{\partial}_\mu \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} y$$

$$= (\partial_\mu z^\alpha - A_\mu^A K_A^\alpha) (\partial_\alpha y - a^{-1} \partial_\alpha K y) + (\partial_\mu \bar{z}^{\bar{\alpha}} - 4_\mu^A K_A^{\bar{\alpha}}) \partial_{\bar{\alpha}} y$$

$\rightarrow$ Vanishes from $\nabla_\alpha K_\beta + \nabla_\beta K_\alpha = 0$?

$$\begin{aligned} \nabla_\mu y &= (\partial_\mu z^\alpha - A_\mu^A K_A^\alpha) (\partial_\alpha y - a^{-1} \partial_\alpha K y) \\ &\quad + (\partial_\mu \bar{z}^{\bar{\alpha}} - 4_\mu^A K_A^{\bar{\alpha}}) \partial_{\bar{\alpha}} y \\ &= \partial_\mu z^\alpha \partial_\alpha y - A_\mu^A K_A^\alpha \partial_\alpha y - \partial_\mu z^\alpha a^{-1} \partial_\alpha K y \\ &\quad + A_\mu^A K_A^\alpha a^{-1} \partial_\alpha K y \\ &\quad + \partial_\mu \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} y - 4_\mu^A K_A^{\bar{\alpha}} \partial_{\bar{\alpha}} y \\ &= \partial_\mu y - A_\mu^A K_A^\alpha \partial_\alpha y - \partial_\mu z^\alpha a^{-1} \partial_\alpha K y + A_\mu^A K_A^\alpha a^{-1} \partial_\alpha K y \\ &\quad - 4_\mu^A K_A^{\bar{\alpha}} \partial_{\bar{\alpha}} y \\ &= \partial_\mu y - A_\mu^A K_A^\alpha \partial_\alpha y - (\partial_\mu K) a^{-1} y + a^{-1} A_\mu^A K_A^\alpha \partial_\alpha K y \\ &\quad - 4_\mu^A K_A^{\bar{\alpha}} \partial_{\bar{\alpha}} y \end{aligned}$$

Not sure.

T Gauge and Peccei-Quinn

• Choosing a rep of PK^n requires gauge fixing for both Dilatation & T-trans.

T-Gauge: $y = \bar{y}$

Note $K(z, \bar{z}) = a \ln(y \bar{y}) = 2a \ln(y)$

$\therefore$ $y = \exp(K/2a)$

Recall the K-Gauge $b_\mu = 0$, then $\delta b_\mu = 0$

$$\delta b_\mu = 0 = \partial_\mu \lambda_D + 2\lambda_D^\alpha e_{\mu\alpha} + \frac{1}{2} \bar{E} \phi_\mu - \frac{1}{2} \bar{\psi}_\mu \psi$$

$$2\lambda_D^\alpha e_{\mu\alpha} = -\partial_\mu \lambda_D - \frac{1}{2} \bar{E} \phi_\mu + \frac{1}{2} \bar{\psi}_\mu \psi$$

* Note this is a super symmetric extension of the classical Conformal Gravity

theory where we found $K_{\mu\nu} = -\frac{1}{2} \partial_\mu \lambda_D$.

$\rightarrow$ "decomposition law" for the K-symmetry.

D-Gauge: $N = -\frac{3}{K^2}$

S-Gauge: $N^\pm \Omega^\pm = 0$

The D-Gauge is invariant under all other Gauge conditions.

$\rightarrow$ Note $\delta N = 2\lambda_D N$

However RHS $\delta(-\frac{3}{K^2}) = 0$, so we must have

$\lambda_D = 0$

Since y is not invariant under Killing symm:

$$\delta y = a^{-1} \theta^\mu y \Gamma_\mu(z)$$

Also not invariant under the Kähler reparametrizations

$$y' = y e^{f(z)/a}$$

To find the decomposition law for $y(\tau)$

$$\delta(y - \bar{y})$$

. Note y has chiral weight 1, so

$$\delta_Q \psi = \frac{1}{\sqrt{2}a} \bar{\psi} \tilde{E} \gamma^a \partial_a K$$

$$\delta_K \psi = a^{-1} \theta^A \psi \sigma_A(z)$$

$$\delta_{\bar{K}} \psi = (f(z)/a) \psi$$

$$\begin{aligned} \delta(\psi - \bar{\psi}) &= i\lambda_T(\psi + \bar{\psi}) + \frac{1}{\sqrt{2}a} \tilde{E}(\psi \gamma^a \partial_a K - \bar{\psi} \gamma^a \partial_a K) \\ &\quad + a^{-1} \theta^A (\psi \sigma_A(z) - \bar{\psi} \bar{\sigma}_A(\bar{z})) \\ &\quad + \frac{1}{a} (f(z)\psi - \bar{f}(\bar{z})\bar{\psi}) = 0 \end{aligned}$$

Now, we impose our gauge choice $\psi = \bar{\psi}$ and we can solve for λ_T :

$$2i\lambda_T + \frac{1}{\sqrt{2}a} \tilde{E}(\psi \gamma^a \partial_a K - \psi \gamma^a \partial_a K) + a^{-1} \theta^A (\psi \sigma_A(z) - \psi \bar{\sigma}_A(\bar{z})) + \frac{1}{a} (f(z) - \bar{f}(\bar{z})) = 0$$

$$\lambda_T = \frac{i}{2} \left(\frac{1}{\sqrt{2}a} \tilde{E}(\psi \gamma^a \partial_a K - \psi \gamma^a \partial_a K) + \frac{\theta^A}{a} (\sigma_A(z) - \bar{\sigma}_A(\bar{z})) + \frac{1}{a} (f - \bar{f}) \right)$$

$$\lambda_T = \frac{i\theta^A}{2a} (\sigma_A(z) - \bar{\sigma}_A(\bar{z})) + \frac{i}{2\sqrt{2}a} \tilde{E}(\psi \gamma^a \partial_a K - \psi \gamma^a \partial_a K) + \frac{i}{2a} (f - \bar{f})$$

* A T-transf. must be included in the eff. symmetry of any field that transforms under T in $\mathbb{H}^{n+1}$

The projection gauge sym transf. in the proj. is related to the Gauge trans. of the embedding with via

$$\delta_{\text{proj}}(\theta) = \delta_{\text{emb}}(\theta) + \delta_T(\lambda_T(\theta))$$

z^a are invariant under λ_T i.e.

$$\delta_T z^a = 0 = \delta_{\text{proj}} z^a$$

Recall this comes from the fact that

Note, fermions still transform under these transf.

$$\chi^\alpha = \bar{y} \Omega^\Gamma G_{\Gamma\bar{J}} \delta^{\alpha\bar{B}} \bar{\nabla}_{\bar{B}} \bar{z}^{\bar{J}}$$

Ω^Γ has chiral weight $n-3/2$

$$\text{So } \delta_T \Omega^\Gamma = -\frac{1}{2} i \lambda_T \Omega^\Gamma.$$

$$\therefore \delta_T \chi^\alpha = \delta \bar{y} \Omega^\Gamma G_{\Gamma\bar{J}} \delta^{\alpha\bar{B}} \bar{\nabla}_{\bar{B}} \bar{z}^{\bar{J}} + \bar{y} \delta_T \Omega^\Gamma G_{\Gamma\bar{J}} \delta^{\alpha\bar{B}} \bar{\nabla}_{\bar{B}} \bar{z}^{\bar{J}}$$

$$\boxed{\delta_T \chi^\alpha = -\frac{3}{2} i \lambda_T \chi^\alpha}$$

Ex: After Gauge fixing we have

$$\delta_{\text{Kähler}} \chi^\alpha = \delta_{\text{ens}} \chi^\alpha + \delta_T(f) \chi^\alpha$$

In $\mathcal{H}^{n+1}$ $\delta_{\text{ens}} \chi^\alpha = 0$ and we are left with

$$\delta_{\text{Kähler}} \chi^\alpha = -\frac{3}{2} i (\lambda_T)_{\text{Kähler}} \chi^\alpha$$

$$= \left(-\frac{3}{2} i\right) \left(\frac{i}{2a}\right) (f(z) - \bar{f}(\bar{z})) \chi^\alpha$$

$$\boxed{\delta_{\text{Kähler}} \chi^\alpha = -\frac{3}{2} i a^{-1} \chi_\alpha(f) \chi^\alpha}$$

* The Gauge and supersymmetry terms can be written as follows:

$$\frac{1}{2a} i \theta^A (r_A - \bar{r}_A) : \quad P_A = i (K_A^\alpha \partial_\alpha K - r_A) = -i (K_A^{\bar{\alpha}} \partial_{\bar{\alpha}} K)$$

Let $\delta \equiv \delta_Q + \delta_\theta$, then

$$\delta z^\alpha = \theta^A K_A^\alpha + \frac{1}{\sqrt{2}} \bar{E} \chi^\alpha$$

$$2P_A = i (K_A^\alpha \partial_\alpha K - K_A^{\bar{\alpha}} \partial_{\bar{\alpha}} K - (r_A - \bar{r}_A))$$

$$(r_A - \bar{r}_A) = 2P_A i + (K_A^\alpha \partial_\alpha K - K_A^{\bar{\alpha}} \partial_{\bar{\alpha}} K)$$

$$\frac{1}{2a} i \theta^A [2P_A i + (K_A^\alpha \partial_\alpha K - K_A^{\bar{\alpha}} \partial_{\bar{\alpha}} K)]$$

$$+ \frac{1}{2a} i (\delta_Q z^\alpha \partial_\alpha K - \delta_Q \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K)$$

$$= \frac{1}{a} \theta^A P_A + (i/2a) [\delta z^\alpha \partial_\alpha K - \delta \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K]$$

∴

$$\boxed{\mathcal{L}_1 = -a \Delta P_A + \frac{i}{2a} [\delta z^\alpha \partial_\alpha K - \delta \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K]}$$

Claim: An object which transforms w/ chiral weight obtains a Kähler weight $(\omega_+, \omega_-) = (c/2, -c/2)$

Proof: Recall, $V' \rightarrow V \exp(\frac{1}{2} [\omega_+ f_\alpha + \omega_- \bar{f}])$

Thus if V has chiral weight c then

$$\delta_T V = i c \Delta_T V$$

$$\begin{aligned} \delta_{K\bar{a}} V &= i c \frac{1}{2} [f_{\bar{a}} - \bar{f}(a)] V \\ &= \mathcal{Q}^{-1} \left[\frac{c}{2} f - \frac{c}{2} \bar{f} \right] V \end{aligned}$$

$$\Rightarrow (\omega_+, \omega_-) = (\frac{c}{2}, -\frac{c}{2})$$

* For χ^α : $c = -3/2$

$\therefore \chi^\alpha$ has Kähler weight $(-3/4, 3/4)$

$\bar{\chi}^{\bar{\alpha}}$ has $(1/4, -1/4)$

- * Since gauge symm and T -symm mix the Gauge symm no longer commute w/ supersym in supergravity when r_A .

Vectors A_μ^A commute w/ susy in the superconf. theory
But not in Super Poincaré.

After Gauge fixing, the T symm. acts as gauge transf, susy, and Kähler transf. Therefore, the group in the Super Poincaré theory can act partly as a gauged R -symmetry:

Since $U(1) \subset SU(2)_R \subset Sp(2) \subset SU(4)$

↙ Gauge fixing

Super Poincaré w/ G'

G' : G w mixing of superconf. T-symm.

$A_\mu \mapsto A_\mu + \partial_\mu \lambda_T$ and now acts as a Gauge field for imaginary part of Kähler transformations:

$$\therefore \delta_{\text{gauge}} A_\mu = \partial_\mu \lambda_T = a^{-1} \partial_\mu \text{Im}(f)$$

where

$$A_\mu = \frac{1}{2} i a^{-1} (\hat{\partial}_\mu z^\alpha \partial_\alpha K - \hat{\partial}_\mu \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K) - \frac{1}{2} i \hat{\partial}_\mu \ln\left(\frac{V}{Y}\right)$$

$$\hat{\partial}_\mu Y = \partial_\mu Y - a^{-1} A_\mu^A r_A Y, \quad \hat{\partial}_\mu z^\alpha := \partial_\mu z^\alpha - A_\mu^A r_A z^\alpha$$

$$A_\mu = \frac{1}{2} i a^{-1} (\hat{\partial}_\mu z^\alpha \partial_\alpha K - \hat{\partial}_\mu \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K) - a^{-1} A_\mu^A (r_A - \bar{r}_A) \frac{i}{2}$$

• Covariant derivatives constructed before & after fixing the T-Gauge $y = \bar{y} \Leftrightarrow y = e^{K/2a}$ are equivalent.

Q.11: (i) Under the addition of A_μ to the other symm. we find that the Kähler weights of Y

$$(\omega_+, \omega_-) = (-2, 0) \text{ are now}$$

$$(\tilde{\omega}_+, \tilde{\omega}_-) = (-1/2, 1/2)$$

$$(ii) \quad \hat{D}_\mu Y = \left[\partial_\mu - \frac{1}{2} a^{-1} (\hat{\partial}_\mu K) \right] Y = 0$$

This def. incorporates the modified Kähler weights

$$(\tilde{\omega}_+, \tilde{\omega}_-)$$

Proof: Initially, in IK^{n+1} Y has conf. weight $c=2$ and Kähler $(\omega_+, \omega_-) = (-2, 0)$ thus, in

$$U(1) \cong \pi^{-1}(P \in PK^n)$$

$$\delta Y' = \frac{f(z)}{a} Y$$

$$\delta Y = \frac{f(z)}{a} Y - \frac{1}{2a} (f - \bar{f}) Y$$

$$= \bar{a}(f(z) + \bar{f}(\bar{z}))\gamma = -a(\frac{1}{z}f - \frac{1}{\bar{z}}\bar{f})\gamma$$

$$\therefore \gamma: (\bar{\omega}_+, \bar{\omega}_-) = (-1/2, -1/2) \quad \checkmark$$

(ii) Note $\hat{\partial}_m K = \partial_m K - 2a A_m^A r_A$

$$\partial_m + \frac{1}{2}a^{-1}(\hat{\partial}_m z^\alpha \partial_\alpha K - \hat{\partial}_m \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K) + \frac{1}{2}a^{-1}A_m^A(r_A - \bar{r}_A)$$

$$- a^{-1} \partial_m z^\alpha \partial_\alpha K$$

$$\rightarrow \hat{\partial}_m \gamma = \frac{\hat{\partial}_m K}{2a} \gamma = (\partial_m \bar{z}^\alpha \partial_\alpha K + \partial_m \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K) \gamma \frac{1}{2a}$$

$$\partial_m \gamma = a^{-1} A_m^A r_A \gamma - \frac{1}{2}a^{-1}(\partial_m z^\alpha \partial_\alpha K + \partial_m \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K)$$

$$= \partial_m \gamma - a^{-1}(A_m^A P_{Ai} + A_m^A K_A^\alpha \partial_\alpha K) - \frac{1}{2}a^{-1}(\quad)$$

$$\rightarrow -\frac{1}{2}a^{-1}(\partial_m z^\alpha \partial_\alpha K + \partial_m \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K) \quad *$$

$$+ \frac{1}{2}a^{-1}(A_m^A K_A^\alpha \partial_\alpha K + A_m^A K_{A\bar{\alpha}} \partial_{\bar{\alpha}} K)$$

* Kähler weight terms

$$\partial_m \gamma = a^{-1} A_m^A P_{Ai} = (A_m^A (K_A^\alpha \partial_\alpha K - K_A^{\bar{\alpha}} \partial_{\bar{\alpha}} K)) a^{-1}$$

$$\partial_m \gamma + \frac{1}{2}a^{-1}(\hat{\partial}_m z^\alpha \partial_\alpha K - \hat{\partial}_m \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} K) + \frac{1}{2}a^{-1}A_m^A(r_A - \bar{r}_A)\gamma$$

$$- a^{-1} \partial_m z^\alpha \partial_\alpha K$$

$$= \partial_m z^\alpha \partial_\alpha \gamma + \partial_m \bar{z}^{\bar{\alpha}} \partial_{\bar{\alpha}} \gamma$$

Combine $\partial_m z^\alpha$ terms:

$$\bar{\partial}_\alpha \gamma = \partial_\alpha \gamma - \frac{1}{2}a^{-1} \partial_\alpha K \gamma \quad ?$$

* Now consider the S-Gauge condition $N_I \Omega^\pm = 0$

Under $\lambda_D, \lambda_T, \text{ Lorentz}$:

$$\delta_T(N_I \Omega^\pm) = i\lambda_T C(N_I \Omega^\pm)$$

$$\delta_D(N_I \Omega^\pm) = \lambda_D H(N_I \Omega^\pm)$$

$$\delta_L(N_I \Omega^\pm) = a N_I \Omega^\pm$$

* We will show that tw3 is also invariant under gauge

δ_D

$$\delta X^\pm = \theta^A K_A^\pm(x), \quad \delta \Omega^\pm = \theta^A \partial_\mp K_A^\pm \Omega^\pm$$

Recall: $X^\pm \mapsto L(X^\pm)$ then $\Omega^\pm \rightarrow \tilde{L}(\Omega^\pm)$

$$\delta_\theta L^\pm = \frac{\partial L^\pm}{\partial x^\mp} \delta_\theta x^\mp = \frac{\partial L^\pm}{\partial x^\mp} \frac{1}{\sqrt{2}} \epsilon \Omega^\mp$$

$$\therefore \tilde{L}(\Omega^\pm) = \partial L^\pm / \partial x^\mp \Omega^\mp$$

Infinitesimally:

$$L(X)^\pm = X^\pm + \theta^A K_A^\pm$$

$$\tilde{L}(\Omega)^\pm = \frac{\partial}{\partial x^\mp} (X^\pm + \theta^A K_A^\pm) \Omega^\mp = (\delta_\mp^\pm + \theta^A \partial_\mp K_A^\pm) \Omega^\mp$$

$$\therefore \delta_\theta \Omega^\pm = \theta^A \partial_\mp K_A^\pm \Omega^\mp$$

$\rightarrow N_I \Omega^\pm$ is not invariant under Q/S-sym.

$$\delta_{Q/S} \Omega^\pm = \frac{1}{\sqrt{2}} P_L (\not{D} X^\pm + F^\pm) \epsilon + \sqrt{2} \psi X^\pm P_L \eta$$

$$\begin{aligned} \therefore \delta_{Q/S} (N_I \Omega^\pm) &= \delta N_I \Omega^\pm + N_I \delta \Omega^\pm \\ &= (N_{I\mp} \delta X^\mp + N_{I\mp} \delta X^\mp) \Omega^\pm + N_I \delta \Omega^\pm \\ &= \left(\frac{1}{\sqrt{2}} \epsilon N_{I\mp} \Omega^\mp \Omega^\mp \right) + \frac{1}{\sqrt{2}} \epsilon \Omega^\mp N_{I\mp} \Omega^\mp \\ &\quad + N_I \left(\frac{1}{\sqrt{2}} P_L (\not{D} X^\pm + F^\pm) \epsilon + \sqrt{2} \psi X^\pm P_L \eta \right) = 0 \end{aligned}$$

From this we can determine N is terms of the other field parameters:

$$0 = N_I P_L (\not{D} X^\pm + F^\pm) \epsilon - 2a P_L \eta + \Omega^\pm \bar{\epsilon} (N_{I\mp} \Omega^\mp + N_{I\mp} \Omega^\mp)$$

• ψ_n term in $\not{D} X^\pm$ is $\sim \bar{\psi}_n \Omega^\pm$ and variables

$$\text{Recall } N_I D_n X^\pm = \frac{1}{2} (2a_n - 2b_n) N$$

But $b_n = 0$ and $N = -a$ so $N_I D_n X^\pm = 0$

• All that remains is A_m^F term

$$-i A_m^F X^\pm$$

$$-i N_I \frac{1}{\sqrt{2}} P_L \gamma^m A_m^F X^\pm \epsilon$$

$$A_m^F = (i/4N) (N_{I\mp} \bar{\Omega}^\mp \gamma_m \Omega^\mp + \frac{3}{2} (\ell_e(f_{AB})) \bar{\chi}^A \gamma_m \chi^B)$$

$$\bar{F}^\mp = -N^\mp I W_I + \frac{1}{2} N^\mp I N^\mp \bar{R}^\mp \bar{\Omega}^\mp \Omega^\mp - a \dots$$

$$F^T = -N^{\pm\bar{I}} N^{\pm\bar{J}} + \frac{1}{2} \Gamma_{\kappa\lambda}^{\pm\bar{I}} \bar{\Omega}^{\kappa} \Omega^{\lambda} + \frac{1}{4} N^{\pm\bar{I}} f_{AB\bar{I}}^{\kappa}$$

$$\frac{1}{4N} P_L \gamma^{\mu} [N^{\pm\bar{I}} \bar{\Omega}^{\pm} \gamma_{\mu} \Omega^{\pm} + \frac{3}{2} (\text{Re}(f_{AB})) \lambda^A \gamma_{\mu} \lambda^B]$$

$$\frac{1}{\sqrt{2}4} P_L \gamma^{\mu} [\bar{\Omega}^{\pm} N^{\pm\bar{I}} \gamma_{\mu} \Omega^{\pm} + \frac{3}{2} \text{Re}(f_{AB}) \lambda^A \gamma_{\mu} \lambda^B] \in *$$

$$* N_{\pm} P_L (-N^{\pm\bar{I}} \bar{\Omega}^{\pm} + \frac{1}{2} \Gamma_{\kappa\lambda}^{\pm\bar{I}} \bar{\Omega}^{\kappa} \Omega^{\lambda} + \frac{1}{4} N^{\pm\bar{I}} f_{AB\bar{I}}^{\kappa} \lambda^A \lambda^B) \in$$

$$= (-N_{\pm} N^{\pm\bar{I}} \bar{\Omega}^{\pm} P_L + \frac{1}{2} N_{\pm} \Gamma_{\kappa\lambda}^{\pm\bar{I}} \bar{\Omega}^{\kappa} \Omega^{\lambda} + \frac{1}{4} N_{\pm} N^{\pm\bar{I}} f_{AB\bar{I}}^{\kappa} \lambda^A \lambda^B)$$

$$\rightarrow = -N_{\pm} \bar{\Omega}^{\pm} N^{\pm\bar{I}} \bar{\Omega}^{\pm} P_L = -3 \bar{\Omega}^{\pm} P_L \in (-1)$$

$$= -3 \bar{\Omega}^{\pm} P_L \in$$

$$\rightarrow X^{\bar{I}} (f_{AB})_{\bar{I}} = 0 \text{ Since } X^{\bar{I}} f_{AB\bar{I}} = 0$$

$$\rightarrow N_{\pm} N^{\pm\bar{I}} N_{\pm} \bar{\Omega}^{\pm} \Omega^{\pm}$$

$$= P_L X^{\bar{I}} N_{\pm} \bar{\Omega}^{\pm} \Omega^{\pm} \in$$

$$* = P_L \bar{\Omega}^{\pm} \Omega^{\pm} \in -3 \bar{\Omega}^{\pm} P_L \in$$

$$0 = \left(\frac{1}{\sqrt{2}4}\right) (P_L \gamma^{\mu} \bar{\Omega}^{\pm} N^{\pm\bar{I}} \gamma_{\mu} \Omega^{\pm}) + \frac{3}{\sqrt{2}2} P_L \gamma^{\mu} \text{Re}(f_{AB}) \lambda^A \gamma_{\mu} \lambda^B \in$$

$$+ P_L \bar{\Omega}^{\pm} \Omega^{\pm} N^{\pm\bar{I}} \in -3 \bar{\Omega}^{\pm} P_L \in -2a P_L \Omega$$

$$+ \Omega^{\pm} \bar{\Omega}^{\pm} (N_{\pm} \Omega^{\pm} + N_{\pm} \bar{\Omega}^{\pm})$$

$$\frac{1}{\sqrt{2}4} P_L \gamma^{\mu} \bar{\Omega}^{\pm} N^{\pm\bar{I}} \gamma_{\mu} \Omega^{\pm} \in + \frac{3}{\sqrt{2}2} P_L \gamma^{\mu} \text{Re}(f_{AB}) \lambda^A \gamma_{\mu} \lambda^B \in$$

$$-3 \bar{\Omega}^{\pm} P_L \in + \Omega^{\pm} \bar{\Omega}^{\pm} N_{\pm} \Omega^{\pm} - 2a P_L \Omega$$

$$2\kappa^{-2} P_L \Omega = -\bar{\Omega}^{\pm} P_L \in + \gamma_a P_L \in \left(\frac{1}{4} N_{\pm} \bar{\Omega}^{\pm} \gamma^a \Omega^{\pm} + \frac{1}{8} \text{Re}(f_{AB}) \lambda^A \gamma^a \lambda^B\right)$$

* The S-charge condition $N_{\pm} \Omega^{\pm} = 0$ is invariant under Gauge transf.

$$\begin{aligned} & \partial_\mu N_I \Omega^I - \partial_\mu N_{\bar{I}} \Omega^{\bar{I}} + N_I \partial_\mu \Omega^I - N_{\bar{I}} \partial_\mu \Omega^{\bar{I}} = 0 \\ & \rightarrow N_I K_A^{\bar{I}} + N_{\bar{I}} K_A^I = 0 \\ & N_I \partial_J K_A^{\bar{I}} + N_{\bar{I}} \partial_J K_A^I + N_I \cancel{\partial_J K_A^{\bar{I}}} + N_{\bar{I}} \cancel{\partial_J K_A^I} \\ & \quad + K_A^{\bar{I}} N_{\bar{J}} \Omega^J + K_A^I N_J \Omega^{\bar{J}} \\ & \quad + N_I \partial_K K_A^{\bar{I}} \Omega^K + N_{\bar{I}} \partial_K K_A^I \Omega^{\bar{K}} = 0 \\ & \rightarrow \Omega^{\bar{J}} N_{\bar{J}} K_A^I + \Omega^J N_J \partial_J K_A^{\bar{I}} + \Omega^{\bar{I}} N_{\bar{I}} K_A^{\bar{J}} \\ & = 0 \text{ as claimed. } \checkmark \end{aligned}$$

An Explicit example: $SU(2, 2)/U(2)$ model:

* This model has one single chiral mult: $\dim(\mathcal{H}) = n+1 = 2$, thus $X^0 = y$, $X^1 = y z^2$
 $Z^0 = 1$, $z^1 = z$

Here $z^1 = z$ is the physical scalar field.

* Let $N = X^{\bar{I}} g_{\bar{I} \bar{J}} \bar{X}^{\bar{J}}$

$$N = i a \bar{X}^0 X^1 - X^0 \bar{X}^1$$

$$\text{so } g_{\bar{I} \bar{J}} = i a \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Therefore our gauge fixing means

$$N = -a = i a (\bar{y} y z - y \bar{y} \bar{z}) = -2 a y \bar{y} \text{Im}(z)$$

So this constraint picks out $\mathbb{C}^+$, the upper half plane of $\mathbb{C}$.

$$0 = \ln(y \bar{y}) + \ln(-i(z - \bar{z}))$$

$$\therefore K(z, \bar{z}) = -a \ln(-i(z - \bar{z}))$$

$$\text{For } a = 3\kappa^{-2}$$

$$K(z, \bar{z}) = -3\kappa^{-2} \ln(-i(z - \bar{z}))$$

(Kähler Potential)

$$\partial_{\bar{z}} K = -3\kappa^{-2} \frac{1}{-i(z - \bar{z})} (-i) = -3\kappa^{-2} \frac{1}{(z - \bar{z})}$$

$$\therefore g_{z\bar{z}} = -3k^2 \frac{1}{(z-\bar{z})^2} \quad \left(\begin{array}{l} \text{Kähler} \\ \text{Metric} \end{array} \right)$$

$$ds^2 = -3k^2 \frac{1}{(z-\bar{z})^2} dz d\bar{z}$$

This is a scaled version of the Poincaré plane.

$$R_{\alpha\bar{\beta}} = -\partial_\alpha \partial_{\bar{\beta}} (\log(\det(g_{\gamma\bar{\delta}})))$$

$$R_{\alpha\bar{\beta}} = g^{\gamma\bar{\delta}} R_{\alpha\bar{\gamma}\bar{\beta}\delta}$$

$$R = g^{\alpha\bar{\alpha}} R_{\alpha\bar{\alpha}} = z g^{1\bar{1}} R_{1\bar{1}} = -\frac{4}{3} k^2$$

Consider the modified Kähler potential:

$$N = a X^0 \bar{X}^0 \left(i \frac{X^1}{X^0} - i \frac{\bar{X}^1}{\bar{X}^0} \right)^b$$

$$= a y \bar{y} \left(i \frac{y \bar{z}}{y} - i \frac{\bar{y} \bar{z}}{\bar{y}} \right)^b$$

$$= a y \bar{y} (i(z - \bar{z}))^b = -a$$

$$\ln(y\bar{y}) + \ln(-(i(z - \bar{z}))^b) = 0$$

$$K(z, \bar{z}) = -a \ln((-1)^{b+1} (i(\bar{z} - z))^b)$$

$$= -a \ln(-1)^{b+1} + a b \ln(i(\bar{z} - z))$$

$$\therefore g_{z\bar{z}} = -3b k^{-2} \frac{1}{(\bar{z} - z)^2}, \quad R = -\frac{4}{3b} k^2$$

→ Modified embedding metric gives us the freedom of an arbitrary scale for the projective field.

Isometry group: $N = i a (\bar{X}^0 X^1 - X^0 \bar{X}^1)$

N is invariant under $U(1)$ symm:

$$X^\pm \mapsto e^{i\theta} X^\pm, \quad \bar{X}^\pm = e^{-i\theta} \bar{X}^\pm$$

as well as $SL(2, \mathbb{R})$ transf.

$$X^\pm \mapsto t^\pm X^\pm, \quad \bar{X}^\pm = t^\pm \bar{X}^\pm$$

$$\rightarrow t^I_S N^I_I \bar{t}^I_S = N^I_S$$

$\therefore$ Isometry group: $U(1) \times SL$

$$\delta x^I = -\theta^A t^I_A x^J$$

The generators of this Group:

$$t_0 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, t_1 = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, t_2 = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, t_3 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

which act on $\begin{pmatrix} x^0 \\ x^1 \end{pmatrix}$

$$u/ \quad f_{12}^1 = 1, \quad f_{13}^2 = 2, \quad f_{23}^3 = 1$$

$$SU(1,1) \cong SU(2,1) \cong SL(2)$$

$$\delta y = \frac{1}{a} y \theta^A \Gamma_A, \quad \delta z = \theta^A K_A^z$$

$$\theta^A \Gamma_A = a \left(-i\theta^0 - \frac{1}{2}\theta^2 - \theta^3 z \right)$$

$$\theta^A K_A^z = \theta^1 + \theta^2 z + \theta^3 z^2$$

From Chiral to Poincaré supergravity:

* The kinetic term of chiral m.f. defines a P/K^n .

$$\Omega^I = y \gamma^\alpha \nabla_\alpha z^I$$

Let us express the Lagrangian in terms of physical fields:

$$a \equiv 3\kappa^{-2}$$

$$y = \exp\left(\frac{1}{6}\kappa^2 K\right)$$

Recall $x^I W_I = 3W$, which can be solved

$$\text{via } W(x) = (x^0)^3 W\left(\frac{x^\alpha}{x^0}\right), \quad \alpha = 1, \dots, n$$

$$\therefore W(x) = y^3 W(z^\alpha), \quad \alpha = 1, \dots, n$$

$$\therefore W = \exp\left[\frac{1}{2}\kappa^2 K(z, \bar{z})\right] W(z)$$

Since $W = y^3 W(z)$, then it has Kähler weight $(w_+, w_-) = (3, 0)$

$$\nabla_\alpha W(z) = \partial_\alpha W(z) + 3 a^{-1} \partial_\alpha K W(z)$$

$$\therefore \boxed{\nabla_\alpha \bar{W}(z) = \partial_\alpha \bar{W}(z) + K^2 \partial_\alpha K W(z)}$$

Note this follows from $\nabla_\alpha W = \cancel{\nabla_\alpha} y \bar{W} + y \nabla_\alpha W$
 $X^I W_I = 3W$:

$$y z^I W_I = 3 y^3 \bar{W} \Rightarrow \boxed{z^I W_I = 3 y^2 \bar{W}}$$

$$\nabla_\alpha (y z^I W_I) = 3 y^3 \nabla_\alpha \bar{W} = 3 \nabla_\alpha W$$

$$\nabla_\alpha z^I W_I y + z^I y \nabla_\alpha W_I = 3 y^3 \nabla_\alpha \bar{W} = 3 \nabla_\alpha W$$

$$\nabla_\alpha W_I = \nabla_\alpha X^K W_{KI} = y \nabla_\alpha z^K W_{KI}$$

Note that

$$W_J + X^I W_{IJ} = 3 W_J$$

$$X^I W_{IJ} = 2 W_J$$

$$\Rightarrow \nabla_\alpha z^I W_I y + 2 W_I \nabla_\alpha z^I = 3 y^3 \nabla_\alpha \bar{W} = 3 \nabla_\alpha W$$

$$\therefore \boxed{y W_I \nabla_\alpha z^I = \nabla_\alpha W = y^3 \nabla_\alpha \bar{W}}$$

* W and $\bar{W}$ and K combine to form the
Kähler-invariant function:

$$\boxed{\mathcal{G}(z, \bar{z}) := K^2 K + \log(K^6 W \bar{W})}$$

Will see the superpotential is not invariant
if $f_A(z) \neq 0$

Gauge invariance: $\boxed{W_I K_A^I = 0}$

$$K_A^T = \gamma (K_A^\alpha \nabla_\alpha Z^T + i a^{-1} P_A Z^T)$$

$$W_I K_A^\alpha \nabla_\alpha Z^T + i a^{-1} P_A Z^T W_I = 0$$

$$\gamma W_I K_A^\alpha \nabla_\alpha Z^T + i K^2 P_A W = 0$$

$$K_A^2 \gamma^3 \nabla_\alpha W + i K^2 P_A \gamma^3 \bar{W} = 0$$

$$\therefore \boxed{K_A^\alpha \nabla_\alpha \bar{W} + i K^2 P_A \bar{W} = 0}$$

Expanding $\nabla_\alpha \bar{W}$ and P_A

$$K_A^\alpha (\partial_\alpha \bar{W} + K^2 \partial_\alpha K \bar{W}) = K^2 (K_A^\alpha \partial_\alpha K - r_A) \bar{W} = 0$$

$$\boxed{K_A^\alpha \partial_\alpha \bar{W} + K^2 r_A \bar{W} = 0}$$

* From this, we can conclude that a constant superpotential must vanish if the theory contains gauged symm w/ $r_A \neq 0$:

→ If $\bar{W}$ is constant then trivially

$$\partial_\alpha \bar{W} = 0 \text{ and } K^2 r_A \bar{W} = 0$$

If $r_A \neq 0$ then $\bar{W} = 0$.

→ If the theory transforms under scalars and $K_A^\alpha = 0$ but $r_A \neq 0$ then again $\bar{W} = 0$.

The Potential:

Consider the Scalar Potential $N=1$ sugra

$$V = V_F + V_D = G^{\bar{I} \bar{J}} W_{\bar{I}} W_{\bar{J}} + \frac{1}{2} (\text{Re}(f))^{-1} P_A P_B$$

• f_{AB} is a positive f.m., V_F is a bit different as the compensating multiplet in the conf. approach

has negative norm in $\mathbb{H}^{n+1}$ w/ $b \in \mathbb{R}$
 $(\bar{I} \bar{J} \dots \bar{n} \bar{n}) = -1$ $2 \bar{I} \bar{J} \dots \bar{n} \bar{n} = 2 \bar{I} \bar{J} \dots \bar{n} \bar{n} = 2 \bar{I} \bar{J}$

$$\begin{aligned}
 & W_I \bar{W}_{\bar{J}} (\gamma \bar{\gamma}) (-a^{-1} z^I \bar{z}^{\bar{J}} + g^{\alpha\bar{\beta}} \nabla_{\alpha} z^I \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{J}}) \\
 &= -a^{-1} W \bar{W} + (\gamma \bar{\gamma}) W_I \nabla_{\alpha} z^I g^{\alpha\bar{\beta}} (\bar{\gamma} \bar{W}_{\bar{J}} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{J}}) \\
 &= -a^{-1} |\gamma|^6 W(z) \bar{W}(\bar{z}) + |\gamma|^6 \nabla_{\alpha} W g^{\alpha\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W}
 \end{aligned}$$

$$V_F = e^{K/2} \left[-3\kappa^2 W(z) \bar{W}(\bar{z}) + \nabla_{\alpha} W g^{\alpha\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} \right]$$

• So we see where the negative norm aspect comes from ~

* $V_D = \frac{1}{2} (\text{Re}(f))^{-1AB} P_A P_B$, this is the same for the global & local theories

We conclude the full potential for $N=2$ susy can be split into negative and positive definite parts:

$$V = V_+ + V_-$$

$$V_- := -3\kappa^2 e^{K/2} W \bar{W}$$

$$V_+ := e^{K/2} \nabla_{\alpha} W g^{\alpha\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} + \frac{1}{2} (\text{Re}(f))^{-1AB} P_A P_B$$

Fermion terms:

For fermions of the chiral mult. we must impose $\Omega^I = \gamma \chi^{\alpha} \nabla_{\alpha} z^I$

Note $\delta_T \chi^{\alpha} = -\frac{3}{2} i \lambda_T \chi^{\alpha}$ and χ^{α} transforms as a tangent vector V^{α} of $\mathbb{P}^1/K^n$ under gauge trans. and holomorphic reps. $e^{f(z)/\kappa}$

Thus, we require:

$$\nabla_{\alpha} \chi^{\alpha} = 0 - \frac{1}{2} i \lambda_{\alpha} \chi^{\alpha} + \frac{3}{2} i \lambda \chi^{\alpha} \nabla_{\alpha} \ln e^{f(z)/\kappa} \chi^{\alpha}$$

$$+ \Gamma_{\beta}^{\alpha} \gamma \chi^{\beta} \hat{\partial}_m z^{\beta}$$

* This comes from $\hat{D}_m^{(0)} \Omega^I = D_m^{(0)} \Omega^I + \Gamma_{JK}^I \Omega^K \hat{\partial}_m \chi^J$

$$D_m^{(0)} (\gamma \chi^{\alpha} \nabla_{\alpha} z^I) + \gamma^2 \Gamma_{JK}^I \chi^{\alpha} \nabla_{\alpha} z^K \hat{\partial}_m z^J \hat{\partial}_n z^{\beta}$$

Note:

$$\nabla_m \chi^I = (\partial_m - i t_m) \chi^I = \gamma (\partial_m z^{\alpha}) \nabla_{\alpha} z^I$$

$$\hat{D}_n \chi^I = \nabla_m \chi^I - A_n^A K_A^I$$

$$= \gamma \partial_m z^{\alpha} \nabla_{\alpha} z^I - A_n^A (\gamma K_A^{\alpha} \nabla_{\alpha} z^I + i \alpha^{-1} p_A z^I)$$

→ Term does not survive since $\Gamma_{KL}^I z^K = 0$

$$\begin{aligned} \hat{D}_n^{(1)} \Omega^I &= D_m^{(0)} (\Omega^I) + \Gamma_{JK}^I \Omega^K \gamma (\hat{\partial}_m z^{\alpha}) \nabla_{\alpha} z^J \\ &= D_m^{(0)} (\gamma \chi^{\alpha} \nabla_{\alpha} z^I) + \Gamma_{JK}^I (\gamma \chi^{\alpha} \nabla_{\alpha} z^K) \hat{\partial}_m z^J \nabla_{\alpha} z^I \\ &= \gamma (D_m^{(0)} \chi^{\alpha}) \nabla_{\alpha} z^I + \gamma \chi^{\alpha} \underline{D_m^{(0)} (\nabla_{\alpha} z^I)} + \cancel{p} \\ &\rightarrow \gamma \chi^{\alpha} (\nabla_m (\nabla_{\alpha} z^I)) \end{aligned}$$

$$\begin{aligned} \hat{D}_m^{(0)} \Omega^I &= \gamma (D_m^{(0)} \chi^{\alpha}) \nabla_{\alpha} z^I + \gamma \chi^{\alpha} (\hat{\partial}_m z^{\alpha} \nabla_{\alpha} + \hat{\partial}_m \bar{z}^{\bar{\alpha}} \bar{\nabla}_{\bar{\alpha}}) \nabla_{\alpha} z^I \\ &\quad + \gamma^2 \Gamma_{JK}^I \chi^{\alpha} \nabla_{\alpha} z^K \hat{\partial}_m z^J \nabla_{\alpha} z^I \end{aligned}$$

$$\hat{D}_n^{(0)} \Omega^I = \gamma (D_m^{(0)} (\chi^{\alpha})) \nabla_{\alpha} z^I + \gamma \chi^{\alpha} \hat{\partial}_m \bar{z}^{\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \nabla_{\alpha} z^I$$

since

$$[\nabla_{\alpha}, \bar{\nabla}_{\bar{\beta}}] V(z, \bar{z}) = c^{-1} (\nu_{-} + \omega_{+}) g_{\alpha \bar{\beta}} V$$

$$\nabla_{\bar{\beta}} \nabla_{\alpha} z^I = c^{-1} g_{\alpha \bar{\beta}} z^I$$

$$\hat{D}_m^{(0)} \Omega^I = \gamma (D_m^{(0)} \chi^{\alpha}) \nabla_{\alpha} z^I + \frac{1}{3} \gamma \kappa^2 \chi^{\alpha} \hat{\partial}_m \bar{z}^{\bar{\alpha}} g_{\alpha \bar{\beta}} z^I$$

Second term does not contribute in $\mathcal{L}_H$ since

$$\bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{\beta}} G_{\bar{\beta} \bar{\gamma}} \bar{z}^{\bar{\gamma}} = 0$$

$$\mathcal{L}_{1/2} = -\frac{1}{2} g_{\alpha\bar{\beta}} [\bar{\chi}^{\alpha} \not{D}^{(0)} \chi^{\bar{\beta}} + \bar{\chi}^{\bar{\beta}} \not{D}^{(0)} \chi^{\alpha}]$$

Claim: $\mathcal{L}_{1/2}$ is invariant under Kähler transform.

Proof: $\delta h \bar{e}^{\alpha} \chi^{\alpha} = \frac{3}{2} i a^{-1} (\text{Im}(f)) \chi^{\alpha}$

$\delta_{\text{K\"ahler}} \chi_{\alpha} = -a^{-1} \partial_{\alpha} (\text{Im}(f))$

we find this is $\mathcal{L}_{1/2} + \partial_{\alpha}(\dots)$.

• We can also expand for mass term for χ^{α} :

$$\gamma^2 \nabla_{\bar{I}} W \nabla_{\alpha} \bar{Z}^{\bar{I}} \nabla_{\bar{\rho}} \bar{Z}^{\rho} = \gamma^2 \nabla_{\bar{I}} W \nabla_{\alpha} \bar{Z}^{\bar{I}} \chi^{\rho} \nabla_{\bar{\rho}} \bar{Z}^{\rho}$$

$$\mathcal{L}_{\text{mass}} = -\frac{1}{2} \nabla_{\bar{I}} W \bar{\Omega}^{\bar{I}} \Omega^{\bar{I}} = -\frac{1}{2} \gamma^2 \nabla_{\bar{I}} W \cdot \bar{\chi}^{\alpha} \nabla_{\alpha} \bar{Z}^{\bar{I}} \chi^{\rho} \nabla_{\bar{\rho}} \bar{Z}^{\rho}$$

$$= \gamma^2 \bar{\chi}^{\alpha} \chi^{\rho} \nabla_{\bar{\rho}} \bar{Z}^{\rho} \nabla_{\alpha} W$$

• Note: $f_{AB\alpha} = \nabla_{\alpha} f_{AB} = \gamma f_{AB\bar{I}} \nabla_{\alpha} \bar{Z}^{\bar{I}}$ (for λ)

(For $\lambda\bar{\lambda}$): $f_{AB\bar{I}} \bar{Z}^{\bar{I}} = 0$, then:

$$G^{\bar{I}\bar{J}} \bar{W}_{\bar{J}} f_{AB\bar{I}} = \bar{\gamma}^3 g^{\alpha\bar{\beta}} f_{AB\alpha} \nabla_{\bar{\beta}} \bar{W}$$

• For mixed $\lambda\chi$: $f_{AB} \mapsto D_A$

• For fermion term:

$$A_{\mu}^F = i\kappa^2 \left[\frac{1}{12} g_{\alpha\bar{\beta}} \bar{\chi}^{\alpha} \gamma_{\mu} \chi^{\bar{\beta}} - \frac{1}{8} \text{Re}(f_{AB}) \bar{\lambda}^A \gamma_{\mu} \lambda^B \right]$$

$$\rightarrow \frac{1}{4} (R \bar{\gamma} \gamma \bar{\delta} - \frac{1}{2} \kappa^2 g_{\alpha\bar{\gamma}} g_{\rho\bar{\delta}}) \bar{\chi}^{\alpha} \chi^{\bar{\beta}} \bar{\chi}^{\bar{\gamma}} \chi^{\delta}$$

Review & Preview:

* Once we construct the final Lagrangian $\mathcal{L}$ w/ $K(z, \bar{z})$ we can ignore where we obtained the theory and use any Kähler potential $K(z, \bar{z})$.

• Moduli space of K^n is eq. to the moduli space $\mathcal{M}(K^n)$.

$$N = (-a) \gamma \bar{\gamma} e$$

Note N (and so $G_{I\bar{J}}$) is an embedding in $(- + \dots +)$

* In a flat Kähler metric: $g_{\alpha\bar{\beta}} = \delta_{\alpha\bar{\beta}}$ w/ one physical scalar z^1 : This can be obtained from the Kähler Potential

$$N = (-a) X^0 \bar{X}^0 \exp\left(-\frac{1}{a} \frac{X^1 \bar{X}^1}{X^0 \bar{X}^0}\right)$$

$$N = -a = -a \bar{X}^0 X^0 \exp\left(-\frac{1}{a} z^1 \bar{z}^1\right)$$

$$0 = \ln(X \bar{Y}) + \left(-\frac{1}{a} z^1 \bar{z}^1\right) =$$

$$\therefore K(z, \bar{z}) = z^1 \bar{z}^1$$

$$\partial_\alpha \partial_{\bar{\beta}} K = 1 = \delta_{\alpha\bar{\beta}} \quad \checkmark$$

Projective & Kähler-Hodge manifolds:

We have a $U(1)$ line bundle w/ connection A_m

The field strength $F_{mn} = \partial_m A_n - \partial_n A_m$

is the pullback of the fundamental Kähler form:

$$\Omega = -2ig_{\alpha\bar{\beta}} dz^\alpha \wedge d\bar{z}^\beta$$

Kähler-Hodge: A_m transforms as a $U(1)$ connection

$$\begin{aligned} A'_m &\rightarrow g^{-1}(d + A)g \\ \underline{A'_m(x') = \frac{\partial x^y}{\partial x'^m} A_y(x) + \partial_m \theta(x)} &\quad \left| \begin{array}{l} E \cong M \times F \\ G \times F \rightarrow F \end{array} \right. \end{aligned}$$

- Sections of this bundle $\psi: M \rightarrow F$ are fields $\psi(x)$ w/ electric charge q

$$\psi(x) \rightarrow e^{iq\theta(x)} \psi(x)$$

$$\text{Here } G = U(1) \ni e^{iq\theta(x)}$$

- Global issues: $A_m(x)$ may not be a continuous

globally continuous θ over all of M . We only have A_m defined in each chart. In an overlap we have $A'_m(x) = A_m(x) + \partial_m \theta(x)$

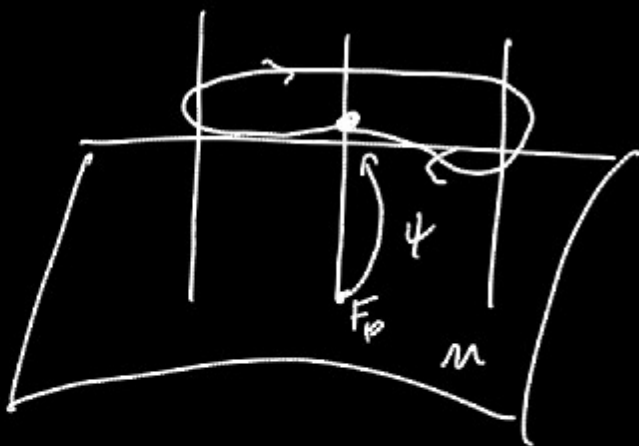
Dirac Quant. Condition:

$$D_m \psi = (\partial_m - iqA_m) \psi$$

$$\psi \rightarrow e^{iq\theta} \psi, \quad A_m \rightarrow A_m + \partial_m \theta$$

$$D_m \psi \rightarrow e^{iq\theta} D_m \psi$$

* Therefore, globally we would expect $e^{iq\theta}$ to be globally defined over any spacetime pt.



$\therefore e^{iq\theta}$ as we follow θ over a closed path the initial & final values should differ by $2\pi n, n \in \mathbb{Z}$.

* Consider a closed 2-surface S^2 :

$$\int \int \frac{1}{2} F_{mn} dx^m dx^n$$

$$= \left(\int_{\partial N} + \int_S \right) \frac{1}{2} F_{mn} dx^m dx^n$$



Now, from Stokes theorem:

$$\left(\int_{\partial N} + \int_S \right) dF = \int_P (A_m^N - A_m^S) dx^m$$

where we take the orientation of p to be $(+)$ on N .

Now, recall $A'_m(x) - A_m(x) = \partial_m \theta$

$$\|A(x) - A(x')\|_p = 0$$

$$\int_{S^2} dF = \int_p d\theta = \theta \Big|_{\theta_i}^{\theta_f} = 2\pi n/q$$

$$\therefore \boxed{q \int_{S^2} dF = 2\pi n}$$

→ Consider a magnetic monopole solution:

$$\vec{B} = p \frac{\vec{r}}{4\pi r^3}$$

where $F_{ij} = \epsilon_{ijk} B^k$. Here $r=0$ is singular.

$$\frac{1}{2} F_{\mu\nu} dx^\mu dx^\nu = \frac{1}{2} \epsilon_{ijk} B^k dx^i dx^j$$

In spherical coords:

$$\frac{1}{2} \epsilon_{ijk} x^k dx^i dx^j = r^3 \sin\theta d\theta d\phi$$

$$q\left(\frac{p}{4\pi}\right) \int_{S^2} \sin\theta d\theta d\phi = pq = 2\pi n$$

Dirac Quantization Condition: $\boxed{pq = 2\pi n}$

* Here, we required $e^{i\theta q}$ to be the same as we followed along a closed loop in the fibres. For Kähler mfd's we know that $K(z, \bar{z})$, the Kähler potential is only locally defined and we must require $K \rightarrow K + f + \bar{f}$ for each local patch.

• In sugra, fermions transform under this symm since $\chi^a \propto \psi$ and $\psi \mapsto \psi e^{f/a}$

$$\text{Note, } \partial_a K \mapsto \partial_a K + \partial_a f$$

$$\partial_{\bar{a}} K \mapsto \partial_{\bar{a}} K + \partial_{\bar{a}} \bar{f}$$

or, in terms of $\delta\psi$:

$$\delta(\partial_{\alpha} K) = \partial_{\alpha} f = -2i \partial_{\alpha} (\chi_m(f))$$

$$J = \begin{pmatrix} i\delta_a^b & 0 \\ 0 & -i\delta_{\bar{a}}^{\bar{b}} \end{pmatrix}$$

$$\text{Im}(f) = \frac{f - \bar{f}}{2i} \Rightarrow \partial_{\alpha} \bar{f} = 0.$$

We have:

$$\omega_a = \{\omega_a = -\frac{1}{2}i\partial_{\alpha} K, \omega_{\bar{a}} = \frac{1}{2}i\partial_{\bar{a}} K\} = -\frac{1}{2}J^{\alpha}_{\bar{a}}\partial_{\alpha} K$$

where ω_a is the gauge field analogous to A_{μ} above. and so

$$\delta\omega_a = \partial_a(\chi_m f)$$

This has non-vanishing curvature:

$$\begin{aligned} R^{(2)} &\equiv \frac{1}{2}(\partial_a \omega_b - \partial_b \omega_a) dz^a \wedge dz^b = (\partial_a \bar{\omega}_{\bar{b}} - \partial_{\bar{b}} \omega_a) dz^a \wedge d\bar{z}^{\bar{b}} \\ &= i\partial_a \partial_{\bar{b}} K dz^a \wedge d\bar{z}^{\bar{b}} = -\frac{1}{2}\Omega \end{aligned}$$

Here Ω is the fundamental 2-form.

• We saw that the $\chi_m(f)$ of f remains in gauge fixing of the T-transf.

$$\lambda_T \propto \frac{i}{2a}(f(z) - \bar{f}(\bar{z})).$$

Thus, as for the Dirac monopole:

$$D_c = \partial_a - ig' \omega_a$$

∴ The Argument states: $g' \iint_{C_2} R^{(2)} = 2\pi n$

or

$$g' \iint \Omega = 4\pi n \quad (\text{Kähler-Hodge Condition})$$

all Kähler charges must be a multiple of g' ∇ non-contractible 2-cycles. Such a mfd. is called Kähler-Hodge.

• Ω is a closed 2-form. Hence, it defines a cohomology called the Kähler Class.

All metrics which differ by an exact form are of the same class.

$\therefore$ An integral over a closed 2-form gives the characteristic # for this class:

$$\int \Omega = \alpha = \int (g' - g^2)$$

• Kähler trans. act on other objects as U(1) trans. and so they define a line bundle.

→ Each line bundle has a unique first Chern class: Set of curvature 2-forms defined up to a set of an exact differential

$$R^{(2)} = dW + W \wedge W$$

[Note, we would usually have

$$R^{(2)ab} = dW^{ab} + W^a{}_c \wedge W^{cb}$$

However the symm. is U(1) so we do not have $[ab]$ indices.

• The 2-forms are $g^* R^{(2)}$

The Chern Class of any U(1) line bundle belongs to the integral cohomology class

Def: A Kähler mfd M is a Kähler-Hodge mfd iff $\exists$ a bundle $L \rightarrow M$ such that $c_1(L) = [K]$, where c_1 is the first Chern class and $[K]$ is the Kähler class.

Consider, now, a U(1) Bundle w/ $F_{\text{cur}} = 2\pi i F_{\text{cur}}$

Projective mlds for $N=1$ susym are Kähler Hodge mlds.

In a purely geometric setting:

$$iA_m = \frac{1}{2} a^{-1} (\partial_m \bar{z}^\alpha \partial_{\bar{\alpha}} K - \partial_m \bar{z}^\alpha \partial_\alpha K) + \frac{1}{2} \partial_m (\ln(V/\bar{V}))$$

$$\partial_\nu A_m = \frac{1}{2} a^{-1} (\partial_\nu \partial_m \bar{z}^\alpha \partial_{\bar{\alpha}} K + \partial_m \bar{z}^\alpha \partial_\nu \partial_{\bar{\alpha}} K - (1) \\ * \frac{1}{2} \partial_\nu \partial_m \ln(V/\bar{V}))$$

We note, the first and last terms will vanish by anti sym/sym

$$\partial_\nu A_m = \frac{1}{2} a^{-1} (\partial_m \bar{z}^\alpha [\partial_\nu z^\alpha \partial_{\bar{\alpha}} K + \partial_\nu \bar{z}^\beta \partial_{\bar{\beta}} K])$$

$$\partial_\nu A_m = \frac{1}{2} a^{-1} (\partial_m \bar{z}^\alpha \partial_\nu z^\alpha \partial_{\bar{\alpha}} K - \partial_m \bar{z}^\alpha \partial_\nu \bar{z}^\beta K_{\alpha\bar{\beta}})$$

$$\begin{aligned} i(\partial_\nu A_m - \partial_m A_\nu) &= \frac{1}{2} a^{-1} (1) - \frac{1}{2} a^{-1} (\partial_\nu \\ &= a^{-1} (\partial_m \bar{z}^\alpha \partial_\nu z^\alpha K_{\alpha\bar{\alpha}} - \partial_m \bar{z}^\alpha \partial_\nu \bar{z}^\beta K_{\alpha\bar{\beta}}) \\ &= -\frac{2}{a} K_{\alpha\bar{\alpha}} [\partial_m \bar{z}^\alpha \partial_\nu \bar{z}^\beta] \end{aligned}$$

$$F_{m\nu} = -\frac{2i}{a} \partial_\alpha \partial_{\bar{\beta}} K (\partial_m \bar{z}^\alpha \partial_\nu \bar{z}^\beta)$$

* The same arguments show

$$\iint \frac{1}{2} F_{m\nu} dx^m \wedge dx^\nu = \frac{2\pi n}{g}$$

g is the "minimum" coeff. of the coupling of A_m to physical fields in the theory.

The Gaugino, λ^A , has coupling $g = 3/2$

$$\text{Since } D_m^{(0)} \lambda^A \sim -\frac{3}{2} i A_m \gamma_\mu \lambda^A$$

$$\text{Gaugino: } \psi_m: \sim -\frac{3}{2} i A_\mu \gamma_\mu \psi_p$$

$\therefore$ For $N=1$ susym, $g = 3/2$

$$PPLE \dots \dots \dots$$

Space-time S^2

2-cycle in $\mathbb{R}^{n+1}$

$$= \frac{1}{2a} \int \Omega = \frac{2\pi n}{4}$$

Kodaira Embedding Thm: A compact Kähler Hodge manifold can be embedded into a projective space.

→ T-sym has non-trivial $U(1)$ action on the scalar manifold: $U(1)_R$ generates Kähler after gauge fixing.

ex: $\mathbb{CP}^1$ w/ one chiral mlt.:

$$K = b \ln(1 + z\bar{z}), \quad g_{z\bar{z}} = \frac{b}{(1+z\bar{z})^2}, \quad R = \frac{4}{b}$$

$$\Omega = -2i \frac{b}{(1+z\bar{z})^2} dz \wedge d\bar{z}$$

$$\therefore N = -a Y \bar{Y} \exp\left(-\frac{K}{a}\right)$$

$$= -a Y \bar{Y} (1 + z\bar{z})^{-b/a}$$

$$= -a X^0 \bar{X}^0 \left(1 + \frac{X^1 \bar{X}^1}{X^0 \bar{X}^0}\right)^{-b/a}$$

$$= -a (X^0 \bar{X}^0)^{1+b/a} (X^0 \bar{X}^0 + X^1 \bar{X}^1)^{-b/a}$$

- The Matrix of second deriv. of N , for $a, b > 0$ has indefinite signature.

$$\int dz \wedge d\bar{z} (2i) \frac{1}{(1+z\bar{z})^2} = 4\pi$$

$$\therefore \frac{3b}{2a} \leq \frac{bK^2}{2} = n$$

"Quantization of Newton's Constant" in terms of b .

All $\mathbb{CP}^n$ have closed homology 2-cycles. Hence Hodge-Kähler is relevant in all cases.

the 1990s, the number of people in the world who are under 15 years of age is expected to increase from 1.1 billion to 1.5 billion.

As the world's population grows, the demand for food and other resources will increase. This will put pressure on the environment and on the world's food supply.

One way to meet this demand is to increase the amount of food that is produced. This can be done by using more land for agriculture.

Another way to meet this demand is to increase the efficiency of food production. This can be done by using better farming techniques.

Both of these methods have their own problems. Increasing the amount of land used for agriculture can lead to deforestation and loss of biodiversity.

Increasing the efficiency of food production can lead to the use of more pesticides and fertilizers, which can be harmful to the environment.

One solution is to use a combination of these methods. This would involve using better farming techniques on existing farmland.

Another solution is to use more land for agriculture, but in a way that is sustainable. This would involve using techniques that protect the environment.

There are many other ways to meet the world's growing demand for food. The key is to find a way that is sustainable and that protects the environment.

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